

**Advanced Combustion via Microgravity Experiments ([ACME](#))
Cool Flames Investigation with Gases ([CFI-G](#))**

ACME/CFI-G Final Report (2022*)

Prepared by:

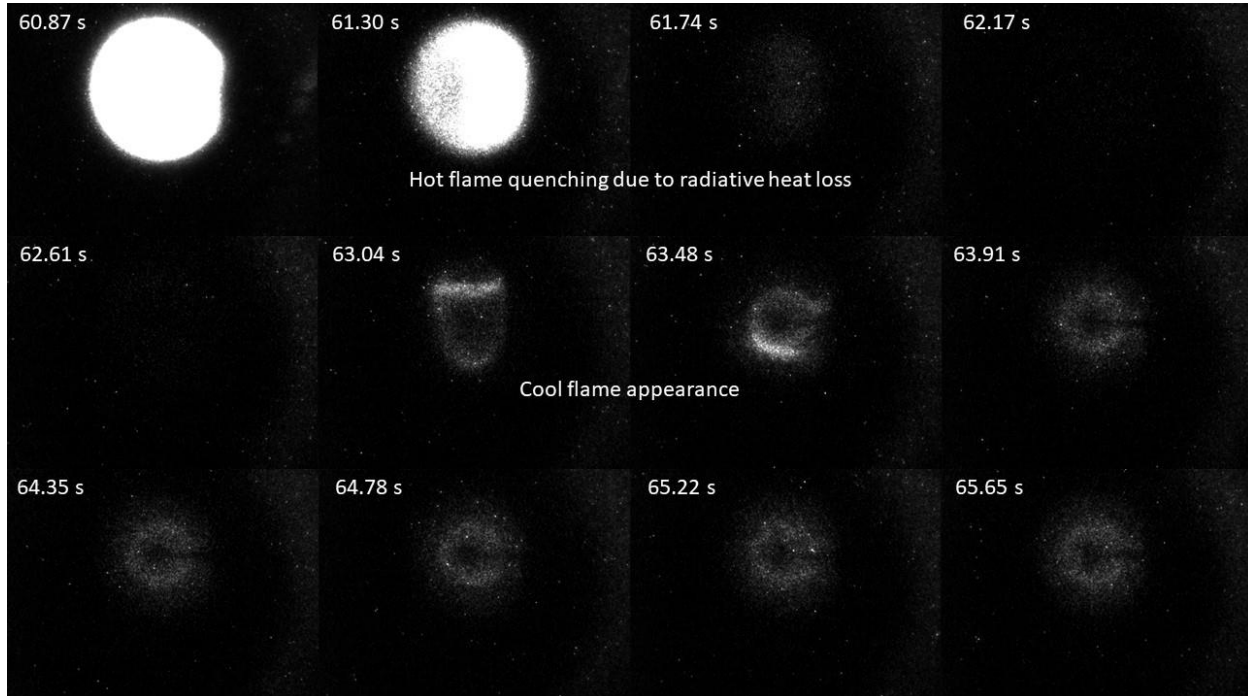
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Cool flame appearance after hot flame quenching in a composite time sequence of intensified images.

**Data analysis and the publication of results is still underway, so it is expected that this report will be updated in the future with at least a revision of the list of papers and presentations.*

Table of Contents

	Page
1. Overview	3
2. Science Overview	4
A. Background	4
B. Rationale for Microgravity	5
C. Science Objectives	6
3. Hardware Overview	6
A. General Description	6
B. Hardware Shortcomings	15
4. Preliminary Results	17
A. Changed Test Plans	17
B. Overview of Experiment Test Operations	18
C. Summary of Preliminary Results	23
D. Surprises and New Discoveries	23
E. Accomplishment of Approved Objectives	23
F. Additional Objectives Achieved	24
G. Impacts, Benefits, and Products	24
H. Patents	24
5. Appendices	25
A. Abbreviations and Acronyms	25
B. 30-Day Operational Accomplishments Report for ISS Increment 64	28
C. 30-Day Operational Accomplishments Report for ISS Increment 65	35
D. Operational Timeline	48
E. Astronauts who Conducted ISS Operations for CFI-G	50
F. ISS Operations per the Payload Weekly Science Summary	51
G. Crew Activity per Space Station Science Highlights	68
H. Crew Activity per the ISS Daily Summary Report	72
I. Crew Activity per the NASA Space Station Blog	79
J. Crew Activity in Publicly Available NASA Photos	80
K. ACME Operations per @ISS_Research on Twitter	81
L. ACME Operations per Astronauts on Twitter	85
M. Summary of the ISS data collection	86
N. Publications, Presentations, Awards, and Outreach	87
1. Publications and Presentations <u>with ISS Results</u>	87
2. Publications and Presentations (comprehensive list)	87
3. Awards	88
4. Outreach	88

1. Overview

The Cool Flames Investigation with Gases ([CFI-G](#)) experiment is one of six independent experiments that make up the Advanced Combustion via Microgravity Experiments ([ACME](#)) project for which the tests were conducted within the Combustion Integrated Rack ([CIR](#)) on the International Space Station ([ISS](#)).

The experiment's Principal Investigator (PI) is Prof. Peter B. Sunderland of the [Department of Fire Protection Engineering](#) at the [University of Maryland](#). The two Co-Investigators (Co-Is) are Profs. Richard L. Axelbaum and Forman A. Williams who are respectively from the [Washington University of St. Louis](#) and the University of California, San Diego ([UCSD](#)).

Meanwhile, Dennis P. Stocker of the NASA Glenn Research Center is the Project Scientist (PS) for ACME including CFI-G and the rest of the ACME investigations. In ACME's preparatory drop testing, etc., Dr. Fumiaki Takahashi of the Case Western Reserve University ([CWRU](#)) served as the Deputy Project Scientist. For ACME's spaceflight operations, the NASA science team included Drs. Uday G. Hegde and Vedha Nayagam who are also with CWRU.

The ISS operations for CFI-G were conducted from March to Oct. 2021 in two sequential rounds of testing. The two rounds occurred during Increments 64-65 and 65-66 and respectively featured normal and inverse flames. The inverse flame tests were conducted with the chamber filled with a mixture of nitrogen and gaseous fuel while an oxygen/nitrogen mixture flowed from the burner. As a result, the inverse flames were effectively inside out, e.g., where the soot formed outside rather than inside of the flames. Although it was in the second round, the final day of CFI-G testing was carried out in the normal flame configuration, with gaseous fuel (with or without nitrogen) flowing into an oxygen/nitrogen atmosphere. This was an added test day carried out under conditions in which low-temperature combustion occurred during the first round of testing.

CFI-G was a late addition to ACME, where the experiment was jointly selected for spaceflight by the National Science Foundation ([NSF](#)) and the Center for the Advancement of Science in Space ([CASIS](#)) rather than NASA. The selection was notably announced after the core ACME hardware had launched to the ISS and just weeks before ACME's on-orbit operations began. It is effectively an extension of ACME's [Flame Design](#) experiment for which the PI and Co-I are respectively Axelbaum and Sunderland. And it is likewise a follow-on to the Cool Flames Investigation ([CFI](#)), led by Williams, in which liquid droplets were burned rather than gaseous fuel.

Although CFI-G wasn't selected for spaceflight by NASA, its ISS operations and the development of its unique hardware, software, and procedure were carried out by the same teams of personnel as the other ACME experiments, primarily from ZIN Technologies, Inc.

2. Science Overview

A. Background

This background is the Project Summary from the experiment's 2017 proposal in response to a solicitation from the NSF/CASIS Collaboration on Combustion and Thermal Transport Processes Research on the International Space Station to Benefit Life on Earth.

Overview:

Cool diffusion flames were discovered in 2012 aboard the ISS. This spawned a rapidly growing research field, but only a small number of measurements at limited conditions are available. Consequently, cool flame kinetics mechanisms cannot be adequately tested or advanced. This could be impeding the development of cleaner and more efficient internal combustion engines.

It is proposed here to observe cool diffusion flames supported by porous spherical burners fed with propane, n-butane, and n-pentane. Normal and inverse flames will be considered for a broad range of dilution conditions and pressures. The ISS is the ideal platform for observing cool diffusion flames because they typically require 20 s to establish and they could burn indefinitely.

Tests will be performed using the ACME insert in the CIR. Minimal hardware development will be required because the spherical porous burners and gas delivery system will be those used by the Flame Design flight experiment, while the diagnostics will be those used by the CFI flight experiment.

The objectives are to observe diverse cool diffusion flames aboard the ISS, to simulate the tests with an advanced CFD model, and to identify or develop a chemical kinetics mechanism that accurately models these conditions.

Intellectual Merit:

Most internal combustion engines are designed using computer models that neglect cool flame chemistry, but ignition and flame propagation in engines depend on cool flame chemistry. Cool flame chemistry also has a significant impact on fuel octane and cetane numbers, whose understanding has large economic consequences.

The proposed experiments will take advantage of many state-of-the-art diagnostics that are already available on the ISS. The flame system to be studied will allow steady spherical cool diffusion flames to be observed for the first time. An advanced CFD model will simulate the flames and will aid the testing and development of cool flame kinetics models.

Observing cool flames burning as steady spherical diffusion flames will yield valuable insight into cool flame chemical kinetics. The measurements proposed cannot be performed in normal gravity or in ground-based microgravity facilities.

Broader Impacts:

The research will be disseminated widely to the combustion research community and to industry. It should advance the state-of-the-art in cool flame kinetics mechanisms, which in turn could lead to cleaner, more efficient internal combustion engines.

This project will have several educational impacts. Graduate students will conduct much of the

research. The research results will be integrated into three combustion courses taught by the PI. The PI will team with two organizations at UMD - the Women in Engineering Program and the Center for Minorities in Science and Engineering - to foster interest in science and engineering among female and minority secondary school students. A day-long flame laboratory will be developed for secondary school students that will foster their interest in combustion and scientific inquiry.

B. Rationale for Microgravity

This is the 'Justification for Testing Aboard the ISS' section (3.1) of the experiment's 2017 proposal in response to a solicitation from the NSF/CASIS Collaboration on Combustion and Thermal Transport Processes Research on the International Space Station to Benefit Life on Earth. The included equation is formatted differently in this copy.

Previous ground-based microgravity experiments have demonstrated the scientific value of spherical, burner-supported diffusion flames. These one-dimensional flames offer a simple, fundamental configuration to improve the understanding of diffusion flame behavior and to explore such effects as transport, kinetics, soot formation, and radiation, which can be more pronounced under microgravity conditions (Atreya et al. 1992; Law et al. 2001; Lecoustre et al. 2012).

Spherically-symmetric flames are not generally practical in normal gravity. This can be explained by considering the Richardson number, which is the ratio of the buoyancy force to the inertial force,

$$Ri = g (\Delta T L) / (T u^2), \quad (1)$$

where g is the acceleration of gravity, T is temperature, L is the characteristic length scale, and u is the characteristic velocity. Taking as representative values $L = 50$ mm, $u = 5$ cm/s, $\Delta T = 2000$ K, and $T = 300$ K, $Ri = 1300$ in normal gravity. Ri can be reduced with a judicious choice of fuels and inert; Law et al. (2001) have shown that nearly spherical flames can be created in normal gravity using low pressure hydrogen inverse flames. However for typical spherical diffusion flames the only way to obtain $Ri \ll 1$ is to reduce g by four orders of magnitude. This requires a microgravity environment.

Available microgravity drop tower facilities do not offer sufficient time to observe quasi-steady spherical diffusion flames, let alone their radiative extinction followed by cool diffusion flames. Experiments performed in the 5 s drop facility confirm that typical spherical diffusion flames continue growing during 5 s (see Fig. 2). Observations of cool flames in the FLEX, FLEX-2, and CFI droplet experiments involved or will involve burn times of up to 60 s.

Analytical work supports these observations of slow flame development. Radial velocities in spherical diffusion flames scale with $1/r^2$, where r is radius. Thus, at only a few burner diameters distance from the sphere, diffusion is the dominant transport mechanism. The characteristic diffusion time is r^2/D , where D is mass diffusivity. For $r = 25$ mm and $D = 20$ mm²/s, this diffusion time is 30 s. Decreasing the flame size can reduce this time, but will not permit the observation of radiative extinction followed by cool diffusion flames. Our analytical and numerical calculations indicate that approximately 30 s of jitter-free (less than 10^{-5} g) microgravity is needed to approach quasi-steady state for these flames. The ISS is the only microgravity platform that can provide this environment.

C. Science Objectives

The objectives text presented here is the 'Hypotheses and Objectives' section (1.3) of the experiment's 2017 proposal in response to a solicitation from the NSF/CASIS Collaboration on Combustion and Thermal Transport Processes Research on the International Space Station to Benefit Life on Earth.

The hypotheses of this research are that testing aboard the ISS can demonstrate the following for the first time, contributing to an improved understanding of cool flame kinetics.

1. Steady-state cool diffusion flames on spherical porous burners.
2. Cool diffusion flames burning propane, n-butane, and n-pentane.
3. Cool diffusion flames for a wide range of Z_{st} , T_{ad} , and pressures.
4. Inverse cool diffusion flames.
5. Cool diffusion flames for which a decrease in fluent flow rate will cause extinction, whereas an increase will cause a transition to a hot diffusion flame, with oscillation possible between hot and cool flames.

The research objectives of the proposed work are as follows.

1. Plan and support flight testing on the ISS.
2. Determine the conditions for which cool diffusion flames occur for propane, n-butane, and n-pentane; normal and inverse flames; and a broad range of Z_{st} , T_{ad} , and fluent flow rate.
3. Support the flight tests with advanced CFD simulations.
4. Identify or develop a chemical kinetics mechanism that yields good agreement between the CFD predictions and the ISS measurements.
5. Disseminate the results as widely as possible.

In contrast to objective 2 (and hypothesis 3), it was not feasible with the Combustion Integrated Rack (CIR) and ACME chamber insert to conduct tests with n-pentane as proposed (because of its low vapor pressure) and testing was instead conducted with ethane as an alternate.

3. Hardware Overview

A. General Description

This section is copied from section 2.A of the April 2021 version of 'ACME Engineering Implementation' which should be consulted for additional details. References to other sections within the boxed text of this section refer to the ACME Engineering Implementation document.

This section provides a general description of the ACME engineering from the '3.0 Experimental' section of the following 2014 conference paper:

Dennis P. Stocker, Fumiaki Takahashi, J. Mark Hickman, and Andrew C. Suttles, "Gaseous Non-Premixed Flame Research Planned for the International Space Station," Spring Technical Meeting of the Central States Section of the Combustion Institute, Tulsa, OK, 16–18 March 2014.

IMPORTANT: The 2104 paper from which this section was extracted proceeded completion of the spaceflight hardware which was mostly launched in 2017. As a result, it does not accurately describe the final engineering in some regards. Therefore updates, clarifications, additional details, shortcomings, references, etc. have been included in boxed notes like this following the original description. As a minor note, the sizes of figures were adjusted in this report, but none were cropped.

As previously discussed, the ACME experiments will be conducted within the Combustion Integrated Rack (CIR) using a chamber insert, depicted in Figure 2, which was designed to accommodate gas delivery to a variety of exchangeable burners. Besides the chamber insert, other ACME-unique hardware includes a high-definition color camera and an avionics package, which is for data acquisition and control.

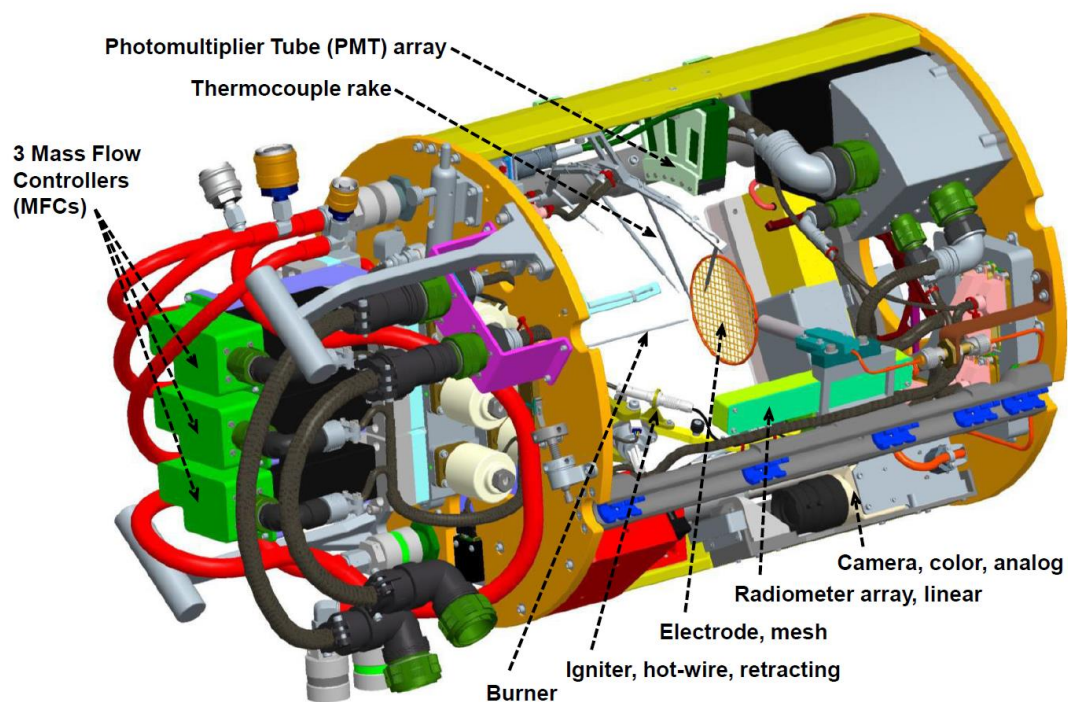


Figure 2: Design of the Advanced Combustion via Microgravity Experiments (ACME) chamber insert that will be used in the Combustion Integrated Rack (CIR) on ISS.

ACME's chamber insert, which is ~0.4 m in diameter, provides common mounting for a variety of burners on the chamber axis (but only one at a time) and includes a retracting resistive igniter. As depicted in Figure 3, there will initially be gas-jet burners (with inner diameters ranging from 0.4 to 3.2 mm), a small coflow burner (where the two concentric tubes have inner diameters of 2.1 and 25 mm), porous spherical burners (with diameters ranging from 6.35 to 12.7 mm), and two flat porous burners for the BRE experiment (with inner diameters of 25 and 50 mm). Most burners will be equipped with a surface thermocouple, and each BRE burner will have two heat flux transducers. The internal pressure of the porous burners will be measured to allow monitoring of the pressure differential across the outlet surface. The igniter will have a crew-exchangeable tip and arm (to accommodate the various burners) and its stepper-motor drive will allow for precise positioning.

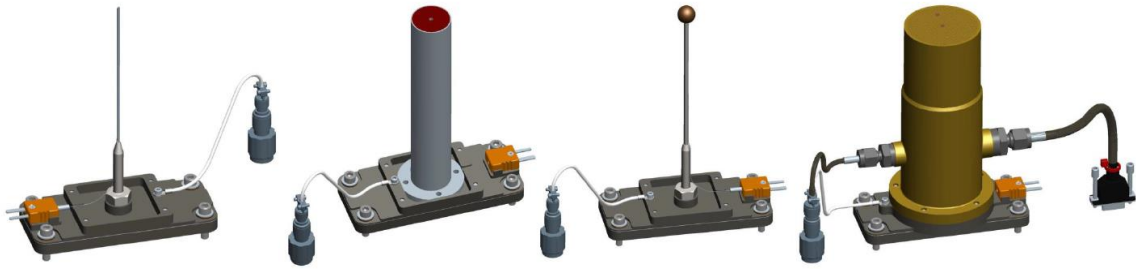


Figure 3: ACME burner types, including from left to right, gas-jet burners, a co-flow burner, porous spherical burners, and flat porous burners for the BRE experiment.

Ultimately, the spherical burners were limited to a single diameter of 6.35 mm, and the fabricated gas-jet burners were limited to two sizes with nominal inner diameters of 1.3 and 2.1 mm. However, only the larger gas-jet burner was used during the spaceflight tests. The burners are addressed in detail in section 3.C.

Unfortunately, the spherical burners have not always yielded spherically symmetric flames, presumably as a result of (1) greater burner flow through the joint between the porous sphere and supply tube, and (2) heat loss to the support tube (encompassing the supply tube) causing local flame quenching.

Two types of igniter tips have been used during ACME testing. Most tests have been conducted using a loop igniter tip like those which had been successfully used in the droplet combustion experiments which preceded ACME research in the CIR. However, some tests were conducted with a coil igniter tip, where that included the CFI-G investigation and the second-round testing for both the Burning Rate Emulator (BRE) and Structure and Response of Spherical Diffusion Flames (s-Flame) experiments, with the exception of the latter's first day testing. For more information about the igniter tips, see section 3.D.

The chamber insert includes three mass flow controllers (visible on the left of Fig. 2) for the delivery of fuel, oxidizer or inert (e.g., helium or carbon dioxide), and nitrogen to the burner. The maximum fuel flow is 2 slpm (on a nitrogen basis), whereas much higher flows, e.g., 10 slpm (on a nitrogen basis), are possible for the oxidizer and nitrogen lines. The crew can exchange the mass flow controllers to provide appropriate ranges that maximize the flow accuracy. The gas delivery system enables on-orbit dilution of the fuel or oxidizer (but not both simultaneously), coflow, inverse flames (e.g., oxidizer delivery to the center of a coflow burner), and premixed flames, although the latter are not planned for any of the current ACME experiments. Gases can also be delivered premixed to the space station where that is necessary in some circumstances, such as the blending of different fuel gases. CIR's standard bottle sizes are 1, 2.25, and 3.8 liters but ACME will sometimes use smaller bottles as a hazard control to limit the deficient reactant (e.g., fuel) that can be released into the combustion chamber.

Thus far, ACME tests were not conducted with carbon dioxide as a diluent. Furthermore, the fuel diluted with helium was premixed prior to launch rather than on orbit. Ultimately, oxidizer flows of up to 20 slpm (on a nitrogen basis) were possible, while the nitrogen flow was limited to a

maximum of 5 slpm. ACME has not used gas bottles smaller than 1 liter as indicated above; the bottle fill pressure was instead limited as an additional hazard control.

ACME's primary flame diagnostics are a suite of imaging systems including five video cameras and two illumination systems, of which three cameras are part of the CIR facility and are already in use on orbit. All but one of the cameras is mounted outside of the chamber. Collectively, the imaging systems will enable color, CH*, and OH* imaging; pyrometry; measurement of the soot volume fraction; and operational support. However, it is unlikely that all functions would be exercised at any one time as will become apparent in the subsequent discussion.

Color imaging is provided by two cameras: a high-definition digital camera mounted outside of the chamber and a standard analog camera mounted on the chamber insert (with a turning mirror). Both cameras are part of the ACME hardware that is now in development. The analog camera signal will be downlinked in near real time to support the experiment operations. Light-emitting diodes within the chamber can be used to illuminate the igniter and burner to both support the operations and provide reference images. The 12-bit digital camera records at up to 30 full frames (1360x1024) per second, and is equipped with a motorized lens with independent control of the zoom, iris, and focus. The digital camera will be equipped with a filter assembly with four options, where selection of a clear "filter" allows for color imaging. Color imaging can also be carried out with a blue-green BG7 Schott glass filter, which balances the color planes to avoid saturation in the red while maintaining sensitivity in the blue and green. The natural flame color can be approximately reconstructed, but the primary purpose of the filter is pyrometry and measurement of the soot volume fraction, again via the techniques of [23-24]. Alternately, CH* imaging can be conducted with this camera because the filter assembly includes a 430-nm bandpass filter with a 10-nm full width at half-maximum transmission (FWHM). A 450-nm bandpass filter (again with a 10-nm FWHM) can be used with the CH* filter to effectively subtract interfering broadband soot emission. However, the filter assembly rotates slowly so such subtraction effectively requires a separate test with each filter and of course reproducible flame conditions.

The video from the analog operations (i.e., ops) camera, also referred to as the internal camera, is not archived in space. The video is instead recorded on the ground in the operations center. Its gain was sometimes adjusted for improved imaging of dim flames, e.g., approaching extinction.

The sensitivity of the color digital camera, often referenced as the ACME DataCam (i.e., Data Camera), to the dim microgravity flames was such that it was often used with long exposure times and thus low framing rates, for example on the order of 1 frame per second. The clear "filter" is of fused silica. Unfortunately, the thickness of the blue-green BG7 Schott glass filter and the added antireflective (AR) coating caused too great a reduction in the red, so it does not balance the intensity of the three color planes as intended and thereby can limit the accuracy of temperature determination. Meanwhile, rather than the 450-nm bandpass filter described above, the high definition color camera was equipped with a bandpass filter for imaging formaldehyde emissions. For more information, see the Data Camera Characterization Report (ACME-RPT-010 Rev. A).

Imaging of the dim hydrogen flames in the Structure and Response of Spherical Diffusion Flames (s-Flame) experiment was particularly challenging with these, if not all, cameras.

Pyrometry can alternately be accomplished with CIR's monochrome High-Bit Depth Multispectral (HiBMs) camera, equipped with a Liquid-Crystal Tunable Filter (LCTF), which allows for imaging at multiple spectra, but requires a quasi-steady flame for measurement at each desired band. This camera has a telecentric optical system and is recorded at 512x512 and 30 frames per second (fps). Although it has a 1024x1024 pixel array, binning has been necessary because of signal loss through the LCTF. Both soot pyrometry and Thin Filament Pyrometry (TFP) are possible, where ACME is equipped with a crew-replaceable array of 5 silicon carbide (SiC) fibers. The fibers are orthogonal to the flame axis and the array can be translated along the flame axis to the desired position downstream of the burner, e.g., based on the downlink of the analog video. During set up, the crew can manually orient the array so that it is orthogonal to either ACME's color digital camera or the HiBMs camera with the LCTF. An alternate fiber array including a platinum wire will be used for inflight characterization, which in combination with preflight calibration will enable determination of the soot volume fraction [24].

In contrast to the preceding paragraph, ACME did not use a HiBMs camera for pyrometry, but instead relied on use of ACME's color DataCam for such temperature measurements. For an example of how this can be done, see the techniques described in the following papers:

P.B. Kuhn, B. Ma, B.C. Connelly, M.D. Smooke, and M.B. Long, "Soot and Thin-filament Pyrometry Using a Color Digital Camera," Proceedings of the Combustion Institute, 33 (2011) 743-750.

B. Ma, M. B. Long, "Absolute light calibration using S-type thermocouples," Proceedings of the Combustion Institute, 34 (2) (2013) 3531-3539.

Thus far, TFP tests have been conducted with 14-micron fibers spaced apart by 5 mm for a total span of 20 mm (in the case of 5 intact ceramic fibers), neglecting optional translation of the array. Tests have been conducted using arrays with and without the platinum wire in the furthest downstream position. Tests have also been conducted with some of the fibers broken and subsequently removed, leaving only three fibers in the final testing for the BRE and Coflow Laminar Diffusion Flame (CLD Flame) experiments and for all of the CFI-G testing.

CIR has a second HiBMs camera, without a filter, that will be used in combination with CIR's illumination package for alternate measurements of the soot volume fraction. This will be accomplished via an extinction technique previously employed in microgravity [3,6,26], where the camera and the collimated light source (with its two fiber-coupled laser diodes) are positioned at opposite windows. The images are recorded at 1024x1024 and 30 fps. Thus far, the primary purpose of this combined system has been the measurement of the droplet diameter for determination of the droplet burning rate.

A HiBMs camera without the Liquid Crystal Tunable Filter (LCTF) has been used throughout much, but not all, of ACME's operations. In the first round of the CLD Flame experiment, it was used without a filter for flame imaging, but this proved to be largely unsuccessful because the camera is very insensitive to blue. Afterward, it was equipped with a red filter corresponding to the CIR illumination package for the rest of the ACME testing. Unfortunately, that filter is sufficiently broad that significant incandescence can pass through the filter limiting its utility for determinations of soot volume fraction via light extinction, e.g., through the technique described in the following papers:

D.L. Urban, Z.G. Yuan, P.B. Sunderland, K.C., Lin, Z. Dai, and G.M. Faeth, "Smoke-Point Properties of Nonbuoyant Round Laminar Jet Diffusion Flames," Proceedings of the Twenty-Eighth Symposium (International) on Combustion 28 (2000) 1965-1972.

F.J. Diez, C. Aalburg, P.B. Sunderland, D.L. Urban, Z.-G. Yuan, G.M. Faeth, "Soot properties of laminar jet diffusion flames in microgravity," Combustion and Flame, 156 (2009) 1514-1524.

P.S. Greenberg and J.C. Ku, "Soot Volume Fraction Imaging," Applied Optics, 36 (22) (1997) 5514-5522.

Nonetheless, soot volume fraction can also be determined from the color DataCam images.

During ACME operations, the HiBMs camera is typically archived at 15 fps because of data limits for transfer and downlink operations. And because operations are constrained by the data limits, image data from this camera are sometimes deleted without downlinking to enable additional testing. And in some testing, HiBMs images aren't even acquired given their lesser value.

Unfortunately, the HiBMs backlight illumination is significantly obstructed by the bracket for the electrode mesh further limiting its utility for the E-FIELD Flames experiment. This was a design trade-off necessitated primarily by ACME's large number of cameras and other optical instruments.

CIR's Low-Light-Level Ultraviolet (LLL-UV) package is a monochrome camera with an intensifier (for 280-700 nm) and a 310-nm bandpass filter (with a 10-nm FWHM) for OH* imaging. Again as a result of the signal loss through the filter, binning is employed such that the OH* images are recorded at 512x512 and 30 fps.

During ACME operations, the LLL-UV camera has only been used for the Structure and Response of Spherical Diffusion Flames (s-Flame) experiment.

While not discussed in the preceding section, another intensified camera was used throughout ACME testing, although its image data was sometimes deleted or not even acquired to allow the performance of additional tests. This was the Cool Flames Investigation (CFI) camera which was generally configured with its CH filter, where an exception is some s-Flame testing with particularly dim flames. Due to design trade-offs, its Field of View (FOV) was partially obstructed by the arms of the Thin Filament Pyrometry (TFP) fiber array when it was moved inwards to the flame. More information about this camera can be found in its characterization report, CIR-RPT-4600.*

The ACME chamber insert will also be equipped with non-imaging optical diagnostics including both thermopile radiometers and Photomultiplier Tubes (PMTs). There is a crew-exchangeable linear array of radiometers of which there are two versions supporting the initial set of experiments. The array has a fixed position which is parallel to the chamber axis and centered (along the chamber length) relative to the chamber windows and thus most of the other instrumentation. One array is equipped with a single wide-view radiometer with two narrow-view radiometers that are downstream and two more that are upstream, where all views are orthogonal to the chamber axis, as depicted in Figure 4. The narrow-view radiometers are intended to support studies of gas-jet flames, where ACME can accommodate shorter burners, making the "lower" detectors useful. The other array, also seen in Fig. 4, provides for three wide-view measurements of the flame, where one view is masked to nominally block radiant emission

from the burner. It was specifically designed for the BRE experiment but will be a resource available to other ACME experiments. Given that the radiometer array can be exchanged by the crew, alternate arrays could be developed.

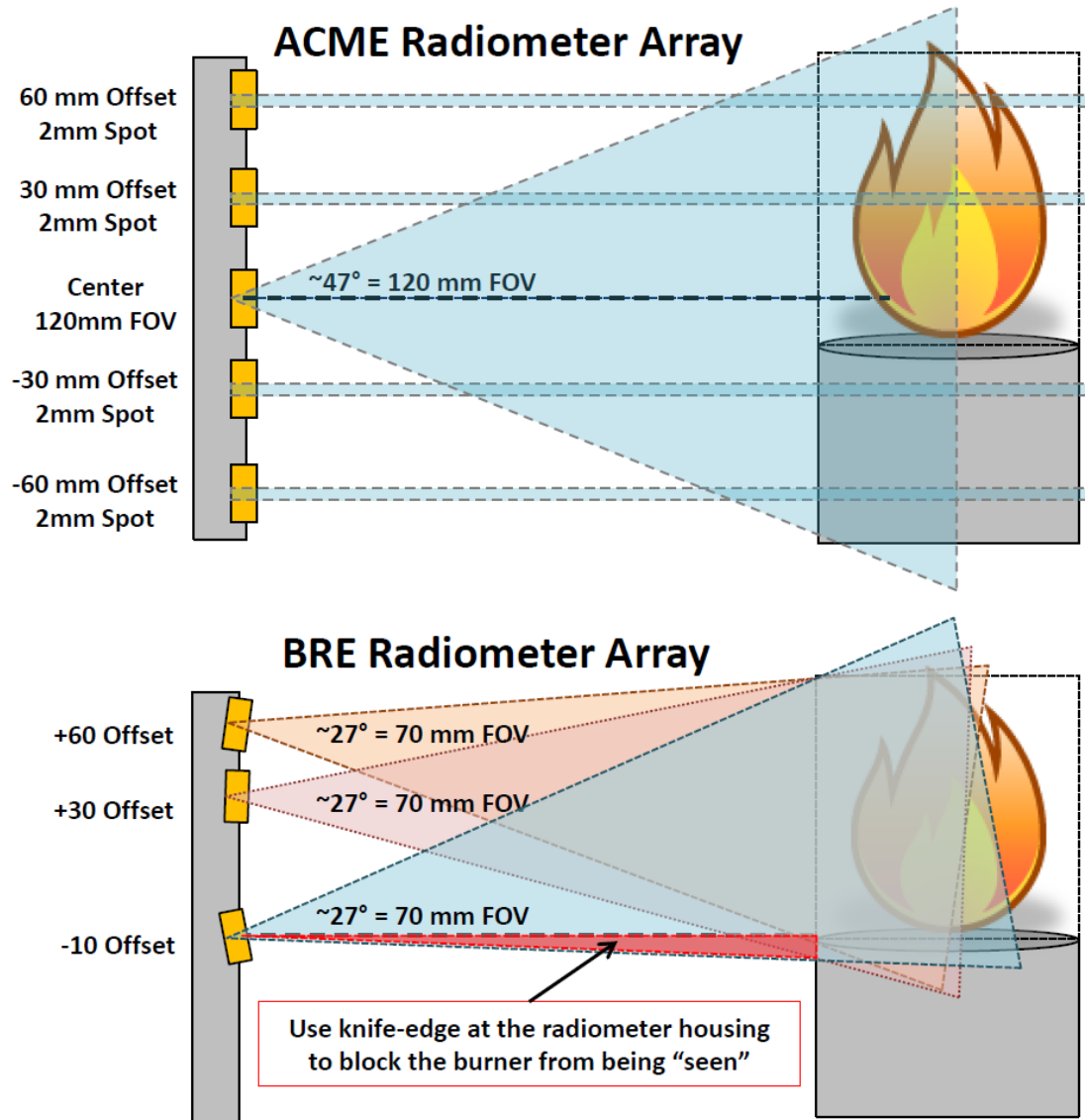


Figure 4: Crew-exchangeable radiometer arrays on the ACME chamber insert: (top) standard (ACME) array of 5 radiometers, and (bottom) array with three radiometers, designed for the BRE experiment, where the lowest detector is masked to avoid view of the burner.

A third radiometer array was used in the CFI-G experiment, where it is important to understand that only a single radiometer array can be used at a time. Unlike ACME's other radiometer arrays, each sensor in the CFI-G array has two fixed gains, i.e., a high gain and low gain. It is described in the Cool Flames Radiometer Characterization Report (CIR-RPT-4600).

The narrow-view radiometers in the standard array are only fully sensitive in small spots, but they are each sensitive in a much greater Field of View (FOV) than represented in the preceding figure.

They do not have lenses; their FOV is a result of an aperture where most of their FOV is within the penumbra which is only visible to a portion of the sensor. Figures representing the measured FOVs of the as-built standard and BRE radiometer arrays are found in section 3.F of this document.

Unfortunately, the radiometers in the BRE array typically saturated in the first round of BRE testing, so the standard array was used in its second round. However, those radiometers similarly saturated in many of round 2 tests. In this regard, each of ACME's BRE and standard radiometers have a fixed gain.

For more information about the standard radiometer array used for most ACME testing, see the Radiometer Characterization Report (ACME-RPT-011). Meanwhile, you can learn about the array used in the first round of BRE testing from the BRE Radiometer Characterization Report (ACME-RPT-012).

An additional wide-view radiometer, that is fixed and not crew exchangeable, is positioned downstream of the burner and flame at 30° from the chamber axis. This detector is primarily intended to support the spherical flame experiments and will have a quicker response than the more sensitive detectors that will be used in the linear arrays. Like the radiometers on the two linear arrays, it will have a spectral range of nominally 0.2-11 micron and its field of view will be established by an appropriately selected aperture.

The FOV of the off-set radiometer is also shown in section 3.F. Learn more about this sensor from the Radiometer Characterization Report (ACME-RPT-011).

To detect the quenching of dim flames, the chamber insert will be equipped with a set of three PMTs. Each PMT will have a wide-angle view, a sight line that is orthogonal to the chamber axis, and will be centered (along the chamber length) relative to the chamber windows so that it is aligned with most of the other instrumentation. The PMTs' spectral range will be 230-700 nm, i.e., visible and near ultraviolet, but one will have a 310-nm bandpass filter and another will have a 430-nm bandpass filter (where both are with a 10-nm FWHM). The pair of detectors will enable measurement of OH* and CH* emission, respectively. The third PMT will be equipped with a neutral density filter to prevent saturation. Like many other components, the PMT array can be removed from the chamber insert by the crew.

Unlike the radiometers, each PMT had a readily adjustable gain. However, the PMT signals sometimes saturated when the gains were set too high. See the PMT Characterization Report (ACME-RPT-013) to learn more about the three Photomultiplier Tube (PMTs).

The chamber insert is also equipped with six far-field thermocouples to measure the spread of the thermal field in the spherical flame experiments. They are of limited utility for other experiments because they are positioned upstream of the burner tip. Therefore, they will generally not be installed by the crew to reduce the chance of accidental damage. There will be two versions of a four-thermocouple rake to allow a choice of near and far measurements. Other arrays could be designed to support future experiments, but they cannot extend downstream of the burner tip without interfering with other (e.g., optical) measurements.

The two lone thermocouples, which are both at nominally 200 mm from the center of the spherical burner, have been installed throughout ACME testing. The tip of one was seen to be slightly bent early in testing, as discussed and depicted in section 3.G of this report.

Only the short (or far) version of the radiometer rake has been used thus far, where factors include the need for crew time to switch between rakes and the heat loss that could result from thermocouples in the long (close) rake. The short (far) rake's thermocouple tips should nominally be at 45, 60, 100, and 100 mm from the spherical burner center. For thermocouples at the same distance from the burner center, the angle between the rays from the burner center to the two thermocouples should be 90 degrees. The bead size of the thermocouples was not measured, but it is undoubtedly at least twice the size of the 0.2 mm diameter wire.

The two thermocouples closest to the burner outlets on each four-thermocouple rake are type B, while the others are type K.

Except for the final test day, CFIG's second-round testing was carried out where the fine tip of the thermocouple closest to the flame was bent at roughly 90 degrees. In the following section (3.B), this can be seen in Figure 5 where, as a size reference, the spherical burner is 6.35 mm in diameter.

Experiment-unique hardware is also mounted on the chamber insert. For example, to accommodate the E-FIELD Flames experiment, the insert is equipped with a high-voltage supply and a crew-removable electrode mesh which can be set between +10 and -10 kV. The circular mesh is orthogonal to the chamber axis and is positioned (during set up) at either ~35 or ~50 mm downstream of the insulated burner. The ion current between the burner and electrode mesh will be determined via measurement of the voltage drop across a known resistor and application of Ohm's Law.

Additional images of the chamber insert can be seen in subsequent sections of this report. The hardware specific to E-FIELD Flames is discussed further in section 2.B.

Additional supporting measurements will be made, where the chamber pressure is a simple example. The ACME experiments will use the CIR's gas chromatograph for pre- and post-test analysis of the chamber atmosphere, e.g., for oxygen, nitrogen, carbon dioxide, carbon monoxide, and fuel gases. Acceleration measurements will be provided by the Space Acceleration Measurement System (SAMS) [27], which is the primary instrument for such measurements on ISS and previously served that role for the space shuttle.

The CIR's Gas Chromatograph Instrumentation Package (GCIP) was not used for the CLD Flame or E-FIELD Flames investigations and thus far was only sometimes used during BRE, Flame Design, and s-Flame testing. Its use requires significant time and as such effectively reduces the number of tests that can be conducted. However, estimates of the chamber contents were computed assuming that the fuel in a normal flame test (for example) was either fully or not at all oxidized, e.g., for unsustained ignition in the latter case.

The CIR facility was equipped with a 3-axis SAMS sensor head and the acceleration measurements, along with relevant documentation, are accessible via the home page for the Principal Investigator Microgravity Services (PIMS), <https://qipoc.grc.nasa.gov/wp/pims/home/>. To be clear, the

acceleration measurements are not archived with the ACME experiment data in the Physical Sciences Informatics (PSI) website, www.nasa.gov/PSI.

All ACME experiments will be constant volume (i.e., rather than constant pressure) studies, where the chamber's free volume will be about 90 liters. Burner flows will generally be constrained to limit the pressure rise, and tests will typically be 0.5-2 minutes in duration.

The chamber's free volume is approximately 83 liters depending on the installed hardware.

The ACME experiments will be configured, but not conducted, by the ISS crew. Instead, the experiments will be commanded from the ground, specifically from the Telescience Support Center (TSC) at the NASA Glenn Research Center in Cleveland, Ohio. They can be conducted in a nominally automated mode using pre-programmed scripts. However, it will also be possible to effectively operate the experiments in a "manual" mode, where changes can be made in response to the analog video downlink and the numerical data available in the telemetry stream. For example, a flow rate could be adjusted to approach a stability or soot limit. Step and ramp changes can be made in the flow rate(s) or electric field potential. The TFP fiber array can be translated uniformly through a soot-free flame to get a continuum measurement of the temperature field. In summary, ACME's experiment control will allow for broad flexibility.

The TSC is now also referred to as the Glenn ISS Payload Operations Center ([GIPOC](#)). Thus far, the ACME tests have always been conducted in a nominally automated mode.

B. Hardware Shortcomings

CFI-G's two primary challenges were in initiating cool flames and discerning their presence. Based on the findings of the earlier studies with droplet combustion, non-premixed cool flames can be directly ignited by limiting the ignition energy or can occur after the self-extinguishment of (normal) hot flames. Tests intended to directly ignite cool flames sometimes unintentionally resulted in hot flames. Meanwhile, evidence of cool flames was first discovered in review of the data a few days after the testing. During that testing, it wasn't known that the cool flames were present after hot flame extinction and the tests were terminated, which unknowingly and sadly shortened the duration of the cool flame combustion.

In later testing with the long/near thermocouple, two of the tests exhibited repeated hot flame quenching and reignition, which resulting from intervening low-temperature combustion. It appears that the closest thermocouple, approximately 1 cm from the burner surface, was a factor in the reignition. The exposed type-B thermocouple includes platinum, a catalyst for the oxidation of hydrocarbons. The catalysis contributed to the rapid reignition limiting the duration of the low-temperature combustion in these tests.

A challenge for cool flame detection was concern about potential damage to the CFI intensified camera by exposing it to bright light, especially from the energized igniter, while the intensifier gain was set high.

There were also issues with the Image Processing and Storage Units (IPSUs) which affected CFI-G operations. On 30 Aug. 2021 (GMT 242), the primary IPSU, used to record the imaging from both the ACME DataCam and CFI intensified camera (IPSU-G, also known as IPSU5), caused issues and

eventually stopped working. Only the first two tests on that day were successful, where two tests aborted due to an IPSU error, and three tests failed to ignite. In closing checkout, it was determined that the hot-wire igniter tip had also failed. With the IPSU-G failure, the test day was nearly a total loss. Except for a small number of sample images downloaded during the midst of testing, the primary image data from even the successful tests was lost. Furthermore, there was no more propane available for use in inverse-flame testing, so the test day could not be repeated. However, astronaut Mark Vande Hei replaced the failed igniter tip on Sept. 8 and the failed hard drive in the IPSU-G on Sept. 10, and CFI-G testing successfully resumed on Sept. 13, just 2 weeks after the double failure.

IPSU1, which is used with the High Bit-depth Multi-spectral (HiBMs) camera and the opposed CIR illumination package, ceased functioning on 29 March 2021 (GMT 088). Tests 21088M1-M3 aborted due to an error with IPSU1 communication, where test M4 was successful when the test was run without use of the HiBMs camera. The troublesome IPSU1 was replaced by astronaut Megan McArthur on May 18 (GMT 138), but it was not available during the intervening test days. However, this wasn't particularly a loss to CFI-G because that camera, with its red filter, is only sensitive to the red backlight from the CIR illumination package and visible thermal radiation, e.g., from glowing soot particles. Furthermore, the illumination could make it more difficult to discern cool flames, so the HiBMs camera, illumination package, and IPSU1 were rarely used for CFI-G. The principal exception was during the second round of testing. With the inverse-flame tests, the soot is generated outside of the flames, so this photography was used to track potential soot deposition on the chamber windows, etc. However, the end-of-day imaging did not appear to change over time, suggesting that the soot was not an issue for the chamber windows. However, the Photomultiplier Tube (PMT) modules and radiometers were closer to the flame than the windows and might have been adversely affected by soot.

When at a small fraction (e.g., less than 10%) of the maximum flow rate, Mass Flow Controller (MFC) performance has been seen to be poor during CFI-G testing, i.e., where actual flow rate differed from that desired.

Meanwhile, at high flow rates of fuel and nitrogen, the burner does not produce spherically symmetric flames. While nominally spherical at moderate flow rates, the flame's center becomes increasingly offset from the burner center. More specifically, the flame is shifted toward the supply tube indicating that the sintered attachment of the porous sphere to the supply tube is more 'open' than the pores of the spherical burner. In other words, the flow from the burner is not spherically symmetric. At sufficiently high flow rates, the flame becomes obviously non-spherical, where it can appear to be ellipsoidal or even parabolic in shape. In the latter case, the flame locally quenches near the supply tube.

During Burning Rate Emulator (BRE) testing on 23 Dec. 2020, it was discovered that the lowest of the 14-micron silicon carbide fibers on the Thin Filament Pyrometry (TFP) fiber array had broken. Recovery plans were developed and enacted on 18 Jan. 2021 to remove the broken fragments from the harp-like array. Unfortunately, the next fiber was broken during the delicate operations where the fibers are only spaced 5 mm apart. Although ACME does have alternate arrays, it also had months of remaining testing, so the decision was made to continue testing with only 3 fibers (instead of 5) for the final BRE testing and the subsequent CLD Flame and CFI-G tests. This was a minor loss, where the fiber array was only occasionally used in CFI-G testing.

Except for the final test day, CFIG's second-round testing was carried out where the fine tip of the thermocouple closest to the flame was bent at roughly 90 degrees. This can be seen in Figure 5 where, as a size reference, the spherical burner is 6.35 mm in diameter.

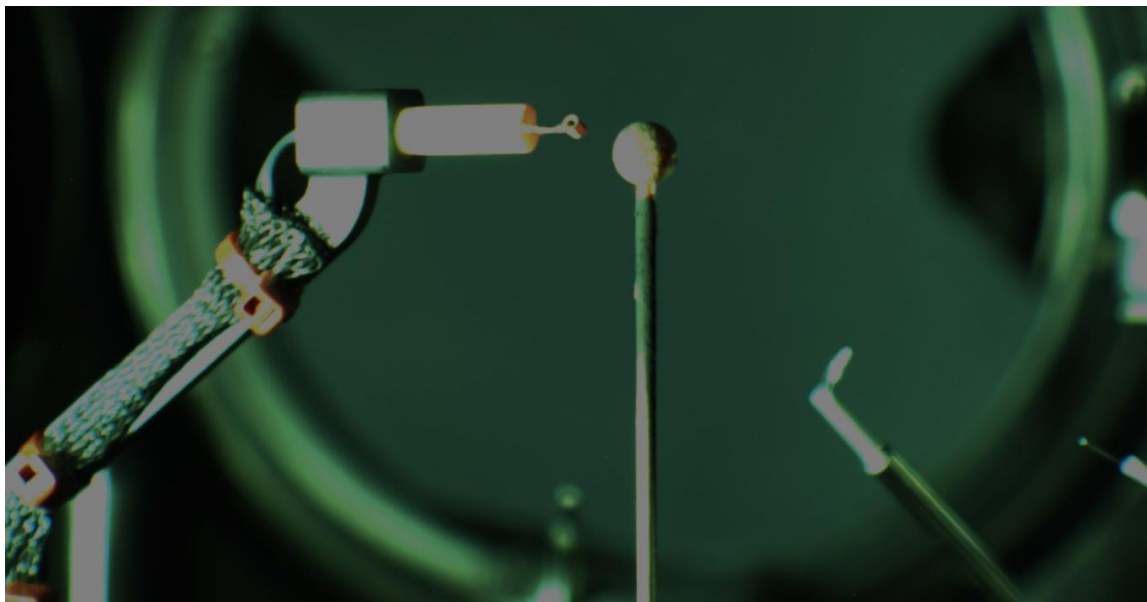


Figure 5: Image of the spherical burner, the coiled hot-wire igniter (just to the left), and two thermocouples (in the lower right) which are part of the short/far thermocouple rake.

In addition to sections 3.A-B, shortcomings are discussed in Appendices B and C of this report and in the *ACME Engineering Implementation* document. The latter document includes discussion of deviations and waivers (in section 4.B), etc.

4. Preliminary Results

A. Changed Test Plans

The most significant change was the substitution of ethane for n-pentane in the test matrix after it was determined through ground testing that the use of the latter was not feasible because of the n-pentane's low vapor pressure.

An additional normal-flames test day was added after the conclusion of the inverse-flame testing. That change was made because it wasn't until after completion of the first round of testing, through analysis of the final day's results, that evidence of cool flames was discovered in the CFI-G results. The added test day, conducted on 18 Oct. 2021 (GMT 291) was carried out with the long (near) thermocouple rake with the goal of measuring the cool flame temperature. Unfortunately, an unwanted outcome was hot-flame reignition (from the cool flames) caused by catalysis from the platinum in the exposed type-B thermocouple closest to the burner. While temperature measurements were made, the cool flames on that additional day of testing were regrettably short-lived. While this was arguably a hardware shortcoming, the use of fine type B thermocouples had been recommended in the ACME science requirements for the closest thermocouples on the long (near) rake.

Other changes to the test plans were minor in comparison.

B. Overview of Experiment Test Operations

While not yet prepared, additional details will be made available in the ACME/CFI-G Experimental Procedure document.

The narrative description of the experimental procedure is an excerpt from the 'Methods / Experimental' section of the following conference paper.

Y.-C. Chien, J. Tinajero, D. Stocker, U. Hegde, and D. Dunn-Rankin, "Electric Field Effects on Flames in Microgravity on the International Space Station," 2018 Spring Technical Meeting of the Central States Section of The Combustion Institute, Minneapolis, MN, 20-22 May 2018.

While the paper was written for the Electric-Field Effects on Laminar Diffusion Flames (E-FIELD Flames) experiment, much of the discussion applies to ACME in general, including the CFI-G investigation. The figure referenced in this excerpt is not included herein. Clarifications and additional information, not in the conference paper, are included in boxed notes like this.

The CIR experiments are configured, but not conducted, by the ISS crew. Thus far, ACME's activity has included the majority of the crew members, regardless of their nationalities. Most often the astronauts and cosmonauts exchange gas bottles, mass flow controllers, and igniter tips. In the latter case, the igniter tip exchange can occur after failure or proactively before failure when the crew already has changes to make changes in the interior of the insert. While these activities may seem mundane, the crew play a critical role in setting up the hardware for each experiment. As an extreme example, there were crew activities on 8 separate days in Sept.-Oct. 2017 to configure the CIR and ACME hardware for the first ACME experiment, namely the Coflow Laminar Diffusion Flame (CLD Flame) experiment. In contrast, only two crew days were required to reconfigure the hardware for the first E-FIELD Flames tests because of the initial use of the same coflow burner. Planning is critical because crew activity is scheduled three weeks in advance, for which detailed procedures must be prepared and uplinked to the ISS.

As a correction, including two maintenance activities, there were 11 crew days in Sept.-Oct. 2017 to prepare for the initial CLD Flame testing on the space station.

All CIR tests are remotely commanded from the ground from NASA's Glenn ISS Payload Operations Center (GIPOC) in Cleveland, Ohio. Typically, four to six people staff four consoles during the testing. The CIR hardware is controlled from the Rack Officer (RO) and Data Management Officer (DMO) consoles, and the ACME hardware is controlled from the Payload Developer (PD) console. Two shifts of personnel are scheduled to staff these three consoles on test days because they typically last 14 hours with the many pre- and post-testing activities. Given that the tests are remotely commanded, the operations are carried out in the day and evening in Cleveland, e.g., from 7 AM to 9 PM. A day's tests, i.e., with flames, are normally conducted in a period of 10 hours or less. Tests are normally conducted on three weekdays per two weeks. On intervening weekdays, the image data is downlinked and the crew often swap one or both gas bottles. Transferring and downlinking data from a day's tests also requires a 2-shift day for the RO and DMO consoles, where it is always necessary to verify that data was downlinked without errors and that data is not missing. Those two positions are also staffed to guide and oversee the crew

activity. As a result, the CIR consoles are normally actively staffed on four weekdays per week. A team of four carry out the detailed planning for the CIR and ACME console staff, and the daily operational plans typically exceed 50 pages on days including testing.

The science console allows for review of the analog video and non-imaging telemetry (at 1 Hz) in near real time but doesn't allow for commanding. In the case of E-FIELD Flames, arrangements were made so that the analog video is streamed to a secure computer at UC Irvine to allow the research team to participate in test operations and to recommend adjustments to the day's planned parameters. That team does not see the telemetry during the testing, but is in communication via phone and e-mail with their NASA research partners at the GRC ISS Payload Operations Center (GIPOC). While there are many preparatory steps, the tests are ultimately conducted using an uploaded software script and parameter files. In advance of the test day, the UC Irvine science team plans the nearly 60 parameters in the voltage sweep script and specifies other parameters that are configured outside of the script, such as camera settings. It is also possible to effectively conduct tests in a 'manual' mode, where parameters are changed in response to the analog video downlink or telemetry. For example, the electric field potential can be adjusted to approach a stability or soot limit.

Provisions were made to enable the investigators from all ACME experiments to remotely view the analog video while they communicated verbally with NASA science personnel in the operations center. All ACME flame tests, including those for CFI-G, were conducted in a nominally automated manner using software scripts and extensive parameter files.

Typically, ten E-FIELD Flames tests are conducted per test day. Despite the short test durations, the low rate results from several unavoidable factors. After a test is conducted, the 125-Hz non-imaging data is downlinked for immediate review by the science team. A small sample of images are often downlinked to ensure that the camera exposures are appropriate, where the times for those images are typically selected based on non-imaging data to provide extreme examples. The review helps to verify parameters or plan for adjustments in subsequent tests, e.g., to select the voltage range after determining where the ion current saturates for the given flame conditions. After a test is conducted, a fan circulates the chamber atmosphere after which the chamber is vented (to space) back to atmospheric pressure, if needed. Midway through a day's E-FIELD Flames tests, the atmosphere is circulated for 15 minutes through a filter cartridge filled with molecular sieves to capture any soot and adsorb at least a portion of the combustion products, e.g., carbon dioxide, water vapor, and incomplete combustion products. The tests are only conducted when both Ku and S band communication is available through NASA's Tracking and Data Relay Satellite (TDRS) System. Depending on the various orbits, there are periodic communication outages with a loss of signal (LOS) that can be a short handover (from one satellite to another) or can last as long as 30 minutes.

Typically, on the order of 12 tests were conducted per CFI-G Flame test day.

In contrast to E-FIELD Flames, recirculation during CFI-G test operations was only conducted prior to venting: (1) at the end of a test day and/or (2) between atmospheres, where the latter was limited to days with normal-flame testing.

A variety of constraints limit the number of tests that can be performed, both overall and on a daily basis. The next combustion experiments in the CIR will feature solid fuels and focus on flame spread and material flammability. That follow-on plan limits the duration of ACME operations, where each ACME experiment is allotted four months for testing. Although it requires additional crew time, these allotments are split into two rounds so that the investigators have a break from testing for careful review of the early results so that they can maximize the scientific outcome of their second round. This paper reports on early results from the first round of E-FIELD Flames.

While nitrogen is available from the space station, the combustion testing is limited by the available gas bottles that were and are launched to support the tests. For E-FIELD Flames and most of ACME's experiments, the test matrices used to plan those gas resources were established in 2010. At that time, NASA limited the scope of the ACME tests matrices with the intent of focused experiments with limited objectives rather than comprehensive experiments. As such, the general target was 50 tests per experiment. For most ACME experiments, the limiting gas constraint is the oxygen/nitrogen bottles used to fill the chamber. This resource constraint is exacerbated by the CIR because it is equipped with a pair of redundant safety timers to limit the fuel that can flow into the combustion chamber (where the timers can be set for the oxidizer flow in the case of inverse flame testing). The timers can only be reset by venting the combustion chamber. Therefore, the timer limit combined with other constraints effectively corresponds to the day's maximum flow time for the fuel (or more generally, the limiting reactant). For E-FIELD Flames, the safety timers are set to the maximum value of 999 seconds, which corresponds to the 10 tests typically completed per day.

<i>The maximum limit of 999 seconds for the valve safety timers was used for all ACME testing.</i>
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Another constraint is the available time in a day, where the console personnel and ISS resources, such as the video downlink, are planned for a limited number of hours. When necessary for the research, NASA does have continuous 24-hour/day operations for some non-CIR experiments. But as discussed, the ACME tests are typically two minutes or less in duration, so the testing is almost always limited to the planned durations.

Data limits are yet another constraint, where the on-orbit hardware can store a limited amount of data. More importantly, 127,000 MB is the most that can be reasonably managed in a two-shift transfer and downlink day. Therefore, tests days are limited to this amount of data, where image data from select cameras is often archived at reduced rates to mitigate this challenge. Nonetheless, the limit sometimes forces decisions to abandon recorded images.

This detailed description of operations and hardware provides the context under which the ACME experiments are conducted. ...

Table 1 and 2 are summary logs of the flames ignited for CFI-G during the two consecutive rounds of testing, including a limited overview of the test conditions. They were copied from this report's Appendix C. 30-Day Operational Accomplishments Report for ISS Increment 65 which comprehensively reflects all CFI-G testing on the ISS.

Table 1. Flames ignited for the CFI-G experiment in the first round of testing, during Increments 64-65, with normal flames (where fuel from the burner flows into oxidizer)

Test Day ¹	Year	Month	Date	Atmosphere ²	Fuel bottle	Hot flames ignited ³
Increment 64						
21088	2021	3	29	~40% O ₂ , 13.7 psia ~21% O ₂ , 14.7 psia ~18% O ₂ , 14.7 psia	100% C ₂ H ₆	14
21102		4	12	~40% O ₂ , 14.7 psia ~21% O ₂ , 14.7 psia	50% n-C ₄ H ₁₀ with N ₂	9
21106			16	~40% O ₂ , 7.35 psia ~21% O ₂ , 7.35 psia		10
Increment 65						
21112	2021	4	22	~40% O ₂ , 13.7 psia ~21% O ₂ , 14.7 psia	100% C ₃ H ₈	13
21125		5	5	~40% O ₂ , 14.7 psia ~21% O ₂ , 14.7 psia	30% n-C ₄ H ₁₀ with N ₂	6
21130			10	~21% O ₂ , 29.4 psia ~21% O ₂ , 44.1 psia	30% C ₃ H ₈ with N ₂	15
21137			17	~40% O ₂ , 14.7 psia ~21% O ₂ , 14.7 psia		12
21153		6	2	~40% O ₂ , 7.35 psia ~40% O ₂ , 29.4 psia	g30% C ₂ H ₆ with N ₂	14
21162			11	~40% O ₂ , 14.7 psia ~21% O ₂ , 14.7 psia		12
21174			23	~40% O ₂ , 7.35 psia ~40% O ₂ , 29.4 psia	30% n-C ₄ H ₁₀ with N ₂	16

¹**Test Day** includes the year as the first two digits followed by the 3-digit day of the year.

²**Atmosphere**, diluted by nitrogen in all cases, is only nominal where the pressure increases and O₂% drops with vitiation.

³**Flames ignited** is the number of sustained or momentary hot flames ignited on that day (neglecting the day change on ISS).

Table 2. Flames ignited for the CFI-G experiment in the second round of testing, during Increments 65-66, with inverse flames (where oxidizer from the burner flowed into a gaseous fuel/nitrogen mixture) plus an added day of normal-flame testing

Test Day ¹	Year	Month	Date	Atmosphere ²	Fluent reactant bottle	Hot flames ignited ³
Increment 65						
21196	2021	7	15	~30% C ₂ H ₆ , 14.7 psia ~30% C ₂ H ₆ , 7.35 psia	85% O ₂ with N ₂	13
21202			21	~6% C ₂ H ₆ , 29.4 psia ~12% C ₂ H ₆ , 29.4 psia ~12% C ₂ H ₆ , 14.7 psia		11
21209			28	~24% C ₂ H ₆ , 7.35 psia		11
21216		8	4	~10% C ₂ H ₆ , 44.1 psia ~10% C ₂ H ₆ , 14.7 psia		12
21222			10	~8% C ₃ H ₈ , 44.1 psia		9
21228			16	~9% C ₃ H ₈ , 29.4 psia ~9% C ₃ H ₈ , 14.7 psia		17
21235			23	~21% C ₃ H ₈ , 14.7 psia ~21% C ₃ H ₈ , 7.35 psia		9
21242			30	~43% C ₃ H ₈ , 7.35 psia		2
21256		9	13	~9% n-C ₄ H ₁₀ , 29.4 psia ~7% n-C ₄ H ₁₀ , 29.4 psia		17
21265			22	~6% n-C ₄ H ₁₀ , 44.1 psia		15
21270			27	~17% n-C ₄ H ₁₀ , 14.7 psia ~14% n-C ₄ H ₁₀ , 14.7 psia		10
21277		10	4	~36% n-C ₄ H ₁₀ , 7.35 psia ~25% n-C ₄ H ₁₀ , 7.35 psia ~18% n-C ₄ H ₁₀ , 7.35 psia ~14% n-C ₄ H ₁₀ , 7.35 psia		11
Increment 66						
21291	2021	10	18	~40% O ₂ , 29.4 psia ~40% O ₂ , 14.7 psia	30% n-C ₄ H ₁₀ with N ₂	22

¹**Test Day** includes the year as the first two digits followed by the 3-digit day of the year.

²**Atmosphere**, diluted by nitrogen in all cases, is only nominal where the pressure increases and O₂% drops with vitiation.

³**Flames ignited** is the number of sustained or momentary hot flames ignited (or reignited) on that day (neglecting the day change on ISS).

C. Summary of Preliminary Results

Abstract for *Spherical Gas-Fueled Cool Diffusion Flames*: An improved understanding of cool diffusion flames could lead to improved engines. CDFs are investigated here using a spherical porous burner with gaseous fuels in the microgravity environment of the International Space Station. Normal and inverse flames burning ethane, propane, and *n*-butane were explored with various fuel and oxygen concentrations, pressures, and flow rates. The diagnostics included an intensified video camera, radiometers, and thermocouples. Spherical CDFs burning gases were observed for the first time. However, these CDFs were not readily produced and were only obtained for normal *n*-butane flames at 2 bar with an ambient oxygen mole fraction of 0.39. The hot flames that spawned the CDFs were 2.6 times as large. An analytical model is presented that combines previous models for steady droplet burning and the partial-burning regime for CDFs. The results identify the importance of burner temperature on the behavior of these CDFs. They also indicate that the observed CDFs reside in rich regions near a mixture fraction of 0.53.

Abstract for *Cool Diffusion Flames in a Stably Stratified Stagnation Flow*: Cool diffusion flames have been a growing research topic since their discovery in 2012. Until now their study has been hindered by the high cost of the experimental systems used to observe them. A method is presented here for observing CDFs inexpensively using a pool of liquid *n*-heptane and parallel plates heated so as to produce a stably stratified stagnation flow. The flames were imaged with a color camera and measurements included gas phase temperatures, fuel evaporation rates, and formaldehyde emissions. These are the first observations of CDFs burning near the surface of fuel pools. The measured peak temperatures were between 705 – 760 K, which corresponds to an increase in gas temperature of 70 K. Autoignition first occurred at 550 K. Cantera autoignition simulations for homogeneous mixtures with the CRECK mechanism predicted a cool flame temperature of 885 K, which is significantly higher than observed.

Publications:

- M. Kim, K.A. Waddell, P.B. Sunderland, V. Nayagam, D.P. Stocker, D.L. Dietrich, Y. Ju, F.A. Williams, P. H. Irace, R.L. Axelbaum, *Spherical Gas-Fueled Cool Diffusion Flames*, Proceedings of the Combustion Institute (2022) in press.
- M. Kim, *Spherical Gas-Fueled Cool Diffusion Flames*, MS Thesis, University of Maryland, 60 pp., May, 2022.
- K.A. Waddell, H. Lee, V. Nayagam, R.L. Axelbaum, P.B. Sunderland, *Cool Diffusion Flames in a Stably Stratified Stagnation Flow*, Combustion and Flame, submitted.
- K.A. Waddell, *Cool Diffusion Flames in Stagnation Flow*, Ph.D. Dissertation, University of Maryland, in preparation.

D. Surprises and New Discoveries

- Spherical cool diffusion flames burning gases were observed for the first time.
- Cool diffusion flames burning near the surface of fuel pools were observed for the first time.

E. Accomplishment of Approved Objectives

All the approved objectives were accomplished, neglecting substitution of ethane for *n*-pentane in the test matrix after it was determined through ground testing that the use of the

latter was not feasible because of the n-pentane's low vapor pressure.

F. Additional Objectives Achieved

Cool diffusion flames burning near the surface of fuel pools in normal gravity were observed for the first time.

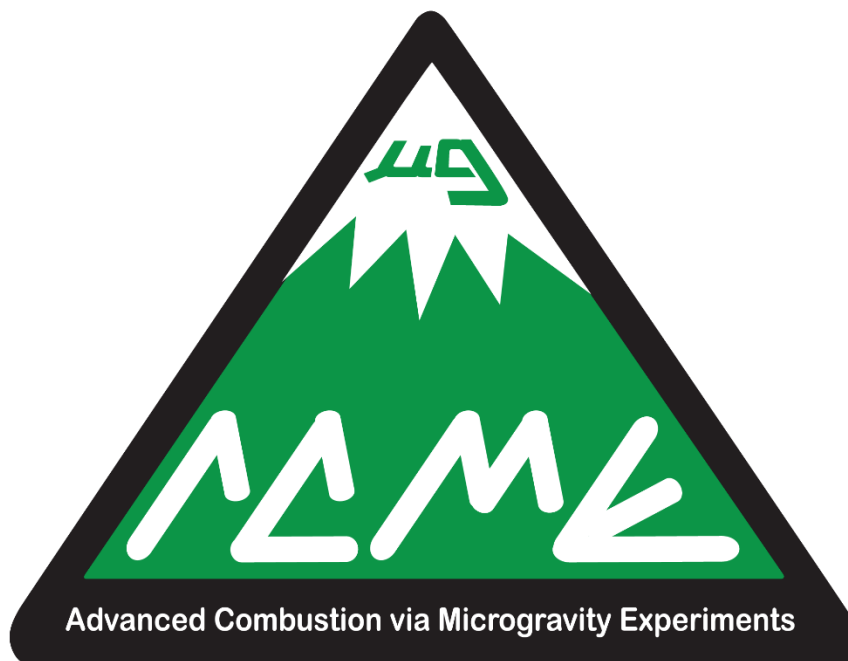
G. Impacts, Benefits, and Products

Combustion provides 86% of the world's energy and an even higher fraction of the energy used for transportation. The internal combustion engines used for transportation generally have thermal efficiencies of 38% or lower because of thermal losses. These efficiencies may increase to 60% if future engines can take advantage of ultra-lean combustion. However, this progress is being impeded by the limited understanding of the chemistry of cool flames. Research into cool diffusion flames (whose peak temperatures are 500 – 1000 K) has grown exponentially since they were discovered in 2012.

The measurements performed for this project will help to test and advance cool flame kinetics mechanisms. This should facilitate the development of more efficient internal combustion engines.

H. Patents

No patents have been filed from the CFI-G research.



Appendix A. Abbreviations and Acronyms

#####	ACME days are identified by year (2 digits) and the day of the year (3 digits, 001-366)
#####x#	ACME tests are identified by day (5 digits; see above), a letter, and a 1-digit number
0-g	microgravity
1-g	normal gravity (on the surface of the Earth)
ACME	Advanced Combustion via Microgravity Experiments (<i>CIR project</i>)
ADP	Acceptance Data Package
ASGSR	American Society for Gravitational and Space Research
AVP	avionics package
BRE	Burning Rate Emulator (<i>an ACME experiment</i>)
C ₂ H ₄	ethylene
C ₃ H ₈	propane
CH ₂ O	formaldehyde
CASIS	Center for the Advancement of Science in Space
CCD	Charge-Coupled Device (<i>such as a camera sensor</i>)
CFI	Cool Flames Investigation (<i>an MDCA experiment preceding ACME in the CIR</i>)
CFI-G	Cool Flames Investigation with Gases (<i>an ACME experiment</i>)
CH ₄	methane
CIA	Chamber Insert Assembly (<i>inserted into the CIR combustion chamber</i>)
CIR	Combustion Integrated Rack (<i>ISS facility in which ACME experiments are conducted</i>)
CLD Flame	Coflow Laminar Diffusion Flame (<i>an ACME experiments</i>)
CM	Combustion Module (<i>space shuttle precursor to the CIR</i>)
cm	centimeter
CO ₂	carbon dioxide
Co-I	Co-Investigator
CRS	Commercial Resupply Service (<i>a designation of ISS cargo missions</i>)
CSA	Canadian Space Agency
CSV	comma-separated values
CWRU	Case Western Reserve University
<i>Da</i>	Damköhler number
DataCam	Data Camera (<i>ACME's primary camera which was characterized for pyrometry, etc.</i>)
E-FIELD Flames	Electric-Field Effects on Laminar Diffusion Flames (<i>an ACME experiment</i>)
EMI	electromagnetic interference
EMS	Experiment Mounting Structure
ERD	Engineering Requirements Document
ESA	European Space Agency
Fiji	Fiji Is Just ImageJ (<i>a version of ImageJ preloaded with many plug-in features</i>)
Flame Design	Flame Design (<i>an ACME experiment</i>)
FOMA	Fuel and Oxidizer Management Assembly (<i>CIR's gas management subsystem</i>)
FOV	field of view
fps	frames per second
GC	Gas Chromatograph (<i>see GCIP</i>)
GCIP	Gas Chromatograph Instrumentation Package (<i>for measuring CIR chamber contents</i>)
GigE	Gigabit Ethernet (<i>an interface standard for high performance cameras</i>)
GPOC	Glenn ISS Payload Operations Center (<i>also known as the TSC</i>)
GMT	Greenwich Mean Time (<i>where most references in this report are the day</i>)
H ₂	hydrogen

HiBMs	High Bit-depth Multi-spectral camera (<i>developed for the CIR facility</i>)
HOBJ	Halcon object (<i>binary uncompressed format; used for the DataCam</i>)
HV	high voltage
HVPS	High-Voltage Power Supply
ICDERS	International Colloquium on the Dynamics of Explosions and Reactive Systems
ID	inner diameter (<i>e.g., of burner tube</i>)
ImageJ	Image Processing and Analysis in Java (<i>open-source image processing software</i>)
IPSU	Image Processing and Storage Unit (<i>CIR device for recording images on ISS</i>)
IPSU-G	IPSU for GigE cameras (<i>including the DataCam and CFI intensified camera</i>)
ISRD	Integrated Science Requirements Document (<i>for a set of experiments</i>)
ISS	International Space Station
JAXA	Japan Aerospace Exploration Agency
kPa	kilopascal
kV	kilovolt
LLL-UV	Low Light Level Ultraviolet camera (<i>developed for the CIR facility</i>)
LSP	Laminar Soot Processes (<i>a shuttle study conducted in the Combustion Module</i>)
MDCA	Multi-user Droplet Combustion Apparatus (<i>a CIR chamber insert</i>)
MFC	Mass Flow Controller
Mist	Water-Mist Fire Suppression Experiment (<i>a Combustion Module experiment</i>)
MIUL	Materials Identification/Usage List
mm	millimeter
MSG	Microgravity Science Glovebox (<i>ISS facility in which SLICE was conducted</i>)
N ₂	nitrogen
NSF	National Science Foundation
NTSC	National Television System Committee
O ₂	oxygen
OMA2	Optical Multi-channel Analyzer 2 (<i>open-source software for image processing</i>)
ops	operations
PARIS	Passive Rack Isolation System (<i>with which the CIR is equipped, but not used for ACME</i>)
PI	Principal Investigator
PIMS	Principal Investigator Microgravity Services
PMT	Photomultiplier Tube (<i>where ACME is equipped with three PMT modules</i>)
PS	Project Scientist
PSI	Physical Science Informatics System
psia	pounds per square inch
PT	pressure transducer
QD	quick disconnect
qty	quantity
RSA	Russian Space Agency, i.e., Roscosmos
s	second
SAMS	Space Acceleration Measurement System
SDP	Safety Data Package
s-Flame	Structure and Response of Spherical Diffusion Flames (<i>an ACME experiment</i>)
SiC	Silicon Carbide (<i>material of the TFP fibers</i>)
SLICE	Structure & Liftoff In Combustion Experiment (<i>CLD Flame precursor</i>)
slpm	standard liters per minute (<i>a measure of the mass or molar flow rate of a gas</i>)
SOFBALL	Structure Of Flame Balls At Low Lewis-number (<i>a Combustion Module study</i>)

SoFIE	Solid Fuel Ignition and Extinction (<i>project to follow ACME in the CIR facility</i>)
SORD	Science Operations Requirements Document (<i>an ACME supplement to the ISRD</i>)
SRD	Science Requirements Document
TDRS	Tracking and Data Relay Satellite (System)
TFP	Thin Filament Pyrometry (<i>a method for gas-phase temperature measurements</i>)
TIFF or TIF	Tagged Image File Format (<i>for the CFI intensified, HiBMs, and LLL-UV cameras</i>)
TM	thermistor
TS	Transport Stream (<i>a video format</i>)
TSC	Telescience Support Center (<i>also known as the GIPOC</i>)
UV	ultraviolet
Ze	Zeldovich number
zero-g	zero gravity, i.e., microgravity
ZIN	ZIN Technologies, Inc. (<i>ACME engineering and operations provider</i>)
μA	microamp

While this list of abbreviations and acronyms is long, it is not comprehensive. Contact ACME Project Scientist Dennis Stocker (216-433-2166, dennis.p.stocker@nasa.gov) for additional clarification.

Appendix B. 30-Day Operational Accomplishments Report for ISS Increment 64

21 June 2021

30-day Operational Accomplishments Report

... for the International Space Station (ISS) Increment 64 **[incorrectly identified as 63]**

ACME Advanced Combustion via Microgravity Experiments CFI-G Cool Flames Investigation with Gases

The CFI-G investigation is one of several independent experiments that are part of the ACME project, which uses a single modular set of hardware to carry out studies of flames burning gaseous fuels in the ISS' Combustion Integrated Rack (CIR). Unlike the other ACME studies, CFI-G was selected for spaceflight by the Center for the Advancement of Science in Space (CASIS) and the National Science Foundation (NSF). CFI-G testing began late in Increment 64 and should be completed in Increment 65.

Prepared by the PI team and Dennis Stocker (dennis.p.stocker@nasa.gov), the ACME Project Scientist, who is the primary contact for s-Flame and ACME information.

Principal Investigator (PI):	Prof. Peter B. Sunderland, University of Maryland, College Park, MD
Co-Investigator (Co-I):	Prof. Richard L. Axelbaum, Washington University in St. Louis, MO
Co-Investigator (Co-I):	Prof. Forman A. Williams, U. of California San Diego, San Diego, CA
NASA Project Scientist (PS):	Dennis P. Stocker, NASA Glenn Research Center, Cleveland, OH
PS team member:	Dr. Vedha Nayagam, Case Western Reserve University, Cleveland, OH

I. Introduction

Tests for the Cool Flames Investigation with Gases (CFI-G) experiment are conducted in the Combustion Integrated Rack (CIR) on the International Space Station, as part of the Advanced Combustion via Microgravity Experiments (ACME) project. As explained in the research proposal:

Cool diffusion flames were discovered during droplet experiments aboard the International Space Station (ISS) in 2012, and this initiated a rapidly growing field of combustion research. Droplet experiments necessarily involve unsteady burning rates, so cool diffusion flames have never been observed as steady spherical flames. A natural progression toward understanding cool diffusion flames is to observe quasi-steady spherical flames on porous burners, and this is the topic of this proposal. The flames to be studied are rich in the physics and chemistry of combustion and thermal transport processes.

Most internal combustion engines are designed using computer models that neglect cool flame chemistry, but ignition and flame propagation in engines depend on cool flame chemistry. Cool flame chemistry also has a significant impact on fuel octane and cetane numbers, whose understanding has large economic consequences.

The flame system to be studied will allow steady spherical cool diffusion flames to be observed for the first time. The tests, performed with gaseous fuel and a porous spherical burner, cannot be performed in normal gravity or in ground-based microgravity facilities.

CFI-G had the following experimental hypotheses and experimental objective when it was proposed in 2017:

Hypotheses

The hypotheses of this research are that testing aboard the ISS can demonstrate the following for the first time, contributing to an improved understanding of cool flame kinetics.

1. Steady-state cool diffusion flames on spherical porous burners.
2. Cool diffusion flames burning propane, *n*-butane, and *n*-pentane.
3. Cool diffusion flames for a wide range of Z_{st} , T_{ad} , and pressures.
4. Inverse cool diffusion flames.
5. Cool diffusion flames for which a decrease in fluent flow rate will cause extinction, whereas an increase will cause a transition to a hot diffusion flame, with oscillation possible between hot and cool flames.

Experimental Objective

- A. Determine the conditions for which cool diffusion flames occur for propane, *n*-butane, and *n*-pentane; normal and inverse flames; and a broad range of Z_{st} , T_{ad} , and fluent flow rate.

where Z_{st} and T_{ad} are respectively the stoichiometric mixture fraction and adiabatic flame temperature.

However, it was later determined that the CIR facility could not support testing with pentane because of the fuel's low vapor pressure so the test matrix was revised to instead include testing with ethane in addition to the propane and *n*-butane.

The additional research objectives are as follows, where CFD stands for Computational Fluid Dynamics.

- B. Support the flight tests with advanced CFD simulations.
- C. Identify or develop a chemical kinetics mechanism that yields good agreement between the CFD predictions and the ISS measurements.
- D. Disseminate the results as widely as possible.

The experiment is carried out using the same spherical burner configuration as ACME's Flame Design and s-Flame (Structure and Response of Spherical Diffusion Flames) experiments. The instrumentation is the same as that used for Flame Design except that an alternate radiometer array was developed for CFI-G. Furthermore, the CFI-G tests were conducted using the

formaldehyde filter on the CFI intensified camera rather than the typically used CH* filter. In this regard, the CFI camera was developed for the Cool Flames Investigation (CFI) which preceded ACME and studied non-premixed cool flames in the combustion of liquid droplets.

Microgravity studies of flames allow for a range of simplifications and provide various advantages over normal-gravity studies. For example, the natural flicker of flames, which is a buoyancy-driven (i.e., gravity-driven) instability, is eliminated yielding quasi-steady flames in microgravity. Characteristic length scales are increased in microgravity flames facilitating analysis of the flame structure. Momentum-dominated flames, which are important for most practical combustion systems, can be studied at low velocities to simplify analysis. Microgravity flames tend to have a much stronger sensitivity to their atmosphere and exhibit a much broader range of characteristics than normal-gravity flames because of the near absence of buoyant entrainment. The long residence times in microgravity flames can lead to strong soot production, but many microgravity flames are soot free. The absence of buoyancy makes microgravity flames advantageous for studies of limit and stability behavior where chemical kinetics are important. For example, chemi-ion species, electronically-excited chemiluminescent species, are at very low concentration in typical flames, making the reduced buoyancy environment valuable for evaluating the minor species chemistry. With these advantages, microgravity testing can lead to new understanding of the complex nature of combustion – regardless of the gravitational environment of the combustion application.

II. Methods/Research Operations

A. Discussion of Method/Protocol

The CFI-G investigation and other ACME experiments are configured, but not conducted, by the ISS crew. The crew's role is to set up the hardware, gas bottles, etc. for each set of tests. The most common changes include the gas bottles on the Combustion Integrated Rack (CIR) and the mass flow controllers on the modular ACME chamber insert.

All testing is commanded from the ground, specifically from the Glenn ISS Payload Operations Center (GIPOC). The science portion of the tests were all conducted in a nominally automated mode using pre-programmed scripts. No CFI-G tests were conducted in a 'manual' mode, where changes are made to a control parameter in response to the analog video downlink and the numerical data available in the telemetry stream. Thus far, that has been the case for all ACME tests, with the mundane exception of 'manually' halting the test after the flame has extinguished or perhaps if a temperature has reached a critical value.

Testing is not conducted when thrusters fire or spacecraft dock or undock from ISS to ensure an acceptable acceleration environment.

Data is downlinked between each set of tests, i.e., day of testing, to allow for preliminary analysis and requisite planning for subsequent testing.

III. Results

A. Baselined Activities Completed – Pre-flight, In-flight, Post-Flight

A total of 33 hot flames were ignited for the CFI-G experiment during Increment 64, where data is still be reviewed to assess the presence of cool flames. This was just the beginning of testing, which is expected to continue through most if not all of Increment 65.

The CFI-G experiment does not have living subjects like a life science experiment, but the tests are independently conducted with various gaseous fuels, including ethane, propane, and *n*-butane. These are simple hydrocarbon gases for which the hot flame chemistry is reasonably well understood, where there is more to be learned about cool flame chemistry.

No samples (e.g., of soot) were collected for CFI-G; all of the ISS results were electronically transmitted to the ground. That is furthermore the case for all currently planned ACME experiments.

A summary log of the flames ignited for CFI-G during Increment 64, including a limited overview of the test conditions, is found in Table 1.

Table 1. Flames ignited for the CFI-G experiment during Increment 64

Test Day ¹	Year	Month	Date	Atmosphere ²	Fuel bottle	Hot flames ignited ³
First round of tests, with normal flames (where fuel from the burner flows into oxidizer)						
21088	2021	3	29	~40% O ₂ , 13.7 psia ~21% O ₂ , 14.7 psia ~18% O ₂ , 14.7 psia	100% C ₂ H ₆	14
21102		4	12	~40% O ₂ , 14.7 psia ~21% O ₂ , 14.7 psia	50% n-C ₄ H ₁₀ with N ₂	9
21106			16	~40% O ₂ , 7.35 psia ~21% O ₂ , 7.35 psia		10

¹**Test Day** includes the year as the first two digits followed by the 3-digit day of the year.

²**Atmosphere**, diluted by nitrogen in all cases, is only nominal where the pressure increases and O₂% drops with vitiation.

³**Flames ignited** is the number of sustained or momentary hot flames ignited on that day (neglecting the day change on ISS).

Pre-flight and post-flight activities are generally not addressed in this report.

B. List of Preflight, In-flight, Postflight Anomalies

1. Mission anomalies

The Image Processing and Storage Unit (IPSU1) used with the High Bit-depth Multi-spectral (HiBMs) camera and the opposed CIR illumination package ceased functioning on the first day, March 29 (GMT 088). Tests 21088M1-M3 all aborted due

to an error with IPSU1 communication, where test M4 was successful when the test was run without use of the HiBMs camera. The troublesome IPSU1 was replaced by astronaut Megan McArthur on May 18 (GMT 138), but it was not available during the intervening test days. However, this wasn't particularly a loss to CFI-G because the camera, with its red filter, is only sensitive to soot. Furthermore, the illumination could make it more difficult to discern cool flames. However, the camera and illumination package will be used during the second round CFI-G testing with inverse flames to track possible soot deposition on the chamber windows.

2. Any other difficulties in obtaining scientific data

CFI-G's two primary challenges are in initiating cool flames and discerning their presence. Based on the findings of the earlier studies with droplet combustion, non-premixed cool flames can be directly by limiting the ignition energy or can occur after the self-extinguishment of (normal) hot flames. Tests intended to directly ignite cool flames sometimes unintentionally resulted in hot flames.

A challenge for cool flame detection is concern about potential damage to the CFI intensified camera by exposing it to bright light, especially from the energized igniter, while the intensifier gain is set high.

When at a small fraction (e.g., less than 10%) of the maximum flow rate, Mass Flow Controller (MFC) performance has been seen to be poor during CFI-G testing, i.e., where actual flow rate differed from that desired.

Meanwhile, at high flow rates of fuel and nitrogen, the burner does not produce spherically symmetric flames. While nominally spherical at moderate flow rates, the flame's center becomes increasing offset from the burner center. More specifically, the flame is shifted toward the supply tube indicating that the sintered attachment of the porous sphere to the supply tube is more 'open' than the pores of the spherical burner. In other words, the flow from the burner is not spherically symmetric. At sufficiently high flow rates, the flame becomes obviously non-spherical, where it can appear to be ellipsoidal or even parabolic in shape. In the latter case, the flame locally quenches near the supply tube.

During Burning Rate Emulator (BRE) testing on 23 Dec. 2020, it was discovered that the lowest of the 14-micron silicon carbide fibers on the Thin Filament Pyrometry (TFP) fiber array had broken. Recovery plans were developed and enacted on 18 Jan. 2021 to remove the broken fragments from the harp-like array. Unfortunately, the next fiber was broken during the delicate operations where the fibers are only spaced 5 mm apart. Although ACME does have alternate arrays, it also has months of remaining testing, so the decision was made to continue testing with only 3 fibers (instead of 5) for the final BRE testing, the subsequent CLD Flame tests, and at least

the start of CFI-G testing. This is viewed as a minor loss, where for example the fiber array is only used in some CFI-G tests.

C. Completeness/Quality of Data

Only the first 3 of 22 CFI-G test days were conducted during Increment 64, so only a small fraction of the test matrix has been completed. In this regard, CFI-G testing is expected to continue through most of Increment 65.

The cameras are sensitive to radiation where pixels gradually become 'hot' (i.e., dead) over time. While these bright spots are addressed in image post-processing, they are a degradation of the image quality.

It is possible that the tests are subtly affected by the acceleration environment, but there is not obvious indication of this; and it has thus far not been evaluated with available measurements from the Space Acceleration Measurement System (SAMS).

Recorded image data is sometimes deleted without downlinking. This is done because of operational constraints which limit the amount of data that can be managed per test day. Given the constraint, the decision is sometimes made to sacrifice some image data to allow additional tests to be conducted. This is intentionally done in order to maximize the scientific return of each day of testing within the project constraints.

D. Status of Data Analysis

[provided by the U. of Maryland investigator team and lightly edited by Stocker]

Data sets of spherical diffusion flames as a function of the fuel mass flow rate, atmospheric O₂ concentration, fuel dilution, and pressure were obtained. The raw test measurements made from devices - such as thermocouples, photomultiplier tubes (PMTs), gas flow meters, and radiometers - were saved in a csv file format per each test. The data rate of the non-imaging measurements was 125 per sec.

The raw data was processed to transform it into scientific measurements. The transformed data were visualized as plots for further analysis. In addition, test images recorded in tiff and hobj formats respectively for the CFI intensified camera and ACME DataCam have been analyzed to find flame diameters. Of these the intensified camera is of most value and all of its images to date have been converted to video files and reviewed. Particular attention has been given to the PMT-2 measurements due to its sensitivity to wavelengths of 230 – 700 nm for CH₂O* to investigate the presence of cool flames. PMT-2 measurements have been analyzed to determine flame durations as well.

Burner temperatures will be analyzed to find their correlation to the cool flame diameters.

E. Preliminary Research Findings [provided by the U. Maryland investigator team]

The first three days of testing have yielded dozens of flames, some lasting several minutes. Based on the output from the photomultipliers, radiometers, and intensified camera images we do not believe any self-sustained cool flames were created. While it is possible that momentary cool flames were created, these are difficult to detect without increasing the gain on the intensified camera before the hot flames extinguish, which is planned.

The video record will furnish excellent measurements of flame diameter versus time for a broad range of fuels and conditions. These will be compared with a simple analytical model and detailed CFD simulations using Sphdif. Thin-filament pyrometry has been performed and the results will soon be analyzed.

F. Communication Imagery

1. Still Imagery

Photos are available online at www.flickr.com/photos/space-flames. Although the Facebook page at www.facebook.com/space.flames is meant to provide updates on ACME and related combustion research, it is good source for recent photos, e.g., of flames, that may not be available on the Flickr site. Additional photographs may be requested from the Project Scientist, Dennis Stocker (dennis.p.stocker@nasa.gov).

2. Video

'The Power of Studying Combustion on the ISS,' which was developed with input from the ACME Project Scientist, is available at <https://science.nasa.gov/science-news/news-articles/the-power-of-studying-combustion-on-the-iss>.

Video(s) describing the ACME project can be found at <https://www1.grc.nasa.gov/space/iss-research/iss-fcf/cir/acme/>.

Videos related to CFI-G and other microgravity combustion research can also be seen on Facebook at www.facebook.com/space.flames.

Contact Dennis Stocker (dennis.p.stocker@nasa.gov) for access to those videos or for additional videos.

IV. Additional Comments from science team

None.

Appendix C. 30-Day Operational Accomplishments Report for ISS Increment 65

14 Dec. 2021

30-day Operational Accomplishments Report ... for the International Space Station (ISS) Increment 65

ACME Advanced Combustion via Microgravity Experiments CFI-G Cool Flames Investigation with Gases

The CFI-G investigation is one of several independent experiments that are part of the ACME project, which uses a single modular set of hardware to carry out studies of flames burning gaseous fuels in the ISS' Combustion Integrated Rack (CIR). Unlike the other ACME studies, CFI-G was selected for spaceflight by the Center for the Advancement of Science in Space (CASIS) and the National Science Foundation (NSF). CFI-G testing began late in Increment 64 and was completed at the very beginning of Increment 66. That final testing is reflected in this report as well as the early testing conducted in Increment 64.

Prepared by the PI team and Dennis Stocker (dennis.p.stocker@nasa.gov), the ACME Project Scientist, who is the primary contact for CFI-G and ACME information.

Principal Investigator (PI):	Prof. Peter B. Sunderland, University of Maryland, College Park, MD
Co-Investigator (Co-I):	Prof. Richard L. Axelbaum, Washington University in St. Louis, MO
Co-Investigator (Co-I):	Prof. Forman A. Williams, U. of California San Diego, San Diego, CA
NASA Project Scientist (PS):	Dennis P. Stocker, NASA Glenn Research Center, Cleveland, OH
PS team member:	Dr. Vedha Nayagam, Case Western Reserve University, Cleveland, OH

V. Introduction

Tests for the Cool Flames Investigation with Gases (CFI-G) experiment are conducted in the Combustion Integrated Rack (CIR) on the International Space Station, as part of the Advanced Combustion via Microgravity Experiments (ACME) project. As explained in the research proposal:

Cool diffusion flames were discovered during droplet experiments aboard the International Space Station (ISS) in 2012, and this initiated a rapidly growing field of combustion research. Droplet experiments necessarily involve unsteady burning rates, so cool diffusion flames have never been observed as steady spherical flames. A natural progression toward understanding cool diffusion flames is to observe quasi-steady spherical flames on porous burners, and this is the topic of this proposal. The flames to be studied are rich in the physics and chemistry of combustion and thermal transport processes.

Most internal combustion engines are designed using computer models that neglect cool flame chemistry, but ignition and flame propagation in engines depend on cool flame chemistry. Cool flame chemistry also has a significant impact on fuel octane and cetane numbers, whose understanding has large economic consequences.

The flame system to be studied will allow steady spherical cool diffusion flames to be observed for the first time. The tests, performed with gaseous fuel and a porous spherical burner, cannot be performed in normal gravity or in ground-based microgravity facilities.

CFI-G had the following experimental hypotheses and experimental objective when it was proposed in 2017:

Hypotheses

The hypotheses of this research are that testing aboard the ISS can demonstrate the following for the first time, contributing to an improved understanding of cool flame kinetics.

1. Steady-state cool diffusion flames on spherical porous burners.
2. Cool diffusion flames burning propane, *n*-butane, and *n*-pentane.
3. Cool diffusion flames for a wide range of Z_{st} , T_{ad} , and pressures.
4. Inverse cool diffusion flames.
5. Cool diffusion flames for which a decrease in fluent flow rate will cause extinction, whereas an increase will cause a transition to a hot diffusion flame, with oscillation possible between hot and cool flames.

Experimental Objective

- A. Determine the conditions for which cool diffusion flames occur for propane, *n*-butane, and *n*-pentane; normal and inverse flames; and a broad range of Z_{st} , T_{ad} , and fluent flow rate.

where Z_{st} and T_{ad} are respectively the stoichiometric mixture fraction and adiabatic flame temperature.

However, it was later determined that the CIR facility could not support testing with pentane because of the fuel's low vapor pressure so the test matrix was revised to instead include testing with ethane in addition to the propane and *n*-butane.

The additional research objectives are as follows, where CFD stands for Computational Fluid Dynamics.

- B. Support the flight tests with advanced CFD simulations.
- C. Identify or develop a chemical kinetics mechanism that yields good agreement between the CFD predictions and the ISS measurements.
- D. Disseminate the results as widely as possible.

The experiment is carried out using the same spherical burner configuration as ACME's Flame Design and s-Flame (Structure and Response of Spherical Diffusion Flames) experiments. The instrumentation is the same as that used for Flame Design except that an alternate radiometer array was developed for CFI-G. Furthermore, the CFI-G tests were conducted using the

formaldehyde filter on the CFI intensified camera rather than the typically used CH* filter. In this regard, the CFI camera was developed for the Cool Flames Investigation (CFI) which preceded ACME and studied non-premixed cool flames in the combustion of liquid droplets.

Microgravity studies of flames allow for a range of simplifications and provide various advantages over normal-gravity studies. For example, the natural flicker of flames, which is a buoyancy-driven (i.e., gravity-driven) instability, is eliminated yielding quasi-steady flames in microgravity. Characteristic length scales are increased in microgravity flames facilitating analysis of the flame structure. Momentum-dominated flames, which are important for most practical combustion systems, can be studied at low velocities to simplify analysis. Microgravity flames tend to have a much stronger sensitivity to their atmosphere and exhibit a much broader range of characteristics than normal-gravity flames because of the near absence of buoyant entrainment. The long residence times in microgravity flames can lead to strong soot production, but many microgravity flames are soot free. The absence of buoyancy makes microgravity flames advantageous for studies of limit and stability behavior where chemical kinetics are important. For example, chemi-ion species, electronically-excited chemiluminescent species, are at very low concentration in typical flames, making the reduced buoyancy environment valuable for evaluating the minor species chemistry. With these advantages, microgravity testing can lead to new understanding of the complex nature of combustion – regardless of the gravitational environment of the combustion application.

VI. Methods/Research Operations

B. Discussion of Method/Protocol

The CFI-G investigation and other ACME experiments are configured, but not conducted, by the ISS crew. The crew's role is to set up the hardware, gas bottles, etc. for each set of tests. The most common changes include the gas bottles on the Combustion Integrated Rack (CIR) and the mass flow controllers on the modular ACME chamber insert.

All testing is commanded from the ground, specifically from the Glenn ISS Payload Operations Center (GIPOC). The science portion of the tests were all conducted in a nominally automated mode using pre-programmed scripts. No CFI-G tests were conducted in a 'manual' mode, where changes are made to a control parameter in response to the analog video downlink and the numerical data available in the telemetry stream. Thus far, that has been the case for all ACME tests, with the mundane exception of 'manually' halting the test after the flame has extinguished or perhaps if a temperature has reached a critical value.

Testing is not conducted when thrusters fire or spacecraft dock or undock from ISS to ensure an acceptable acceleration environment.

Data is downlinked between each set of tests, i.e., day of testing, to allow for preliminary analysis and requisite planning for subsequent testing.

VII. Results

G. Baselined Activities Completed – Pre-flight, In-flight, Post-Flight

A total of 280 hot flames were ignited (or in some cases naturally reignited) for the CFI-G experiment during Increments 64-66, where 33, 225, and 22 flames were ignited in the three increments, respectively. Testing began near the end of Increment 64 and the final test day was conducted at the start of Increment 66. Cool flame evidence was seen in some tests and 15 cool flames, listed and discussed later, have been identified thus far.

The CFI-G experiment does not have living subjects like a life science experiment. The tests were instead conducted burning various gaseous fuels, including ethane (C_2H_6), propane (C_3H_8), and *n*-butane ($\text{n-C}_4\text{H}_{10}$). These are simple hydrocarbon gases for which the hot flame chemistry is reasonably well understood, but where there is much remaining to be learned about the cool flame chemistry.

No samples (e.g., of soot) were collected for CFI-G; all of the investigation's data was electronically transmitted to the ground. That is furthermore the case for all current ACME experiments.

Summary logs of the flames ignited for CFI-G for the two rounds of testing are found in Tables 1 and 2. The first round of testing featured normal flames in which the fuel flowed from the burner into an oxygen/nitrogen atmosphere. In contrast, the second round was largely conducted with inverse-flame tests in which an oxygen/nitrogen mixture flowed from the burner into a mixture of nitrogen and gaseous fuel. The round two exception was the final test day (at the start of Increment 66) which was again conducted in the normal-flames mode.

Table 1. Flames ignited for the CFI-G experiment in the first round of testing, during Increments 64-65, with normal flames (where fuel from the burner flows into oxidizer)

Test Day ¹	Year	Month	Date	Atmosphere ²	Fuel bottle	Hot flames ignited ³
Increment 64						
21088	2021	3	29	~40% O ₂ , 13.7 psia ~21% O ₂ , 14.7 psia ~18% O ₂ , 14.7 psia	100% C ₂ H ₆	14
21102		4	12	~40% O ₂ , 14.7 psia ~21% O ₂ , 14.7 psia	50% n-C ₄ H ₁₀ with N ₂	9
21106			16	~40% O ₂ , 7.35 psia ~21% O ₂ , 7.35 psia		10
Increment 65						
21112	2021	4	22	~40% O ₂ , 13.7 psia ~21% O ₂ , 14.7 psia	100% C ₃ H ₈	13
21125		5	5	~40% O ₂ , 14.7 psia ~21% O ₂ , 14.7 psia	30% n-C ₄ H ₁₀ with N ₂	6
21130			10	~21% O ₂ , 29.4 psia ~21% O ₂ , 44.1 psia	30% C ₃ H ₈ with N ₂	15
21137			17	~40% O ₂ , 14.7 psia ~21% O ₂ , 14.7 psia		12
21153		6	2	~40% O ₂ , 7.35 psia ~40% O ₂ , 29.4 psia	g30% C ₂ H ₆ with N ₂	14
21162			11	~40% O ₂ , 14.7 psia ~21% O ₂ , 14.7 psia		12
21174			23	~40% O ₂ , 7.35 psia ~40% O ₂ , 29.4 psia	30% n-C ₄ H ₁₀ with N ₂	16

¹**Test Day** includes the year as the first two digits followed by the 3-digit day of the year.

²**Atmosphere**, diluted by nitrogen in all cases, is only nominal where the pressure increases and O₂% drops with vitiation.

³**Flames ignited** is the number of sustained or momentary hot flames ignited on that day (neglecting the day change on ISS).

Table 2. Flames ignited for the CFI-G experiment in the second round of testing, during Increments 65-66, with inverse flames (where oxidizer from the burner flowed into a gaseous fuel/nitrogen mixture) plus an added day of normal-flame testing

Test Day ¹	Year	Month	Date	Atmosphere ²	Fluent reactant bottle	Hot flames ignited ³
Increment 65						
21196	2021	7	15	~30% C ₂ H ₆ , 14.7 psia ~30% C ₂ H ₆ , 7.35 psia	85% O ₂ with N ₂	13
21202			21	~6% C ₂ H ₆ , 29.4 psia ~12% C ₂ H ₆ , 29.4 psia ~12% C ₂ H ₆ , 14.7 psia		11
21209			28	~24% C ₂ H ₆ , 7.35 psia		11
21216		8	4	~10% C ₂ H ₆ , 44.1 psia ~10% C ₂ H ₆ , 14.7 psia		12
21222			10	~8% C ₃ H ₈ , 44.1 psia		9
21228			16	~9% C ₃ H ₈ , 29.4 psia ~9% C ₃ H ₈ , 14.7 psia		17
21235			23	~21% C ₃ H ₈ , 14.7 psia ~21% C ₃ H ₈ , 7.35 psia		9
21242			30	~43% C ₃ H ₈ , 7.35 psia		2
21256		9	13	~9% n-C ₄ H ₁₀ , 29.4 psia ~7% n-C ₄ H ₁₀ , 29.4 psia		17
21265			22	~6% n-C ₄ H ₁₀ , 44.1 psia		15
21270			27	~17% n-C ₄ H ₁₀ , 14.7 psia ~14% n-C ₄ H ₁₀ , 14.7 psia		10
21277		10	4	~36% n-C ₄ H ₁₀ , 7.35 psia ~25% n-C ₄ H ₁₀ , 7.35 psia ~18% n-C ₄ H ₁₀ , 7.35 psia ~14% n-C ₄ H ₁₀ , 7.35 psia		11
Increment 66						
21291	2021	10	18	~40% O ₂ , 29.4 psia ~40% O ₂ , 14.7 psia	30% n-C ₄ H ₁₀ with N ₂	22

¹**Test Day** includes the year as the first two digits followed by the 3-digit day of the year.

²**Atmosphere**, diluted by nitrogen in all cases, is only nominal where the pressure increases and O₂% drops with vitiation.

³**Flames ignited** is the number of sustained or momentary hot flames ignited (or reignited) on that day (neglecting the day change on ISS).

Pre-flight and post-flight activities are generally not addressed in this report.

H. List of Preflight, In-flight, Postflight Anomalies

3. Mission anomalies

On Aug. 30 (GMT 242), the primary Image Processing and Storage Unit, used to record the imaging from both the ACME DataCam and CFI intensified camera (IPSU-G, also known as IPSU5), caused issues and eventually stopped working. Only the first two tests on that day were successful, where two tests aborted due to an IPSU error, and three tests failed to ignite. In closing checkout, it was determined that the hot-wire igniter tip had also failed. With the IPSU-G failure, the test day was nearly a total loss. Except for a small number of sample images downloaded during the midst of testing, the primary image data from even the successful tests was lost. Furthermore, there was no more propane available for use in inverse-flame testing, so the test day could not be repeated. However, astronaut Mark Vande Hei replaced the failed igniter tip on Sept. 8 and the failed hard drive in the IPSU-G on Sept. 10, and CFI-G testing successfully resumed on Sept. 13, just 2 weeks after the double failure.

Looking back to Increment 64, the Image Processing and Storage Unit (IPSU1) used with the High Bit-depth Multi-spectral (HiBMs) camera and the opposed CIR illumination package ceased functioning on March 29 (GMT 088). Tests 21088M1-M3 aborted due to an error with IPSU1 communication, where test M4 was successful when the test was run without use of the HiBMs camera. The troublesome IPSU1 was replaced by astronaut Megan McArthur on May 18 (GMT 138), but it was not available during the intervening test days. However, this wasn't particularly a loss to CFI-G because that camera, with its red filter, is only sensitive to the red backlight from the CIR illumination package and visible thermal radiation, e.g., from glowing soot particles. Furthermore, the illumination could make it more difficult to discern cool flames, so the HiBMs camera, illumination package, and IPSU1 were rarely used for CFI-G. The principal exception was during the second round of testing. With the inverse-flame tests, the soot is generated outside of the flames, so this photography was used to track potential soot deposition on the chamber windows, etc. However, the end-of-day imaging did not appear to change over time, suggesting that the soot was not an issue for the chamber windows. However, the Photomultiplier Tube (PMT) modules and radiometers were closer to the flame than the windows and might have been adversely affected by soot.

4. Any other difficulties in obtaining scientific data

CFI-G's two primary challenges are in initiating cool flames and discerning their presence. Based on the findings of the earlier studies with droplet combustion, non-premixed cool flames can be directly ignited by limiting the ignition energy or can occur after the self-extinguishment of (normal) hot flames. Tests intended to directly ignite cool flames sometimes unintentionally resulted in hot flames. Meanwhile, evidence of cool flames was first discovered in review of the data a few days after the testing. During that testing, it wasn't known that the cool flames were present after

hot flame extinction and the tests were terminated, which unknowingly and sadly shortened the duration of the cool flame combustion.

In later testing with the long/near thermocouple, two of the tests exhibited repeated hot flame quenching and reignition, which resulting from intervening low-temperature combustion. It is likely that the closest thermocouple, approximately 1 cm from the burner surface, was a factor in the reignition. The exposed type-B thermocouple includes platinum, a catalyst for the oxidation of hydrocarbons. The catalysis contributed to the rapid reignition limiting the duration of the low-temperature combustion in these tests.

A challenge for cool flame detection was concern about potential damage to the CFI intensified camera by exposing it to bright light, especially from the energized igniter, while the intensifier gain is set high.

When at a small fraction (e.g., less than 10%) of the maximum flow rate, Mass Flow Controller (MFC) performance has been seen to be poor during CFI-G testing, i.e., where actual flow rate differed from that desired.

Meanwhile, at high flow rates of fuel and nitrogen, the burner does not produce spherically symmetric flames. While nominally spherical at moderate flow rates, the flame's center becomes increasing offset from the burner center. More specifically, the flame is shifted toward the supply tube indicating that the sintered attachment of the porous sphere to the supply tube is more 'open' than the pores of the spherical burner. In other words, the flow from the burner is not spherically symmetric. At sufficiently high flow rates, the flame becomes obviously non-spherical, where it can appear to be ellipsoidal or even parabolic in shape. In the latter case, the flame locally quenches near the supply tube.

During Burning Rate Emulator (BRE) testing on 23 Dec. 2020, it was discovered that the lowest of the 14-micron silicon carbide fibers on the Thin Filament Pyrometry (TFP) fiber array had broken. Recovery plans were developed and enacted on 18 Jan. 2021 to remove the broken fragments from the harp-like array. Unfortunately, the next fiber was broken during the delicate operations where the fibers are only spaced 5 mm apart. Although ACME does have alternate arrays, it also had months of remaining testing, so the decision was made to continue testing with only 3 fibers (instead of 5) for the final BRE testing and the subsequent CLD Flame and CFI-G tests. This was a minor loss, where the fiber array was only occasionally used in CFI-G testing.

I. Completeness/Quality of Data

The experiment's proposal to CASIS and NSF stated:

"It is assumed here that approximately 50 ISS flames can be conducted.

These will include approximately 12 chamber fills following evacuation."

Far more testing was carried out for this experiment, where that included the ignition of 280 hot flames in 23 days of ISS testing, some of which included multiple chamber fills. Although some of the flames were short lived, the testing far exceeded the quantity that had been proposed.

The cameras are sensitive to radiation where pixels gradually become 'hot' (i.e., dead) over time. While these bright spots are addressed in image post-processing, they are a degradation of the image quality.

It is possible that the tests are subtly affected by the acceleration environment, but there is not obvious indication of this; and it has thus far not been evaluated with available measurements from the Space Acceleration Measurement System (SAMS).

Recorded image data is sometimes deleted without downlinking. This is done because of operational constraints which limit the amount of data that can be managed per test day. Given the constraint, the decision is sometimes made to sacrifice some image data to allow additional tests to be conducted. This is intentionally done to maximize the scientific return of each day of testing within the project constraints.

J. Status of Data Analysis

[provided by the U. of Maryland investigator team and lightly edited by Stocker]

Data sets of spherical diffusion flames as a function of the fuel mass flow rate (for normal flames), the oxidizer mass flow rate (for inverse flames), O₂ concentration, fuel dilution, and pressure were obtained for all the normal and inverse flame test days. The raw test measurements made from devices - such as thermocouples, photomultiplier tubes (PMTs), gas flow meters, and radiometers - were saved in a csv file format per each test. The data rate of the non-imaging measurements was 125 per sec.

The raw data was processed to transform it into scientific measurements. The transformed data were visualized as plots for further analysis. In addition, test images recorded in tiff and hobj formats respectively for the CFI intensified camera and ACME DataCam have been analyzed to investigate the presence of cool flames and to find flame diameters. Of these the intensified camera is of most value, and all of its images to date have been converted to video files and reviewed. Since the intensified camera images were taken under the maximum possible gain applied, the images were averaged and filtered to decrease the overall noise level, and the intensity of the images was increased by a factor of 50 to facilitate the search of intensity peaks.

Particular attention has been given to the PMT-2 measurements due to its sensitivity to wavelengths of 230 – 700 nm for CH_2O^* to investigate the presence of cool flames. PMT-2 measurements have been analyzed to determine flame durations as well. Because cool diffusion flames were identified through the readings of radiometers after the radiative extinction of hot flames, the radiometer data was carefully analyzed. Burner temperatures were analyzed to find their correlation to the ignition and diameter of cool flames.

K. Preliminary Research Findings

[provided by the U. of Maryland investigator team and lightly edited by Stocker]

A total of 280 hot diffusion flames were observed. Among them, 189 flames experienced radiative extinction, and the rest were extinguished when their flow was terminated, either because the preset flow time was out or the burner overheated. Of those that extinguished radiatively, 15 subsequently reignited as cool diffusion flames. These cool flames were observed only with the following conditions: a) concentrations of 30 vol% n-butane and 40 vol% O_2 , b) normal flames, c) a pressure of 2.03 bar, and d) mass flow rates between 0.75 – 1.5 mg/s.

Cool diffusion flames were initially identified by the readings of radiometers after the radiative extinction of hot flames. Unlike the heat flux shown in Fig. 1(a), Fig. 1(b) shows that the radiant heat flux did not drop to zero after the radiative extinction of a hot flame. Instead, it decreased to a certain level and stayed at that level until the fuel stopped flowing. Figure 1(c) shows multiple hot-cool flame transitions. The first cool flame, the first drop of the radiant heat flux, followed the radiative extinction of the first hot flame. In a few seconds, the heat flux abruptly increased due to reignition of a hot flame. This cycle of radiative extinction, transition to a cool flame, and reignition of a hot flame continued until the flow was terminated. The observed cool flames are summarized in Table 3.

Cool flames were also identified by processed images taken from the intensified camera. When cool flames appeared after the radiative extinction of the hot flames, no flame luminosity was evident for about 1 s, after which cool flames appeared. For test points 21174C3, C4, C6, and F1, the cool flames lasted for approximately 14 s and disappeared as soon as the fuel flow stopped. For the tests of multiple hot-cool flame transitions, 21291A1 and 21291C1, the first cool flame lasted longer than the following cool flames. The intensified camera images of hot and cool flames were used to find flame radii as well. The radii of hot flames were measured at the time just before the flame oscillation.

Representative images of a hot flame and a cool flame are presented in Fig. 2. It is shown that the hot flame is brighter and larger than the cool flame. While a flame sheet is clearly indicated in the hot flame image, the intensity of a cool flame gradually decreases from the peak along radial direction, which implies that the reaction zone of the cool flame is

broadier than that of the hot flame. In addition, it is noted that the hot flame is more spherical and less attached to the burner tube than the cool flame.

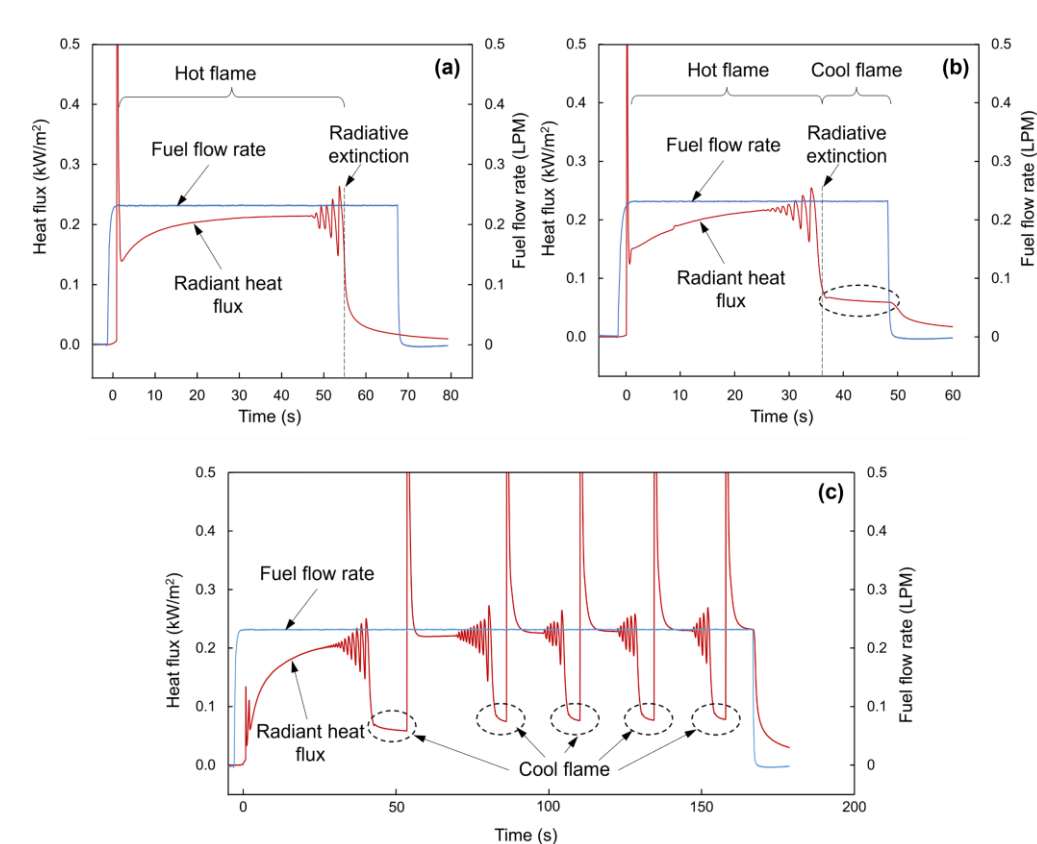


Fig. 1. Radiant heat flux and fuel flow rate versus time for (a) a hot flame that extinguishes, (b) a hot flame that transitions to a cool diffusion flame, and (c) multiple hot-cool flame transitions. These flames involve 30 vol% n-butane flowing at 1.5 mg/s into (a) 37.9, (b) 39.5, and (c) 39.9 vol% O₂ at 300 K. The pressure for (a) is 0.5 bar whereas that for (b) and (c) is 2.0 bar.

Table 3. Summary of the cool flame tests

test pt. ¹	X_{O_2}	X_f	p (bar)	$\dot{m}^2$ (mg/s)	flame duration (sec)	r_{hf} (mm)	r_{cf} (mm)	r_{hf}/r_{cf}
21174C3	0.395	0.300	2.03	1.50	13.9	17.6	7.6	2.3
21174C4	0.394	0.300	2.03	1.00	13.8	14.6	5.3	2.8
21174C6	0.385	0.300	2.03	0.75	14.3	12.4	5.1	2.4
21174D1 ³	0.392	0.300	2.03	1.00	-	13.9	-	-
21174F1	0.390	0.300	2.03	1.50	13.8	17.4	7.5	2.3
21291A1-1	0.399	0.300	2.03	1.50	10.9	17.2	7.1	2.4
21291A1-2	0.399	0.300	2.03	1.50	3.4	17.3	6.5	2.7
21291A1-3	0.399	0.300	2.03	1.50	3.8	17.6	6.3	2.8
21291A1-4	0.399	0.300	2.03	1.50	4	17.6	6.2	2.8
21291A1-5	0.399	0.300	2.03	1.50	3.7	17.7	6.2	2.9
21291C1-1	0.394	0.300	2.03	1.00	6.2	14.4	5.1	2.8
21291C1-2	0.394	0.300	2.03	1.00	5.8	14.9	5.5	2.7
21291C1-3	0.394	0.300	2.03	1.00	5.5	14.7	5.7	2.6
21291C1-4	0.394	0.300	2.03	1.00	5.4	14.9	5.7	2.6
21291C1-5	0.394	0.300	2.03	1.00	5.3	14.4	5.5	2.6

¹ Subnumbers after a dash represent the sequential order of reignited hot flames and following cool flames during a single test.
² Hydrocarbon mass flow rate.
³ The cool flame radius could not be evaluated due to low camera gain.

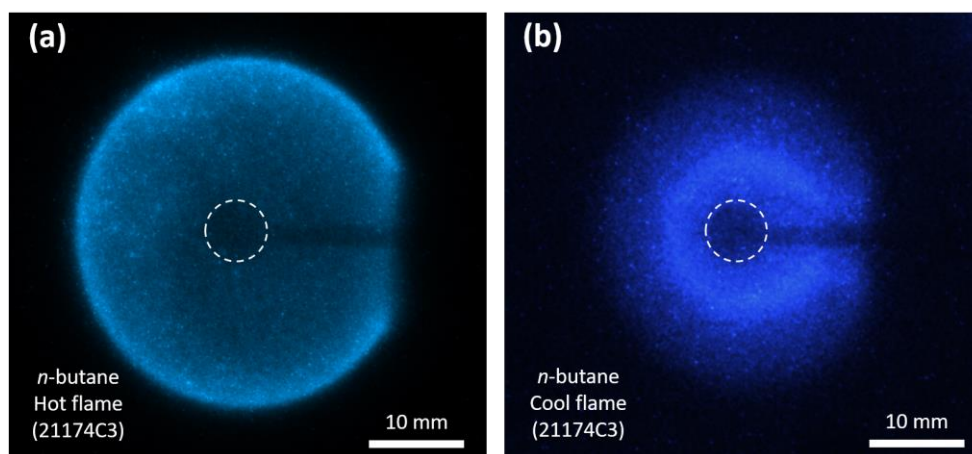


Fig. 2. Images of (a) a hot flame and (b) a cool flame at test point 21174C3. The dashed circle represents the gas-fed porous sphere burner. These images were converted from the grayscale camera output to the expected blue colors shown here.

L. Communication Imagery

1. Still Imagery

Photos are available online at www.flickr.com/photos/space-flames. Although the Facebook page at www.facebook.com/space.flames is meant to provide updates on ACME and related combustion research, it is good source for recent photos, e.g., of

flames, that may not be available on the Flickr site. Additional photographs may be requested from the Project Scientist, Dennis Stocker (dennis.p.stocker@nasa.gov).

2. Video

'The Power of Studying Combustion on the ISS,' which was developed with input from the ACME Project Scientist, is available at <https://science.nasa.gov/science-news/news-articles/the-power-of-studying-combustion-on-the-iss>.

Video(s) describing the ACME project can be found at <https://www1.grc.nasa.gov/space/iss-research/iss-fcf/cir/acme/>.

Videos related to CFI-G and other microgravity combustion research can also be seen on Facebook at www.facebook.com/space.flames.

Contact Dennis Stocker (dennis.p.stocker@nasa.gov) for access to those videos or for additional videos.

VIII. Additional Comments from science team

None.

Appendix D. Operational Timeline

This summary timeline of CFI-G operations was prepared by Dennis Stocker, where the red text indicates the key days when flames were ignited. The dates were based on plans, where it is possible that the dates for some simple tasks, e.g., hardware gather or stowage, may have shifted slightly.

The timeline also includes the launch and landing of spacecraft crews who supported the CFI-G operations.

Date	GMT	Activity
CFI-G (first round, normal flames)		
2020		
Oct. 14	288	Launch of Kud-Sverchkov, Rubins, and Ryzhikov
Nov. 15	320	Launch of Glover, Hopkins, Noguchi, and Walker
2021		
March 9-10	068-069	ACME chamber insert reconfigured for CFI-G by Rubins
March 11	070	Hardware stow by Rubins; Mass Flow Controller (MFC) testing & tuning
March 16	075	Bottle swaps, etc. by Hopkins
March 29	088	CFI-G tests using 100% C2H6
April 9	099	Bottle swap by Hopkins; <i>launch of Dubrov, Novitsky, and Vande Hei</i>
April 12-13	102-103	CFI-G tests using 50% n-C4H10 (with ops planned past midnight EDT)
April 16	106	CFI-G tests using 50% n-C4H10
April 17	107	Landing of Kud-Sverchkov, Rubins, and Ryzhikov
April 19	109	Bottle swaps by Hopkins
April 22	112	CFI-G tests using 100% C3H8
April 23	113	Launch of Hoshide, Kimbrough, McArthur, and Pesquet
April 28	118	Fuel MFC swap by Kimbrough
April 29	119	Bottle swap by Kimbrough
May 2	122	Landing of Glover, Hopkins, Noguchi, and Walker
May 5	125	CFI-G tests using 30% n-C4H10
May 6	126	Hardware gather by Vande Hei
May 7	127	Bottle swaps by Vande Hei
May 10-11	130-131	CFI-G tests using 30% C3H8 (with ops planned past midnight EDT)
May 17	137	CFI-G tests using 30% C3H8
May 18	138	IPSU1 swap by McArthur
May 20	140	Imaging checkout
May 25	145	Bottle swaps by McArthur
June 2	153	CFI-G tests using 30% C2H6 (with ops planned until midnight EDT)
June 8	159	Bottle swap by Kimbrough
June 11-12	162-163	CFI-G tests using 30% C2H6 (with ops planned past midnight EDT)
June 14	165	Bottle swaps by Vande Hei
June 23	174	CFI-G tests using 30% n-C4H10

CFI-G (second round, inverse flames except for the final test day)**2021**

July 2	183	ACME chamber insert reconfigured for inverse-flame tests by Vande Hei
July 7	188	MFC testing & tuning
July 9	190	MFC testing & tuning
July 12	193	Bottle swaps by McArthur
July 15-16	196-197	CFI-G tests using C2H6 (with ops planned past midnight EDT)
July 19	200	Bottle swap by McArthur
July 21-22	202-203	CFI-G tests using C2H6 (with ops planned past midnight EDT)
July 26	207	Bottle swap by McArthur
July 28-29	209-210	CFI-G tests using C2H6 (with ops planned past midnight EDT)
Aug. 2	214	Bottle swaps by Vande Hei
Aug. 3	215	Bottles opened by Kimbrough
Aug. 4-5	216-217	CFI-G tests using C2H6 (with ops planned past midnight EDT)
Aug. 10-11	222-223	Bottle swaps by Vande Hei; CFI-G tests using C3H8 (with ops past midnight)
Aug. 16	228	Bottle swap by Vande Hei; CFI-G tests using C3H8
Aug. 20	232	Bottle swap by Kimbrough
Aug. 23	235	Bottle swap by Kimbrough; CFI-G tests using C3H8
Aug. 30-31	242-243	Bottle swap by Vande Hei; CFI-G tests using C3H8 (with ops past midnight)
Sept. 8	251	Igniter tip replacement by Vande Hei
Sept. 10	253	IPSU-G hard drive replacement by Vande Hei; no-flame check-out
Sept. 13-14	256-257	Bottle swap by McArthur; CFI-G tests using n-C4H10 (ops past midnight)
Sept. 20	263	Bottle swap by McArthur
Sept. 22	265	Bottle swap by McArthur; CFI-G tests using n-C4H10
Sept. 27-28	270-271	Bottle swap by Kimbrough; CFI-G tests using n-C4H10 (ops past midnight)
Oct. 4-5	277-278	Bottle swap by Vande Hei; CFI-G tests using n-C4H10 (ops past midnight)
Oct. 5	278	<i>Launch of Shkaplerov</i>
Oct. 12	285	ACME chamber insert reconfigured for normal-flame tests by McArthur
Oct. 13	286	Bottle swaps by Kimbrough
Oct. 14	287	Bottle swap by McArthur
Oct. 17	290	<i>Landing of Novitsky</i>
Oct. 18	291	CFI-G <u>normal-flame</u> tests using 30% C3H8 (with ops planned until midnight)
Nov. 8	312	<i>Landing of Hoshide, Kimbrough, McArthur, and Pesquet</i>

2022

March 30	089	<i>Landing of Dubrov, Shkaplerov, and Vande Hei</i>
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Appendix E. Astronauts who Conducted ISS Operations for CFI-G

This list was prepared by Dennis Stocker from his log of ACME operations.

The CFI-G experiment was supported by five NASA astronauts, including three men and two women. They are listed by the number of days in which they conducted supporting on-orbit activities and alphabetically. Crew members from partner agencies were not involved (at least directly) in the research.

Crew member	Year	Day(s) of year	# days
Vande Hei, Mark (NASA)	2021	126, 127, 165, 183, 214, 222, 228, 242, 251, 253, 277	11
McArthur, K. Megan (NASA)	2021	138, 145, 193, 200, 207, 256, 263, 265, 285, 287	10
Kimbrough, Shane (NASA)	2021	118, 119, 159, 215, 232, 235, 270, 286	8
Hopkins, Mike (NASA)	2021	075, 099, 109	3
Rubins, Kate (NASA)	2021	068-069, 070	3

The early CFI-G tasks were carried out by Kate Rubins and Mike Hopkins, but most of the on-orbit work was undertaken by Mark Vande Hei, Megan McArthur, and Shane Kimbrough. Additional information about the crew activity can be seen in Appendix D and some of the subsequent appendices.

Appendix F. ISS Operations per the Payload Weekly Science Summary

2021 (for all of CFI-G)

First round (normal flames)

Increment 64

Week 21: March 5-11

On GMT 062, the final test points were performed for ACME CLD Flame (Coflow Laminar Diffusion). The Principal Investigator, Marshall Long from Yale University, considers the experiment to have been very successful. On this day, 11 flames were ignited, which contributed to the over 300 that were ignited for all of CLD Flame. Early results show that the dilution limits are extended in microgravity compared to normal gravity. Steady microgravity flames are also more stable and have lower lift-off heights due to the lack of buoyancy. We have gathered a superior set of experimental data to be used for improving our ability to computationally model highly diluted flames. This capability could be implemented in design tools resulting in more efficient and less polluting combustion here on Earth.

On GMT 068 & 069, Kate Rubins configured the ACME Chamber Insert to support the next ACME experiment, CFI-G Part 1 (Cool Flames Investigation with Gases). CFI-G will study flames with unusually low temperatures known as cool flames. These flames were discovered in 2012 with the first subrack payload to use CIR, but this investigation only used liquid fuels. ACME's hardware will allow CFI-G to study gaseous fuels to gain valuable insight into the combustion processes in engines.

The time-lapse CLD Flame image associated with this submission is not included in this report.

Week 22: March 12-18

On GMT 075, Mike Hopkins lubricated the CIR GC Argon QDs and replaced the gas bottles at CIR Manifold #2 and CIR Manifold #4. These tasks prepared the CIR for supporting the next ACME experiment, CFI-G Part 1 (Cool Flames Investigation with Gases). CFI-G will study flames with unusually low temperatures known as cool flames. These flames were discovered in 2012 with the first subrack payload to use CIR, but this investigation only used liquid fuels. ACME's hardware will allow CFI-G to study gaseous fuels to gain valuable insight into the combustion processes in engines.

Week 24: March 26 – April 1

On GMT 088, the first test points were performed for ACME CFI-G Part 1 (Cool Flames Investigation with Gases) with 14 ethane flames, sometimes diluted with nitrogen, ignited in various oxygen/nitrogen atmospheres. CFI-G studies flames with unusually low temperatures known as cool flames, which were discovered in 2012 with droplet combustion research being conducted in CIR at the time. That experiment burned liquid fuel droplets, which shrink as they burn. Instead, CFI-G will burn gaseous fuel ejected from a porous spherical burner to investigate cool flames under quasi-steady conditions. Cool flame chemical kinetics are important to ignition and the results could lead to cleaner burning and more efficient internal combustion engines.

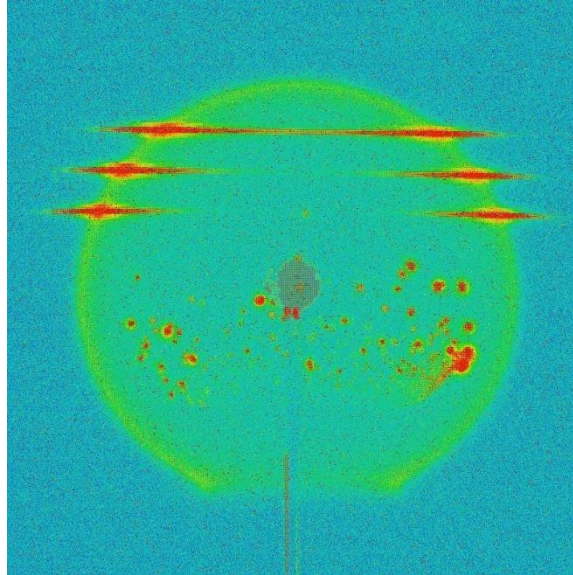


Figure 1. Preliminary False-Colored Image of Hot Ethane Flame for CFI-G

This false-colored image shows a spherical flame, which is only possible in microgravity without the buoyant flows created by Earth's gravity. Spherical flames greatly simplify experimental analysis by reducing the flame structure to a single dimension, which allows computational models to focus on chemical kinetics and transport phenomena.

The hot flame in Figure 1 has a hole at the bottom because of heat loss to the fuel tube, which supplies the ethane to the porous sphere. The three bright lines passing through the flame are 14-micron ceramic fibers, which were translated into the flame to determine temperature measurements from the color of the glowing fibers. Residual soot clusters, like the fibers, are glowing because of their high temperature. These soot clusters were "pushed" to the bottom half of the flame by the insertion of the fibers.

Week 26: April 9-15

On GMT 102, nine nitrogen-diluted butane flames were ignited for ACME CFI-G (Cool Flames Investigation with Gases). Non-premixed (i.e., fuel and oxidizer are not mixed prior to reacting) cool flames were discovered in 2012 with CIR droplet combustion research using liquid fuels. CFI-G's goal is to study non-premixed cool flames using gaseous fuels, which have not been observed before and so far have not been observed in CFI-G testing. It is still possible that they may be detected in a more thorough review of the data. Whether cool flames are present or not, we are still acquiring useful results, such as data about soot processes.

CAPTION TO VIDEO, WHICH IS NOT INCLUDED IN THIS LOG

In this test point video, a bright, sooty spherical flame is ignited with a retracting hot-wire igniter. Due to microgravity, the soot remains in the flame longer and forms clusters. Many clusters escape the flame while trapped soot eventually fades as it cools.

On its right side, the flame starts to open due to heat loss to the burner's fuel supply tube. With increasing heat loss, the flame nearly extinguishes, which creates a larger hole. This large hole in the flame shell allows for the direct mixing of the fuel and oxygen, which causes the flame to

momentarily reestablish. After a few oscillations, the hot flame extinguishes. The results from this and the other tests will be carefully reviewed to look for evidence of cool flame behavior.

Increment 65

Week 2: April 16-22

On GMT 106, additional tests were performed with nitrogen-diluted butane for ACME CFI-G (Cool Flames Investigation with Gases). While last week's tests with this fuel were performed at a chamber pressure of 1 atm, this week's tests were performed at half that pressure. Ten hot flames were ignited, and the data is being analyzed to determine if a cool flame occurred after the hot flame extinguished. In 13 other tests, data is being analyzed to determine if a cool flame was ignited directly with a low-energy ignition. Meanwhile, the data collected from the hot flames contribute to an earlier ACME experiment, Flame Design, which only tested with ethylene fuel. On GMT 109, Mike Hopkins replaced the gas bottles at CIR Manifold #2 and CIR Manifold #4 to support continuation of CFI-G test points.

CAPTION TO VIDEO, WHICH IS NOT INCLUDED IN THIS LOG

This clip from a test point video shows a 50% butane flame ignited in a chamber atmosphere with elevated oxygen at 0.5 atm. The flame is bright and sooty, where the soot is formed when the fuel molecules within the flame combine in the absence of oxygen, which is outside the flame. The flame is like a bubble that separates the two reactants. The flame shell, like the film of the bubble, is a thin sheet where the fuel and oxygen come together and react, producing heat, light, carbon dioxide, and water vapor.

Due to microgravity, the soot remains in the flame longer and forms clusters. Some of these soot clusters escape the flame—being pushed by the outward flow of the fuel from the burner. Others are trapped in the flame as a temperature gradient drives them away from the flame shell and toward the cooler burner. These opposing forces cause some soot clusters to just “dance” in place.

After the video clip ends, the soot continues to fade as the flame temperature drops, where hardly any soot remained when the flame extinguished.

Week 3: April 23-29

On GMT 112, ACME CFI-G (Cool Flames Investigation with Gases) test points were performed using propane fuel. Flames with unusually low temperatures, known as cool flames, were discovered in 2012 with CIR droplet combustion research using liquid fuels. CFI-G's goal is to study cool flames using gaseous fuels to gain valuable insight into the combustion processes in engines. On GMT 118 and 119, Shane Kimbrough replaced an ACME controller and the gas bottle at CIR Manifold #4 to support continuation of CFI-G test points.

Week 4: April 30 – May 6

On GMT 125, ACME CFI-G (Cool Flames Investigation with Gases) test points were performed using nitrogen-diluted butane fuel at both elevated and normal oxygen concentrations. The data is being analyzed to determine if the elusive and hard-to-detect cool flames were burning after the hot flames extinguished or if they ignited directly. Regardless, the testing is providing valuable results that will add to our understanding of flammability limits and soot production, clustering, and oxidation.

CFI-G studies flames with unusually low temperatures known as cool flames, which were discovered in 2012 with droplet combustion research being conducted in CIR at the time. That experiment burned liquid fuel droplets, which shrink as they burn. Instead, CFI-G will burn gaseous fuel ejected from a porous spherical burner to investigate cool flames under quasi-steady conditions. Cool flame chemistry is important to ignition, and the results could lead to cleaner burning and more efficient internal combustion engines.

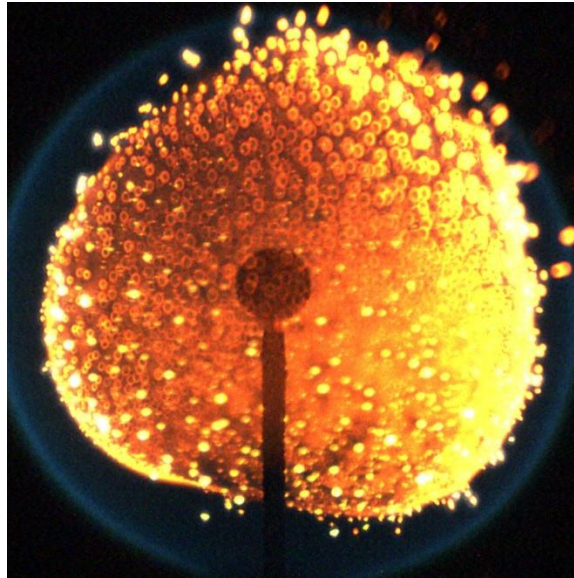


Figure 1. CFI-G Hot Flame of Nitrogen-Diluted Butane

Figure 1 shows a hot flame of nitrogen-diluted butane in an atmosphere of elevated oxygen. The porous spherical burner of sintered stainless-steel sits on top of the tube that supplies the gaseous fuel. The blue flame shell is spherical, which is only possible in microgravity without the buoyant flows created by Earth's gravity. Spherical flames greatly simplify experimental analysis by reducing the flame structure to a single dimension, which allows computational models to focus on the chemistry and transport phenomena rather than the prediction of the flame shape.

The yellow particles are clusters of soot. Their color is the result of thermal radiation and is indicative of their temperature. Soot is formed when the fuel molecules within the flame combine in the absence of oxygen, which is outside the flame. The flame is like a bubble that separates the two reactants. The flame shell, like the film of the bubble, is a thin sheet where the fuel and oxygen come together and react, producing heat, light, carbon dioxide, and water vapor.

Due to microgravity, the soot remains in the flame longer and forms clusters. Some of these soot clusters escape the flame because they are pushed by the outward flow of the fuel from the burner. Others are trapped in the flame as a temperature gradient drives them away from the flame shell and toward the cooler burner.

Week 5: May 7-13

On GMT 130, ACME CFI-G (Cool Flames Investigation with Gases) test points were performed using nitrogen-diluted propane fuel at chamber pressures of both 2 and 3 atm. Fifteen hot flames were ignited.

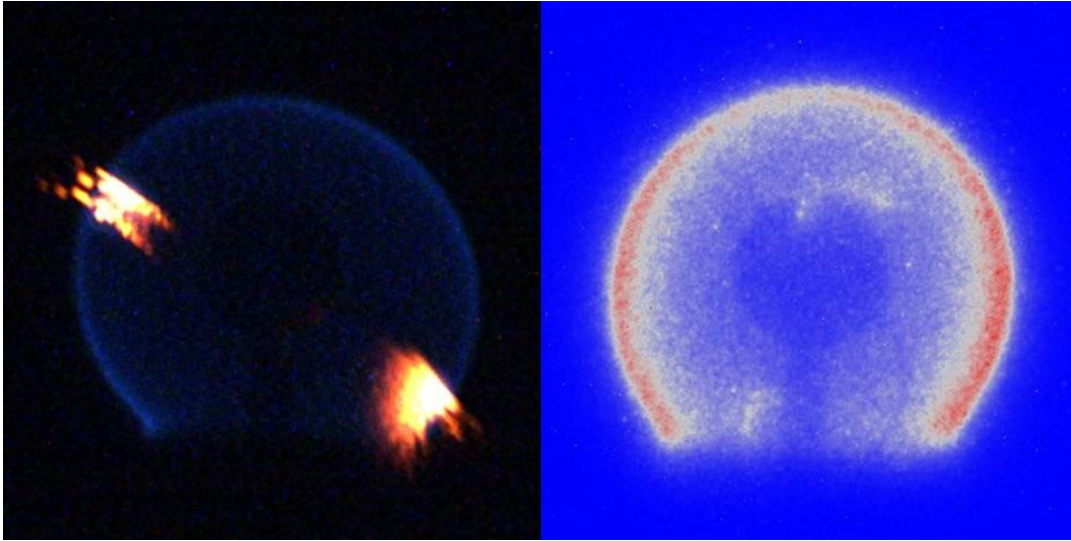


Figure 1. CFI-G Hot Flame of Nitrogen-Diluted Propane

Figure 1 shows images of a CFI-G hot flame from two different cameras. The left image is from a color camera. The right image, which has been false-colored, is from an intensified camera filtered for formaldehyde (CH_2O) emissions, which are associated with cool flame chemistry. When cool flames were first discovered in 1809, Sir Humphrey Davy observed a “pale phosphorescent light.”

Residual soot created during ignition escapes from opposite sides of the flame. In the right image, soot is visible as light spots, such as at the top of the porous sphere.

Week 6: May 14-20

On GMT 137, ACME CFI-G (Cool Flames Investigation with Gases) test points were performed using nitrogen-diluted propane fuel in elevated oxygen concentrations (about 40%). These tests were performed at a chamber pressure of 1 atm, while last week’s tests with this fuel were performed at 2 and 3 atm. Twelve hot flames were ignited.

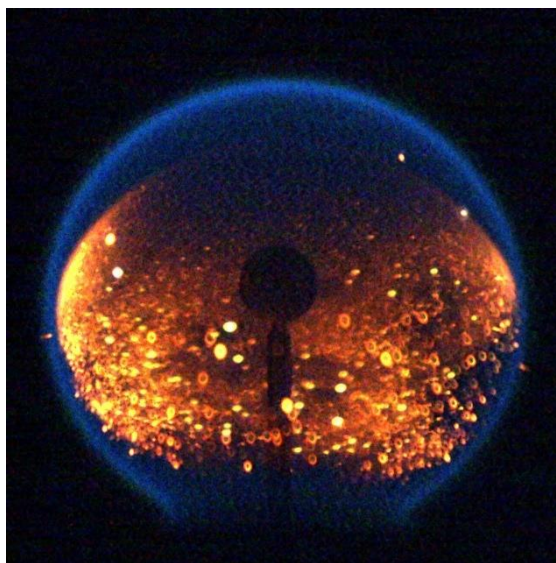


Figure 1. CFI-G Hot Flame of Nitrogen-Diluted Propane

Figure 1 shows a hot flame of nitrogen-diluted propane in an atmosphere of elevated oxygen. At ignition, this flame was bright yellow from the soot, but the soot diminished over time due to radiative heat loss. In the absence of buoyancy, the soot remained in the flame longer and formed the large clusters seen in Figure 1. By the time the flame extinguished, the soot had largely faded away. The porous spherical burner of sintered stainless-steel sits on top of the tube that supplies the gaseous fuel. The opening in the bottom of the flame is due to heat loss to the tube.

Week 7: May 21-27

On GMT 138, Megan McArthur replaced a failing FCF IPSU (Image Processing and Storage Unit) on the back of the CIR Optics Bench. Later in the week, we powered up the new IPSU and its associated camera package. The checkout was successful! On GMT 145, Megan replaced the gas bottles at CIR Manifold #2 and CIR Manifold #4 to support continuation of CFI-G test points.

Week 8: May 28 – June 3

On GMT 153, 14 nitrogen-diluted ethane flames were ignited for ACME CFI-G (Cool Flames Investigation with Gases) in atmospheres ranging between 30-40% oxygen with pressures of both 0.5 and 2 atm.

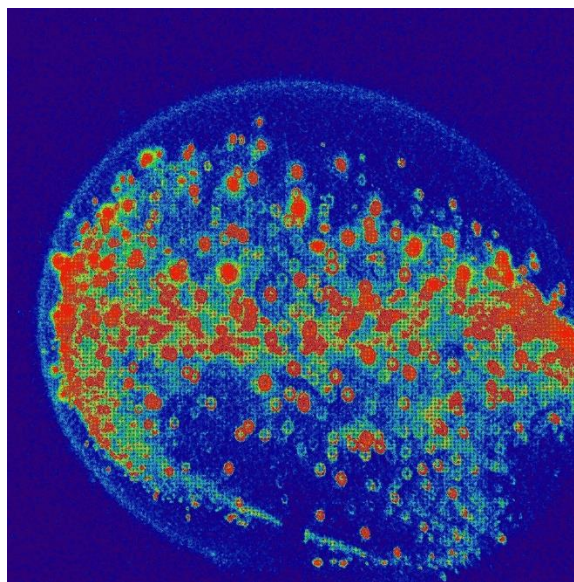


Figure 1. False-colored Image of a CFI-G Flame with Residual Soot Clusters

Figure 1 shows a hot flame of nitrogen-diluted ethane in an atmosphere with elevated oxygen and pressure. Initially, the flame was bright yellow with soot, and over time some of the soot escaped from the flame. This preliminary false-colored image is from later in the test, where residual soot clusters were still visible within the flame when it self-extinguished due to excessive heat loss.

Week 9: June 4-10

On GMT 159, Shane Kimbrough replaced the gas bottle at CIR Manifold #2 to support continuation of CFI-G test points.

Week 10: June 11-17

On GMT 162, 12 nitrogen-diluted ethane flames were ignited for ACME CFI-G (Cool Flames Investigation with Gases) at ambient pressures. Some tests were performed in atmospheres with 21% oxygen, while some tests were performed in higher oxygen concentrations of 35-40%. During two tests, the 14-micron ceramic fibers were translated into the flame to optically measure the flame temperature through a technique known as Thin Filament Pyrometry (TFP). On GMT 165, Mark Vande Hei replaced the gas bottles at CIR Manifold #2 and CIR Manifold #4 to support continuation of CFI-G test points.

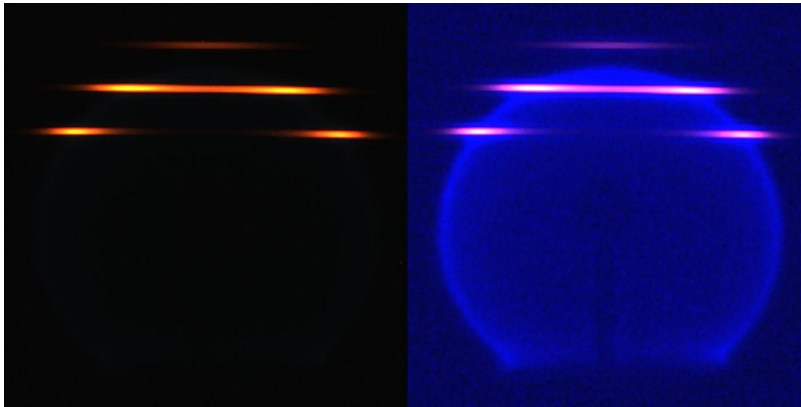


Figure 1. Flame Temperature Measurement of CFI-G Hot Flame

Figure 1 shows two images of the same nitrogen-diluted ethane flame. The right-hand image has been enhanced to show the faint spherical flame. The color of the fibers' glow in the left-hand image is indicative of the temperature, where yellow is hotter than orange, which in turn is hotter than red. Using a camera characterized for quantitative measurements, we can determine a radial temperature distribution across the flame by analyzing the color variation along the length of a fiber.

The top fiber is outside of the flame. Its glow can be used to determine the temperature near the flame, where it is obviously cooler despite its proximity. The fibers that pass through the flame are much brighter except for their centers, which are dimmer and cooler. The flame shell separates the fuel and the oxygen, where the heat, light, carbon dioxide, and water vapor are created at its surface, so the flame is hottest at its shell.

Week 11: June 18-24

On GMT 174, we ignited 16 nitrogen-diluted butane flames in the final tests for the first round of ACME CFI-G (Cool Flames Investigation with Gases). Cool flames remained elusive, where there has been no clear evidence of these low-temperature phenomena during this round of testing. In the coming weeks and months, we will review the data more closely to assess their potential presence. Regardless, we gathered a wealth of valuable data that will improve our fundamental understanding of soot process and flame dynamics. Since CFI-G tested with different fuels, these data expand upon the results gained from the other ACME experiments.

Next week, the ACME Chamber Insert will be configured to support the second and final round of CFI-G testing. This testing will explore the potential existence of cool flames in an inverse configuration. The flow direction will be reversed, where a mixture of nitrogen and oxygen will flow from the burner into the Combustion Chamber, which will be filled with a mixture of nitrogen and gaseous fuel. As a result, the flames will be inside out, where, as an example, the soot will be formed outside of the flame.

CAPTION TO VIDEO, WHICH IS NOT INCLUDED IN THIS LOG

In this test point video, soot that forms at ignition escapes from the flame, especially opposite of the burner's supply tube (toward the left). The escaping soot clusters exacerbate the heat loss experienced by the flame and cause it to locally quench, creating a hole in the flame. This hole allows for direct mixing of the fuel and oxygen, which are normally separated by the flame, thereby enhancing flammability. As a result, the flame reforms, closing the hole, but it opens again due to heat loss. This repeated opening and closing of the flame creates a visually dramatic flame that swirls around the porous spherical burner.

Second round (inverse flames except for the final test day)**Week 12: June 25 – July 1**

Recent data analysis shows evidence of cool flames!

Data from several of the ACME CFI-G (Cool Flames Investigation with Gases) tests on GMT 174 indicate the presence of cool flames. These tests were performed at a chamber pressure of 2 atm with nitrogen-diluted butane fuel. Cool flames give off little heat or light and were too faint to be visible real time during testing. But their presence was revealed by thermal radiation and burner temperature measurements after the hot flames quenched.

Non-premixed (i.e., fuel and oxidizer are not mixed prior to reacting) cool flames were discovered in 2012 with CIR droplet combustion research using liquid fuels. Now, we've achieved non-premixed cool flames using gaseous fuels! Cool flame chemistry is important to ignition and the results could lead to vehicles that experience less wear on engine parts, get better gas mileage, and create less pollution.

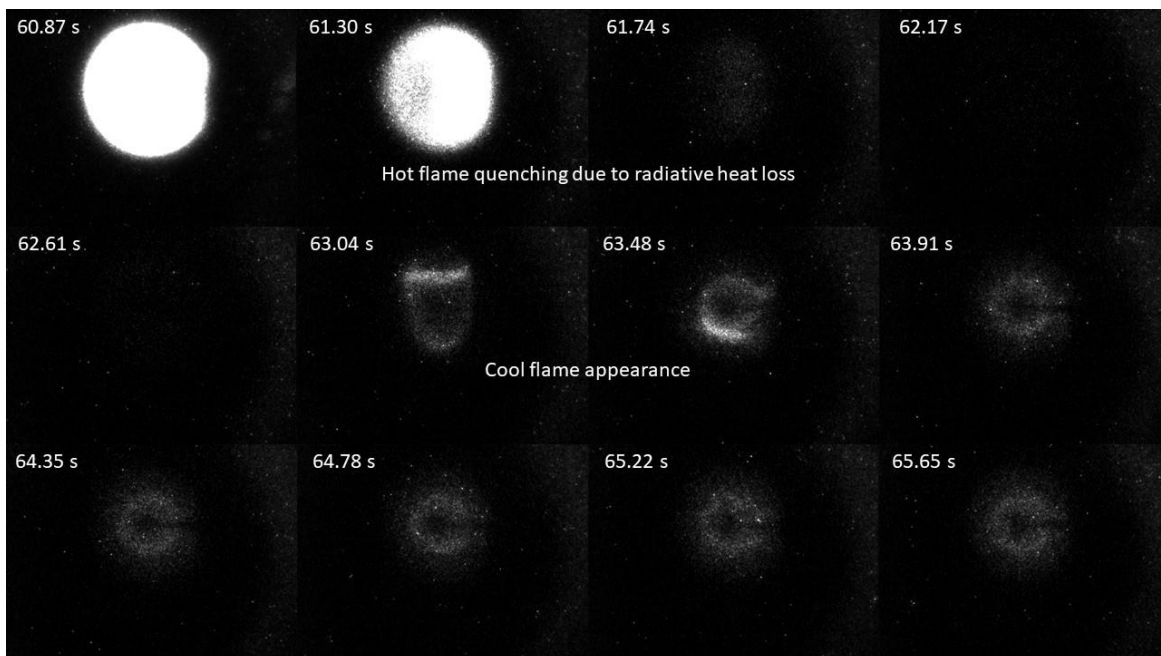


Figure 1. Time Sequence from CFI-G Hot Flame to Cool Flame

The images in Figure 1 are from an intensified camera filtered to look for the very faint emissions of a cool flame (associated with the excited formaldehyde molecule, CH_2O^*). The sequence shows the quenching of the hot flame due to radiative heat loss, followed by a comparatively dark period lasting roughly a second, after which the cool flame becomes evident. As expected, the cool flame is much smaller than the hot flame. The dark center in each cool flame image is the porous spherical burner and its fuel supply tube can be seen extending to the right.

Week 13: July 2-8

On GMT 183, Mark Vande Hei configured the ACME Chamber Insert to support CFI-G Part 2 (Cool Flames Investigation with Gases). CFI-G studies flames with unusually low temperatures known as cool flames. Part 2 testing will explore the potential existence of cool flames in an inverse configuration. The flow direction will be reversed, where a mixture of nitrogen and oxygen will flow from the burner into the Combustion Chamber, which will be filled with a mixture of nitrogen and gaseous fuel. As a result, the flames will be inside out, where, as an example, the soot will be formed outside of the flame. Cool flame chemistry is important to ignition and the results could lead to vehicles that experience less wear on engine parts, get better gas mileage, and create less pollution.

Week 14: July 9-15

On GMT 193, Megan McArthur replaced the gas bottles at CIR Manifold #2 and CIR Manifold #4 to support the start of CFI-G Part 2 (Cool Flames Investigation with Gases) test points.

Week 15: July 16-22

On GMT 196, the first test points were performed for ACME CFI-G Part 2 (Cool Flames Investigation with Gases), where inverse flames were ignited in space for the first time. We ignited seven sustained inverse flames with diluted ethane fuel at both elevated and reduced pressures. With inverse flames, the diluted oxygen flows from the burner into a chamber filled with diluted gaseous fuel. As a result, inverse flames are inside out, where, as an example, the soot is formed outside rather than inside the flame. On GMT 200, Megan McArthur replaced the gas bottle at CIR Manifold #4 to support continuation of CFI-G Part 2 (Cool Flames Investigation with Gases) test points.

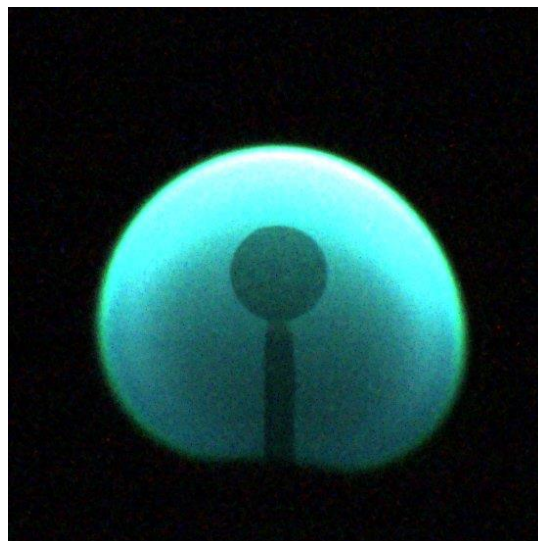


Figure 1. CFI-G Inverse Flame of Nitrogen-Diluted Ethane

Figure 1 shows a bright blue, spherical inverse flame. The porous spherical burner of sintered stainless-steel sits on top of the tube that typically supplies gaseous fuel for normal flames. Instead, for inverse flames it supplies oxygen, which may be diluted with nitrogen.

Week 16: July 23-29

On GMT 202, we ignited 11 inverse flames for CFI-G Part 2 (Cool Flames Investigation with Gases). For these tests, an oxygen/nitrogen mixture flowed from the burner into the Combustion Chamber filled with an ethane/nitrogen mixture. We observed that flames in 12% ethane burned longer than those in 6% ethane. Tests were performed at a chamber pressure of both 1 and 2 atm, and some tests used the array of 14-micron ceramic fibers to make flame temperature measurements. On GMT 207, Megan McArthur replaced the gas bottle at CIR Manifold #4 to support continuation of CFI-G Part 2 test points.

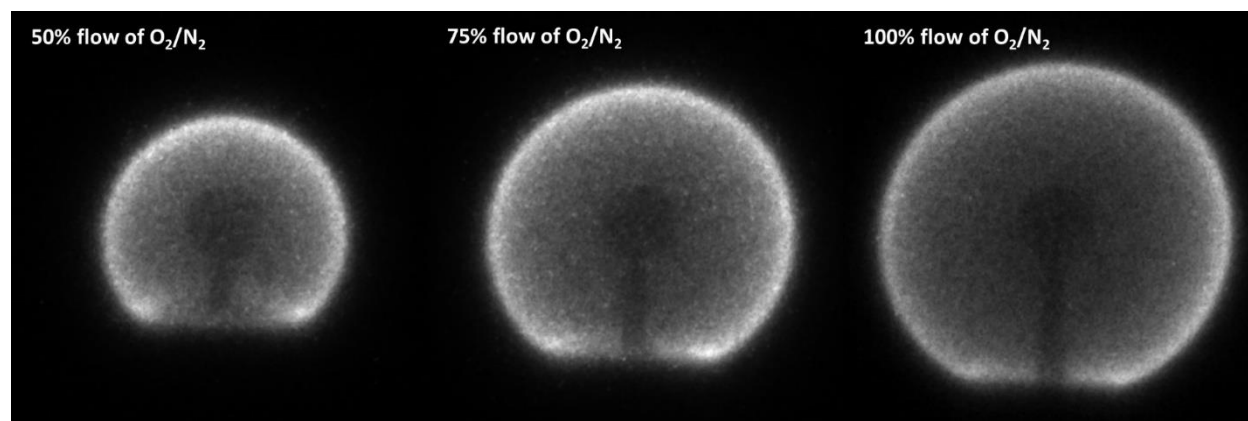


Figure 1. Composite Image of CFI-G Inverse Flames at Increasing Oxidizer Flow Rates

Figure 1 compares the flames from three tests with different flow rates of oxidizer from the burner. The flame size is dependent on the limiting reactant, which for inverse flames is the oxygen.

The porous spherical burner and its supply tube are visible within each spherical flame. Heat loss to the supply tube causes the flame to locally quench and creates an opening. This opening leads to direct mixing of the reactants, which would otherwise be separated by the flame shell. Mixing enhances the combustion and brightens the flame at the opening.

Week 17: July 30 – Aug. 5

On GMT 209 and 216, we ignited 23 inverse flames in a Combustion Chamber filled with ethane and nitrogen for CFI-G Part 2 (Cool Flames Investigation with Gases). These two test days ignited flames in different fuel concentrations and pressures while an oxygen/nitrogen mixture flowed from the porous spherical burner. We gained valuable information about the conditions that influence soot formation. On GMT 214, Mark Vande Hei replaced the gas bottles at CIR Manifold #2 and CIR Manifold #4. On GMT 215, Shane Kimbrough opened the valves on these two bottles. These tasks were performed to support continuation of CFI-G Part 2 test points.

CAPTION TO VIDEO, WHICH IS NOT INCLUDED IN THIS LOG

This test point video of a CFI-G inverse flame is at 3x normal speed. The flame is initially blue but overtime an orange band appears just outside the flame. This orange band is made up of soot particles, which are black but glow because of their high temperature—just like metal in a forge.

The soot is formed when the fuel, at a high temperature, reacts with itself in the absence of oxygen. The temperature is highest at the flame's surface, and for inverse flames, the fuel is outside the flame. Therefore, in this test, the soot forms just outside the flame.

The glowing soot doesn't fully encompass the flame because of heat loss to the burner's supply tube and because the flow from the burner is not quite uniform. However, in this instance, these shortcomings reveal the sensitivity of soot formation to temperature, residence time, and reactant mixture.

Week 18: Aug. 6-12

On GMT 222, Mark Vande Hei replaced the gas bottle at CIR Manifold #4 to support CFI-G Part 2 (Cool Flames Investigation with Gases). We then ignited seven inverse flames in a Combustion Chamber filled with propane and nitrogen. These sustained flames flowed oxygen/nitrogen mixtures over a broad range of concentrations from the spherical burner into the fuel-filled chamber. The testing today was limited to conserve the oxidizer for upcoming tests with a higher propane concentration but lower chamber pressure.

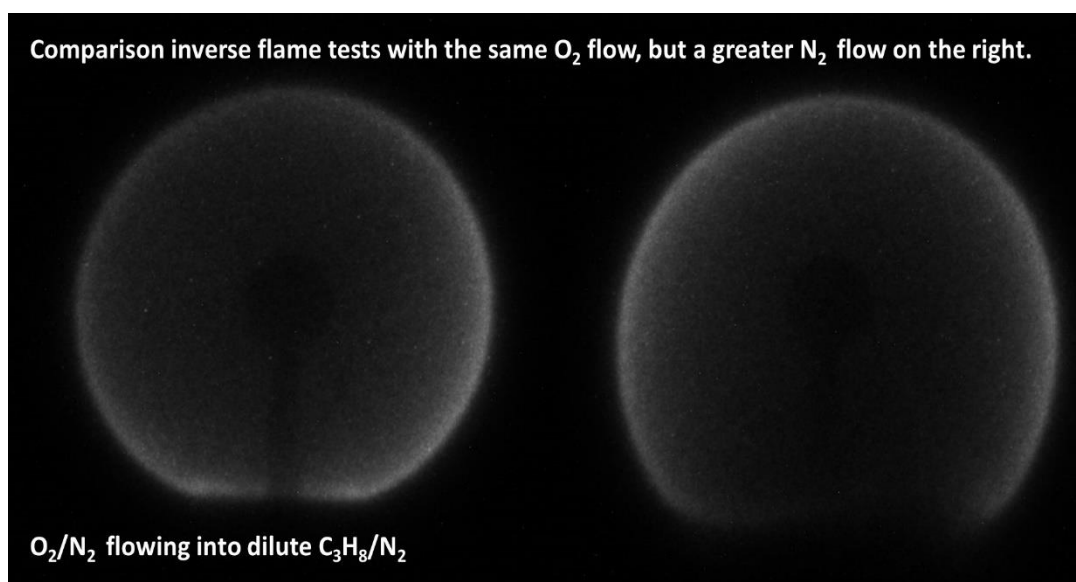


Figure 1. Comparison inverse flame tests with the same oxygen flow, but a greater nitrogen flow on the right.

This image compares two CFI-G inverse flames that had the same oxygen flow rate, but where the flame on the right had a greater nitrogen flow rate, making the oxygen more dilute and the total flow rate greater. The chamber had been filled to 3 atm with propane and nitrogen. The flames, photographed with an intensified camera, are similar in size. Flame size is determined by access to sufficient fuel to consume the oxygen, which was at the same flow rate in both cases.

Week 19: Aug. 13-19

On GMT 228, Mark Vande Hei replaced the gas bottle at CIR Manifold #4 to support CFI-G Part 2 (Cool Flames Investigation with Gases). We then ignited sixteen sustained inverse flames in a Combustion Chamber filled with propane and nitrogen at two different pressures. The test parameters included

varying the concentration ratios of the oxygen/nitrogen mixture flowing from the burner and the flow rates.

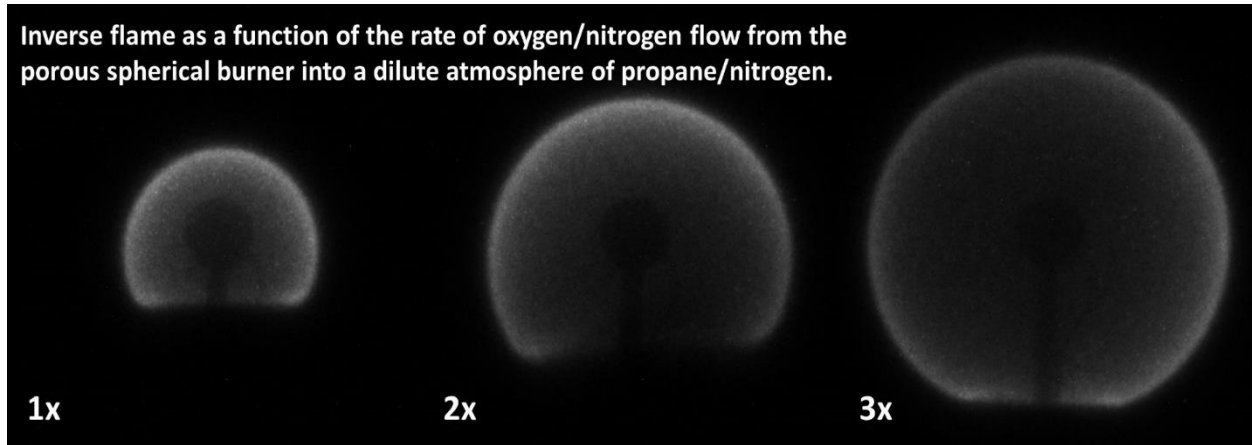


Figure 1. Composite Image of CFI-G Inverse Flames at Increasing Flow Rates

This composite image compares inverse flames with multiples of oxygen/nitrogen flow rates from the porous spherical burner faintly seen within the flames. The oxygen flows outward until it reaches sufficient fuel to be consumed. As such, the tests with the greater oxidizer flow rates result in larger flames. Radiative heat loss scales with flame size, and the larger flames are dimmer than the smaller flames. The oxygen/nitrogen ratio, propane concentration, and Combustion Chamber pressure were consistent in this trio of tests.

These flames are locally quenched by heat loss to the burner's supply tube leaving an opening in the flame. The opening, which allows reactant mixing, often oscillates in size while growing until the flame fully extinguishes. In some tests the extinction occurs abruptly, but in others the oscillations can last for tens of seconds. Meanwhile, some flames are quasi-steady despite the opening. This is one example of how these spherical flames are not as simple as had earlier been imagined.

Week 20: Aug. 20-26

Shane Kimbrough replaced the gas bottle at CIR Manifold #2 on GMT 232 and the gas bottle at CIR Manifold #4 on GMT 235 to support CFI-G Part 2 (Cool Flames Investigation with Gases). We then ignited eight sustained inverse flames in a Combustion Chamber filled with a 20% propane mixture at two different pressures. These tests also varied the flow rate and concentration ratio of the oxygen/nitrogen mixture flowing from the burner.

CAPTION TO VIDEO, WHICH IS NOT INCLUDED IN THIS LOG

This clip from the beginning of a test point video shows an inverse flame that is initially sooty, but the soot quickly fades away as the flame temperature drops. With inverse flames, the fuel is outside the flame, so that is where the soot forms. Soot is created when the fuel in the chamber, heated to a high temperature by the flame, combines in the absence of oxygen, which is inside the flame.

The flame cools due to both radiative heat loss and conductive heat loss to the burner, especially to the tube supplying the oxygen/nitrogen mixture. The soot lasts the longest away from the supply tube where the flame is hottest. To visibly glow due to their temperature, materials must

be about 800 K. Therefore, if it's not hot enough for soot to glow, then it's also not hot enough for the fuel to combine and then form the soot.

Week 21: Aug. 27 – Sept. 2

On GMT 242, Megan McArthur replaced the gas bottle at CIR Manifold #4 to support CFI-G Part 2 (Cool Flames Investigation with Gases). We then ignited only two inverse flames before a hard drive failure in the FCF Image Processor, which processes and stores our image data. Then, during the end-of-the-day check, we discovered that the igniter tip had burned out as well.

This day was our fourth day of testing with inverse propane flames, and we have yet to see evidence of cool flames with this fuel. After the hardware is replaced, we will start inverse-flame testing with butane—the fuel with which cool flames were discovered in earlier normal-flame testing.

CAPTION TO VIDEO, WHICH IS NOT INCLUDED IN THIS LOG

In this test point video, the bright soot shell that surrounds the propane flame hides the spherical burner. The high temperature of the flame is revealed when it extinguishes, and you can see the burner glowing because of its high temperature. The test point automatically stops when the burner reaches 650 °C. Materials start glowing because of their temperature at about 525 °C (the Draper point).

Week 22: Sept. 3-9

On GMT 251, Mark Vande Hei replaced the burnt-out igniter tip for CFI-G Part 2 (Cool Flames Investigation with Gases). Following replacement of the FCF Image Processor failed hard drive later this week, we'll return to performing CFI-G test points.

Week 23: Sept. 10-16

On GMT 253 on the back of the CIR Optics Bench, Mark Vande Hei replaced the failed hard drive in the FCF Image Processor. Afterwards, the CIR was powered up and a checkout of the new hard drive and igniter tip was successful. On GMT 256, Megan McArthur replaced the gas bottle at CIR Manifold #4 to support CFI-G Part 2 (Cool Flames Investigation with Gases). We then ignited 17 inverse flames in a Combustion Chamber filled with nitrogen and butane at two different concentrations. Tests were performed at a chamber pressure of about 2 atm, and we varied the oxygen concentration and oxidizer flow rate from the burner into the fuel-filled chamber. A couple tests also used the array of 14-micron ceramic fibers to make flame temperature measurements.

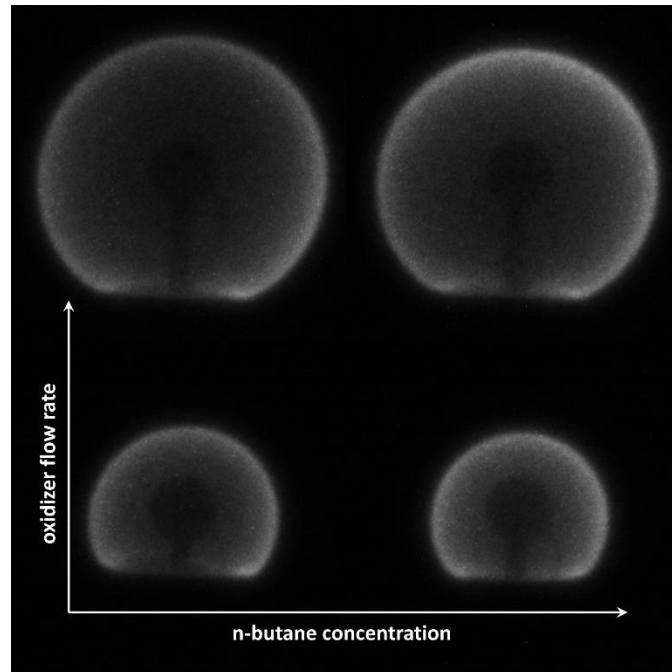


Figure 1. CFI-G Inverse Flames at Increasing Oxidizer Flow Rate and Butane Concentrations

This composite image shows the effects of both the oxidizer flow rate from the burner and the butane concentration on the inverse flames. The oxygen concentration was held constant in these four tests. From bottom to top, the oxidizer flow rate was doubled, which greatly impacted the flame size. From left to right, the ~20% increase in the butane concentration had a more subtle effect. At the higher fuel concentration, the flames are slightly smaller and much brighter.

Week 24: Sept. 17-23

Megan McArthur replaced the gas bottle at CIR Manifold #2 on GMT 263 and the gas bottle at CIR Manifold #4 on GMT 265 to support CFI-G Part 2 (Cool Flames Investigation with Gases). We then ignited 15 inverse flames in a Combustion Chamber filled with butane and nitrogen while an oxygen/nitrogen mixture flowed from the porous spherical burner. Tests were performed at a chamber pressure of 3 atm, and we varied the oxidizer flow rate and oxygen/nitrogen ratio.

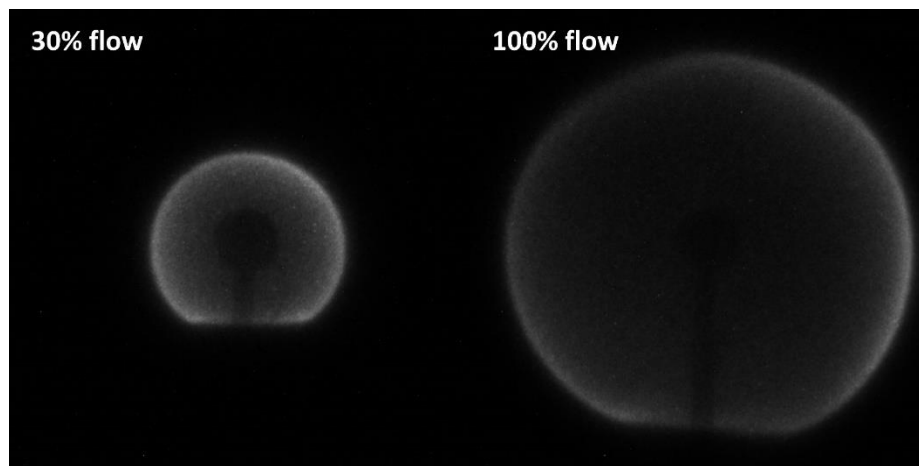


Figure 1. Effects of Oxidizer Flow Rate from the Burner on CFI-G Inverse Flames

This composite image shows the effects of the oxidizer flow rate on inverse flames. In these tests, the flame size depends on the amount of fuel in the atmosphere required to consume the oxygen, so a larger flow rate results in a larger flame. However, a flame's radiative heat loss increases with its size and the larger flame is significantly dimmer. Measurements of thermal radiation from the flame on the right were several times greater than from the flame on the left.

Week 25: Sept. 24-30

On GMT 270, Megan McArthur replaced the gas bottle at CIR Manifold #4 to support CFI-G Part 2 (Cool Flames Investigation with Gases). We then ignited ten inverse flames. For these tests, we varied the concentration of the butane fuel in the Combustion Chamber, the flow rate of the oxygen/nitrogen mixture from the burner, and the oxygen concentration.

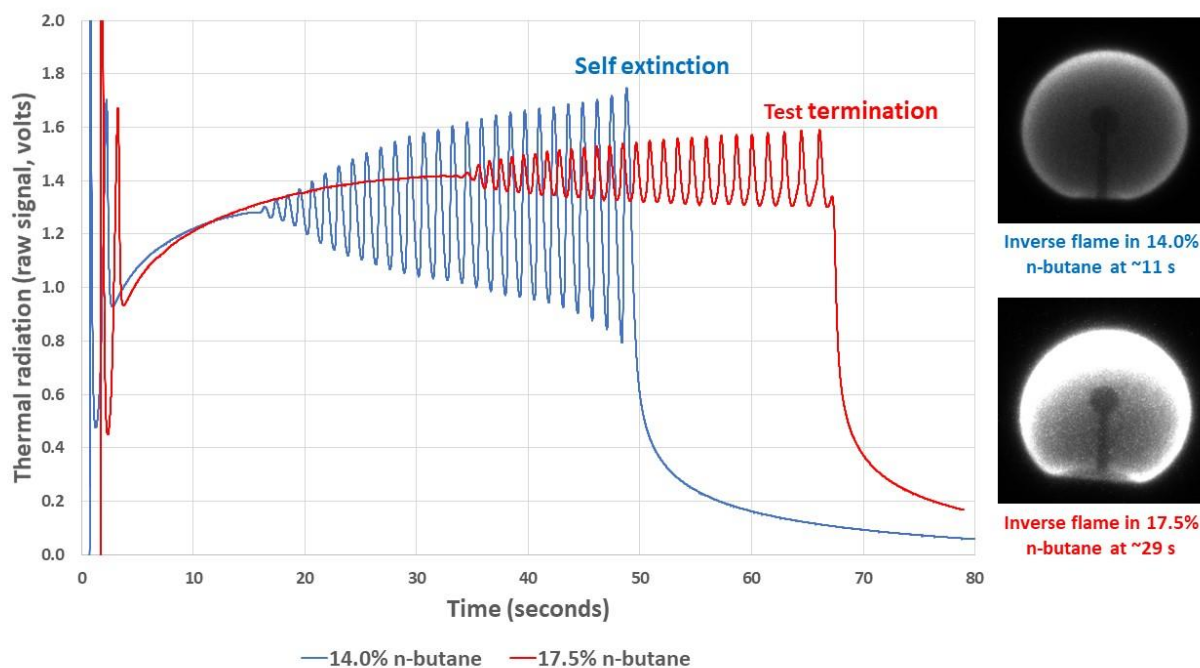


Figure 1. Effects of Butane Concentration on a CFI-G Inverse Flame

Figure 1 is a graph of the thermal radiation emitted by an inverse flame, and it compares two tests with the same oxidizer mixture ratio and flow rate but a different butane concentration. The dim flame in the 14.0% butane experienced large oscillations (shown in blue on the graph) due to heat loss to the burner's supply tube. These oscillations grew in strength until the flame self-extinguished. In contrast, the flame in 17.5% butane was much brighter. It eventually experienced similar oscillations (shown in red on the graph), but they were weaker and didn't cause the flame to extinguish during the test.

Week 26: Oct. 1-7

On GMT 277, Mark Vande Hei replaced the gas bottle at CIR Manifold #4 to support CFI-G Part 2 (Cool Flames Investigation with Gases). We then ignited eleven inverse flames. The tests were carried out at a pressure of 0.5 atm with various butane concentrations ranging from 14 to 36% (by volume), where the

fuel was diluted with nitrogen. The other test parameters included changing the flow rate and ratio of the oxygen/nitrogen mixture issuing from the burner. This marks the completion of CFI-G Part 2!

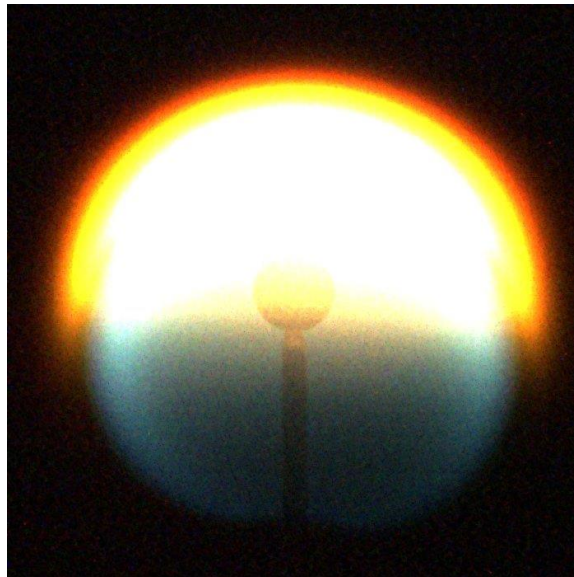


Figure 1. Example CFI-G Inverse Butane Flame

This example image can be readily identified as an inverse flame because the soot is outside, rather than inside, of the flame. The soot is formed when the fuel combines with itself at high temperatures in the absence of oxygen. In this regard, the inverse flame is like a bubble separating the outer fuel from the inner oxygen, where the two reactants are consumed producing light, heat, and chemical products (e.g., carbon dioxide and water vapor). Flames of hydrocarbon fuels are naturally blue, while the high-temperature soot glows red, orange, yellow, or even white depending on its temperature.

At ignition, this flame was all bright and sooty, but the system is not fully symmetric. There is some asymmetry to the burner flow, and there is heat loss to the tube supplying the gas to the porous spherical burner. As a result, the soot faded from the lower hemisphere, which in this image appears to be soot free. Meanwhile, the upper hemisphere remained enveloped in a layer of glowing soot. Tests like this have yielded valuable data for improving our understanding of how to control soot generation.

Week 27: Oct. 8-14

On GMT 285, Megan McArthur configured the ACME Chamber Insert to support an extra day of testing for CFI-G Part 1 (Cool Flames Investigation with Gases). Part 1 studies normal flames as opposed to inverse flames like Part 2. For normal flames, the gaseous fuel flows from the burner into a Combustion Chamber filled with diluted oxygen. The reverse is true for inverse flames, making them inside out. The purpose of this extra day is to recreate the test conditions in which we achieved cool flames in earlier testing. The goal of CFI-G is to study these low temperature flames to gain valuable insight into the combustion processes in engines. Also, to support this extra day of testing, Shane Kimbrough set the CIR Valve Timers and replaced the gas bottles at CIR Manifold #2 and CIR Manifold #4 on GMT 286 and Megan McArthur replaced the gas bottle at CIR Manifold #4 on GMT 287.

Increment 66

Week 1: Oct. 15-21

On GMT 291, we performed the last 12 tests for ACME CFI-G (Cool Flames Investigation with Gases) during an extra day of testing with normal flames. The purpose of this extra day was to recreate the test conditions in which we achieved cool flames in earlier testing, and we successfully achieved cool flames! In two tests, the butane hot flames extinguished due to radiative heat loss, but then seconds later they reignited. This extinction and reignition occurred several times during each of these tests. The butane reignited because a cool flame occurred between the hot flames. Cool flame chemistry is important for ignition in internal combustion engines and an improved understanding would aid the development of vehicles that experience less wear on engine parts, get better gas mileage, and create less pollution.

CAPTION TO VIDEO, WHICH IS NOT INCLUDED IN THIS LOG

This video clip is from the end of a test in which the hot butane flame repeatedly extinguishes and reignites. While the low-temperature cool flames between the hot flames aren't visible in this video, they cause the closest thermocouple to warm enough to visibly glow prior to reignition.

Reignition causes the temperature to abruptly increase, but then it drops while the hot flame burns and weakens. When the hot flame extinguishes and transitions to a cool flame, the temperature starts increasing again until another temperature spike caused by the next reignition.

In another test, we saw evidence of a cool flame where the temperature temporarily rose after the hot flame extinguished but not enough to reignite a hot flame.

Appendix G. Crew Activity per Space Station Science Highlights

The subsequent text and images are from the weekly highlights prepared by the Lead Increment Scientist and made available via e-mail and online via www.nasa.gov/mission_pages/station/research/.

With the major exception of the Space Station Science Highlights, links were generally not included in the excerpts herein because they typically go to secondary sources of information for the ACME project, for which the primary web site is <https://www1.grc.nasa.gov/space/iss-research/iss-fcf/cir/acme/>. Clarifications have been added in red.

2021 (for all of CFI-G)

First round (normal flames)

Week of March 8, 2021

Other investigations on which the crew performed work:

- ACME is a set of six independent studies of gaseous flames. ACME's goals are to advance fuel efficiency and reduce pollutant production in practical combustion on Earth, and to improve spacecraft fire prevention.

Week of April 26, 2021

Observing flames

The Advanced Combustion via Microgravity Experiments (ACME) include six independent studies of flames that seek to improve fuel efficiency and reduce pollution from combustion on Earth and help prevent fires in spacecraft. ACME investigations are conducted in the space station's Combustion Integrated Rack (CIR), which includes a combustion chamber, gas bottles, and five cameras for observing flames. The research aimed at spacecraft fire prevention examines different materials, including how to put them out when burning, conditions needed for a material to continue burning, and the effectiveness of existing methods to screen and select materials for spacecraft. The crew replaced controllers for ACME during the week.

Week of June 7, 2021

Other investigations on which the crew performed work:

- ACME includes six independent studies to improve fuel efficiency and reduce pollution from combustion on Earth and help prevent fires in spacecraft. The spacecraft fire prevention studies examine different materials, conditions needed for them to burn, how to extinguish them, and the effectiveness of methods to screen and select materials for spacecraft.

Note that only one of the six ACME experiments was focused on spacecraft fire safety and that was Burning Rate Emulator (BRE), not CFI-G.

Second round (inverse flames except for the final test day)

Week of July 12, 2021

Other investigations on which the crew performed work:

- ACME is a set of six independent studies of gaseous flames intended to advance fuel efficiency and reduce pollutant production on Earth and improve spacecraft fire prevention. Currently, Cool Flames Investigation with Gases observes the chemical reactions of cool diffusion flames.

Week of July 19, 2021



This Space to Ground video is accessible via the following sites, where the CFI-G discussion appears halfway through the 3 minute clip.

<https://www.youtube.com/watch?v=4q8yN1bSsr8>

https://images.nasa.gov/details-jsc2021m000030_Space%20to%20Ground_380_210723

Week of July 26, 2021

Other investigations on which the crew performed work:

- ACME includes six independent studies of flames that seek to improve fuel efficiency and reduce pollution from combustion on Earth and help prevent fires in spacecraft.

Week of August 2, 2021

Other investigations on which the crew performed work:

- Cool Flames Investigation with Gases, part of the ACME series of studies, observes chemical reactions of cool flames, which burn at lower temperatures. Nearly impossible to create in Earth's gravity, cool flames are easily created in microgravity and studying them may improve understanding of combustion and fires on Earth.

Week of August 9, 2021

Other investigations on which the crew performed work:

- Cool Flames Investigation with Gases, part of the ACME series of studies, observes chemical reactions of cool flames, which burn at lower temperatures. Nearly impossible to create in Earth's gravity, cool flames are easily created in microgravity and studying them may improve understanding of combustion and fires on Earth.

Week of August 16, 2021

Other investigations on which the crew performed work:

- Cool Flames Investigation with Gases, part of the ACME series of studies, observes chemical reactions of cool flames, which burn at lower temperatures. Nearly impossible to create in Earth's gravity, cool flames are easily created in microgravity and studying them may improve understanding of combustion and fires on Earth.

Week of August 23, 2021

Other investigations on which the crew performed work:

- Cool Flames Investigation with Gases, part of the ACME series of studies, observes chemical reactions of cool flames, which burn at lower temperatures. Cool flames were first discovered aboard the space station in 2012 burning liquid fuel; more recently, the investigation created cool flames [burning gaseous fuels](#). Studying these flames may improve understanding of combustion and fires on Earth.

Week of September 6, 2021

Other investigations on which the crew performed work:

- Cool Flames Investigation with Gases, part of the ACME series of studies, observes chemical reactions of cool flames, which burn at lower temperatures. Nearly impossible to create in Earth's gravity, cool flames are easily created in microgravity and studying them may improve understanding of combustion and fires on Earth.

Week of September 20, 2021

Other investigations involving the crew:

- Cool Flames Investigation with Gases, part of the ACME series of studies, observes chemical reactions of cool flames, which burn at lower temperatures. Nearly impossible to create in Earth's gravity, cool flames are easily created in microgravity and studying them may improve understanding of combustion and fires on Earth.

Week of September 27, 2021

Other investigations involving the crew:

- Cool Flames Investigation with Gases, part of the ACME series of studies, observes chemical reactions of cool flames, which burn at lower temperatures. Nearly impossible to create in Earth's gravity, cool flames are easily created in microgravity, and studying them may improve understanding of combustion and fires on Earth.

Week of October 4, 2021

Other investigations on which the crew performed work:

- Cool Flames Investigation with Gases, part of the ACME series of studies, observes chemical reactions of cool flames, which burn at lower temperatures. Nearly impossible to create in Earth's gravity, cool flames are easily created in microgravity, and studying them may improve understanding of combustion and fires on Earth.

Week of October 11, 2021

Coolest flames in space

The Advanced Combustion via Microgravity Experiments (ACME) project is a set of six independent studies of gaseous flames conducted in the Combustion Integrated Rack (CIR). The researchers' goals are to gain information that can lead to improved fuel efficiency, reduced pollutant production in practical combustion on Earth, and improved spacecraft fire prevention. One of the ACME studies currently being conducted is known as the Cool Flames Investigation with Gases. A cool flame is one that burns at about 600 degrees Celsius. A candle is about two times hotter, burning at around 1,400 degrees Celsius. The results of this investigation could lead to cleaner, more efficient internal combustion engines. This past week, a crew member was tasked with replacing a butane bottle in preparation for the experiment.

Week of October 18, 2021

Other investigations involving the crew:

- Cool Flames Investigation with Gases, part of the ACME series of studies, observes chemical reactions of cool flames, which burn at lower temperatures. Nearly impossible to create in Earth's gravity, cool flames are easily created in microgravity and studying them may improve understanding of combustion and fires on Earth.

Appendix H. Crew Activity per the ISS Daily Summary Report

*This appendix was compiled from <https://blogs.nasa.gov/stationreport>. Headings were added identifying the investigation's two rounds of testing, but CFI-G operations listed under the headings "All activities are complete unless otherwise noted" or "Completed Task List Activities" were not included in this report. Some days of crew activity were only listed in those ways and as a result this section does not comprehensively document crew activity in support of the CFI-G experiment. **Minor additions were made in red to correct the names and abbreviations of ACME investigations.***

2021 (for all of CFI-G)

First round (normal flames)

2021 March 9: GMT 068

Combustion Integrated Rack/Advanced Combustion via Microgravity Experiments (CIR/ACME): The crew performed the first part of the work needed to reconfigure the ACME chamber insert from Coflow Laminar Diffusion **Flame** (CLD **Flame**) support to Cool Flames Investigation with Gases (CFI-G) support, the next ACME experiment to be performed. Cool diffusion flames were discovered during droplet combustion experiments aboard the ISS in 2012, and this initiated a rapidly growing field of combustion research. A cool flame is one that burns at about 600 degrees Celsius. A typical candle is about two-and-a-half times hotter, burning at around 1,400 degrees Celsius. Most internal combustion engines are designed using computer models that neglect cool flame chemistry, but ignition and flame propagation in engines depend on cool flame chemistry. Cool flame chemistry also has a significant impact on fuel octane and cetane numbers, whose understanding has large economic consequences.

2021 March 10: GMT 069

Combustion Integrated Rack (CIR)/Advanced Combustion via Microgravity Experiments (ACME): The crew performed the second part of the work needed to reconfigure the ACME chamber insert from Coflow Laminar Diffusion **Flame** (CLD **Flame**) support to Cool Flames Investigation with Gases (CFI-G) support, which is the next ACME experiment to be performed. Cool diffusion flames were discovered during droplet combustion experiments aboard the ISS in 2012, and this initiated a rapidly growing field of combustion research. A cool flame is one that burns at about 600 degrees Celsius. A typical candle is about two-and-a-half times hotter, burning at around 1,400 degrees Celsius. Most internal combustion engines are designed using computer models that neglect cool flame chemistry, but ignition and flame propagation in engines depend on cool flame chemistry. Cool flame chemistry also has a significant impact on fuel octane and cetane numbers, whose understanding has large economic consequences.

2021 March 16: GMT 075

Combustion Integrated Rack (CIR) / Advanced Combustion via Microgravity Experiments (ACME): The crew performed the steps necessary to lubricate the quick disconnect (QD) for the gas chromatograph argon gas bottle. They also replaced the N₂/O₂ gas bottle at manifold 2 and installed an ethane fuel bottle at the manifold 4 location. This was performed in support of the Cool Flames Investigation with Gases (CFI-G) experiment. Cool diffusion flames were discovered during droplet combustion experiments aboard the ISS in 2012, and this initiated a rapidly growing field of combustion research. A cool flame is one that burns at about 600 degrees Celsius. A typical candle is about three-and-a-half times hotter, burning at around 1,400 degrees Celsius. Most internal combustion engines are designed using computer models that neglect cool flame chemistry, but ignition and flame propagation in engines depend on cool flame

chemistry. Cool flame chemistry also has a significant impact on fuel octane and cetane numbers, whose understanding has large economic consequences.

2021 April 9: GMT 099

CIR/ACME/CFI-G (Combustion Integrated Rack/Advanced Combustion via Microgravity Experiments/Cool Flames Investigation with Gases): As a continuation of the on-going CFI-G experiment, the crew removed the 100% ethane fuel bottle and installed a 50% butane, 50% nitrogen fuel bottle. Cool diffusion flames were discovered during droplet combustion experiments aboard the ISS in 2012, and this initiated a rapidly growing field of combustion research. A cool flame is one that burns at about 600 degrees Celsius. A typical candle is about three-and-a-half times hotter, burning at around 1,400 degrees Celsius. Most internal combustion engines are designed using computer models that neglect cool flame chemistry, but ignition and flame propagation in engines depend on cool flame chemistry. Cool flame chemistry also has a significant impact on fuel octane and cetane numbers, whose understanding has large economic consequences.

2021 April 19: GMT 109

Combustion Integrated Rack (CIR) – Fluids and Combustion Facility: The crew replaced the Manifold 2 and 4 gas bottles for upcoming CIR operations. The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion investigations in microgravity.

2021 April 28: GMT 118

ACME (Advanced Combustion via Microgravity Experiments): Crew replaced an ACME controller. The crew opened the rack doors and the combustion chamber of the ACME and then replaced the Fuel Controller. The Advanced Combustion via Microgravity Experiments (ACME) project is a set of six independent studies of gaseous flames to be conducted in the Combustion Integrated Rack (CIR). ACME's primary and secondary goals are (1) improved fuel efficiency and reduced pollutant production in practical combustion on Earth and (2) spacecraft fire prevention through innovative research focused on materials flammability.

2021 May 7: GMT 127

CIR/ACME (Combustion Integrated Rack/Advanced Combustion via Microgravity Experiments): The crew replaced the 40% O₂/60% N₂ manifold bottle with a new bottle of the same composition. They also replaced the 30% butane/70% N₂ bottle with a new bottle of the same composition. This was performed in support of the CFI-G (Cool Flames Investigation with Gases) experiment. Cool diffusion flames were discovered during droplet combustion experiments aboard the ISS in 2012, and this initiated a rapidly growing field of combustion research. A cool flame is one that burns at about 600 degrees Celsius. A typical candle is about three-and-a-half times hotter, burning at around 1,400 degrees Celsius. Most internal combustion engines are designed using computer models that neglect cool flame chemistry, but ignition and flame propagation in engines depend on cool flame chemistry. Cool flame chemistry also has a significant impact on fuel octane and cetane numbers, whose understanding has large economic consequences.

2021 May 18: GMT 138

CIR IPSU Replacement: Today, a crewmember replaced the failed FCF IPSU in the CIR (Combustion Integrated Rack), which included moving the FCF IPSU at position UML2 to replace the FCF IPSU at position

UML5. The Combustion Integrated Rack (CIR) includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion investigations in microgravity.

2021 May 25: GMT 145

ACME (Advanced Combustion via Microgravity Experiments): A crewmember replaced CIR Manifold #4 with a new C₄H₁₀ (Butane) and C₂H₆ (Ethane) bottles and replaced O₂ bottles in Manifold #2. The Advanced Combustion via Microgravity Experiments (ACME) project is a set of six independent studies of gaseous flames to be conducted in the Combustion Integrated Rack (CIR). ACME's primary and secondary goals are (1) improved fuel efficiency and reduced pollutant production in practical combustion on Earth, and (2) spacecraft fire prevention through innovative research focused on materials flammability.

2021 June 8: GMT 159

CIR/ACME (Combustion Integrated Rack/Advanced Combustion via Microgravity Experiments): The crew replaced a 40% O₂, 60% N₂ manifold bottle with another bottle of the same composition. This was performed in support of the CFI-G part 1 (Cool Flames Investigation with Gases) experiment. Cool diffusion flames were discovered during droplet combustion experiments aboard the ISS in 2012, and this initiated a rapidly growing field of combustion research. A cool flame is one that burns at about 600 degrees Celsius. A typical candle is about three-and-a-half times hotter, burning at around 1,400 degrees Celsius. Most internal combustion engines are designed using computer models that neglect cool flame chemistry, but ignition and flame propagation in engines depend on cool flame chemistry. Cool flame chemistry also has a significant impact on fuel octane and cetane numbers, whose understanding has large economic consequences.

2021 June 14: GMT 165

Cool Flames Investigation with Gases (CFI-G): The crew replaced manifold #2 and #4 bottles with bottles of the same composition. This was performed in support of part 1 of the CFI-G experiment. Cool diffusion flames were discovered during droplet combustion experiments aboard the ISS in 2012, and this initiated a rapidly growing field of combustion research. A cool flame burns at about 600 degrees Celsius. A typical candle is about three-and-a-half times hotter, burning at around 1,400 degrees Celsius. Most internal combustion engines are designed using computer models that neglect cool flame chemistry, but ignition and flame propagation in engines depend on cool flame chemistry. Cool flame chemistry also has a significant impact on fuel octane and cetane numbers, whose understanding has large economic consequences.

Second round (inverse flames except for the final test day)

2021 July 2: GMT 183

Combustion Integrated Rack (CIR) / Advanced Combustion via Microgravity Experiments (ACME): The crew performed the steps necessary to access the ACME combustion insert and reconfigured from Cool Flames Investigation with Gases (CFI-G) part 1 to CFI-G part 2. This involved replacing the igniter tip, replacing three ACME flow controllers, etc. Cool diffusion flames were discovered during droplet combustion experiments aboard the ISS in 2012. This initiated a rapidly growing field of combustion research. A cool flame is one that burns at about 600 degrees Celsius. A typical candle is about two times hotter, burning at around 1,400 degrees Celsius. Most internal combustion engines are designed using computer models that neglect cool flame chemistry, but ignition and flame propagation in engines depend

on cool flame chemistry. Cool flame chemistry also has a significant impact on fuel octane and cetane numbers, whose understanding has large economic consequences.

2021 July 12: GMT 193

Combustion Integrated Rack (CIR) – Fluids and Combustion Facility: The crew removed the O₂ Bottle from the CIR Manifold #2 rack and replaced it with a high percentage oxygen bottle containing 85% O₂ and 15% N₂. The crew also replaced the fuel bottle in the CIR Manifold #4 rack with a bottle containing 100% C₂H₆ (Ethane). The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion investigations in microgravity.

2021 July 19: GMT 200

Combustion Integrated Rack (CIR) – Fluids and Combustion Facility: The crew exchanged the manifold #4 bottle with a bottle containing 100% C₂H₆. The CIR includes an optics bench, combustion chamber, fuel, and oxidizer control, and five different cameras for performing combustion investigations in microgravity.

2021 July 26: GMT 207

Combustion Integrated Rack/Advanced Combustion via Microgravity Experiments (CIR/ACME): The crew removed a used 100% ethane (CH₃-CH₃) fuel bottle with a new bottle of the same composition. This will allow the continuation of the Cool Flames Investigation with Gases (CFI-G) part 2 test points. Cool diffusion flames were discovered during droplet combustion experiments aboard the ISS in 2012, and this initiated a rapidly growing field of combustion research. A cool flame is one that burns at about 600 degrees Celsius. A typical candle is about two times hotter, burning at around 1,400 degrees Celsius. Most internal combustion engines are designed using computer models that neglect cool flame chemistry, but ignition and flame propagation in engines depend on cool flame chemistry. Cool flame chemistry also has a significant impact on fuel octane and cetane numbers, whose understanding has large economic consequences.

2021 Aug. 2: GMT 214

Combustion Integrated Rack/Advanced Combustion via Microgravity Experiments (CIR/ACME): The crew exchanged a used 85% O₂/ 15% N₂ bottle with a new one of the same composition; however, during the installation, a loose pressure relief valve body was found. The ground is discussing steps to re-tighten the valve body and proceed with operations. This will allow the continuation of the Cool Flames Investigation with Gases (CFI-G) part 2 test points. Cool diffusion flames were discovered during droplet combustion experiments aboard the ISS in 2012, and this initiated a rapidly growing field of combustion research. A cool flame is one that burns at about 600 degrees Celsius. A typical candle is about two times hotter, burning at around 1,400 degrees Celsius. Most internal combustion engines are designed using computer models that neglect cool flame chemistry, but ignition and flame propagation in engines depend on cool flame chemistry. Cool flame chemistry also has a significant impact on fuel octane and cetane numbers, whose understanding has large economic consequences.

2021 Aug. 10: GMT 222

Combustion Integrated Rack (CIR) – Fluids and Combustion Facility (FCF): The crew replaced the Manifold 4 fuel bottle with 1.00 liters of 100% C₃H₈ (Propane). The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion investigations in microgravity.

2021 Aug. 16: GMT 228

Cool Flames Investigation with Gases: The crew replaced the Manifold 4 fuel bottle with 1.00 liters of 100% C₃H₈ (Propane). Cool diffusion flames were discovered during droplet combustion experiments aboard the ISS in 2012, and this initiated a rapidly growing field of combustion research. A cool flame is one that burns at about 600 degrees Celsius. A typical candle is about three-and-a-half times hotter, burning at around 1,400 degrees Celsius. Most internal combustion engines are designed using computer models that neglect cool flame chemistry, but ignition and flame propagation in engines depend on cool flame chemistry. Cool flame chemistry also has a significant impact on fuel octane and cetane numbers, whose understanding has large economic consequences.

2021 Aug. 20: GMT 232

Cool Flames Investigation with Gases: A crewmember removed Manifold 2 O₂ bottles with a bottle that contains 85% O₂ and 15% N₂. Cool diffusion flames were discovered during droplet combustion experiments aboard the ISS in 2012, and this initiated a rapidly growing field of combustion research. A cool flame is one that burns at about 600 degrees Celsius. A typical candle is about two times hotter, burning at around 1,400 degrees Celsius. Most internal combustion engines are designed using computer models that neglect cool flame chemistry, but ignition and flame propagation in engines depend on cool flame chemistry. Cool flame chemistry also has a significant impact on fuel octane and cetane numbers, whose understanding has large economic consequences.

2021 Aug. 23: GMT 235

Combustion Integrated Rack/Advanced Combustion via Microgravity Experiments/Cool Flames Investigation with Gasses (CIR/ACME/CFI-G): A crewmember exchanged a used 100% propane bottle with a new bottle of the same composition. Cool diffusion flames were discovered during droplet combustion experiments aboard the ISS in 2012, and this initiated a rapidly growing field of combustion research. A cool flame is one that burns at about 600 degrees Celsius. A typical candle is about two times hotter, burning at around 1,400 degrees Celsius. Most internal combustion engines are designed using computer models that neglect cool flame chemistry, but ignition and flame propagation in engines depend on cool flame chemistry. Cool flame chemistry also has a significant impact on fuel octane and cetane numbers, whose understanding has large economic consequences.

2021 Aug. 30: GMT 242

Combustion Integrated Rack/Advanced Combustion via Microgravity Experiments/Cool Flames Investigation with Gasses (CIR/ACME/CFI-G part 2): A crewmember exchanged a used 100% propane bottle with a new bottle of the same composition. Cool diffusion flames were discovered during droplet combustion experiments aboard the ISS in 2012, and this initiated a rapidly growing field of combustion research. A cool flame is one that burns at about 600 degrees Celsius. A typical candle is about two times hotter, burning at around 1,400 degrees Celsius. Most internal combustion engines are designed using computer models that neglect cool flame chemistry, but ignition and flame propagation in engines depend on cool flame chemistry. Cool flame chemistry also has a significant impact on fuel octane and cetane numbers, whose understanding has large economic consequences.

2021 Sept. 8: GMT 251

Advanced Combustion via Microgravity Experiments (ACME): A crewmember removed and replaced the ACME Igniter Tip #2. The ACME project is a set of six independent studies of gaseous flames to be conducted in the Combustion Integrated Rack (CIR). ACME's primary and secondary goals are to (1)

improve fuel efficiency and reduce pollutant production in practical combustion on Earth, and (2) spacecraft fire prevention through innovative research focused on materials flammability.

2021 Sept. 10: GMT 253

Combustion Integrated Rack – Fluids and Combustion Facility (CIR): A crewmember replaced the FCF Hard Drive installed in the FCF Image Processor in the CIR. The crew photo documented the final configuration of the FCF Image Processor. The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion investigations in microgravity.

2021 Sept. 13: GMT 256

Combustion Integrated Rack/Advanced Combustion via Microgravity Experiments/Cool Flames Investigation with Gasses (CIR/ACME/CFI-G part 2): Following successful repairs last week, this week the crew removed an empty propane fuel bottle and installed a butane fuel bottle, allowing the continuation of the experiment. Spherical Cool Diffusion Flames Burning Gaseous Fuels (Cool Flames Investigation with Gases) observes the chemical reactions of cool diffusion flames. Diffusion flames are created by supplying fuel to a burner and mixing in ambient air for combustion. Cool flames, which burn at extremely low temperatures, are nearly impossible to create in Earth's gravity, but are easily created in microgravity; studying them may improve the understanding of combustion and fires on Earth.

2021 Sept. 22: GMT 265

Cool Flames Investigation with Gases: A crewmember replaced the Manifold #4 fuel bottle with a new bottle containing 100% Butane. Cool diffusion flames were discovered during droplet combustion experiments aboard the ISS in 2012, and this initiated a rapidly growing field of combustion research. A cool flame is one that burns at about 600 degrees Celsius. A typical candle is about two times hotter, burning at around 1,400 degrees Celsius. Most internal combustion engines are designed using computer models that neglect cool flame chemistry, but ignition and flame propagation in engines depend on cool flame chemistry. Cool flame chemistry also has a significant impact on fuel octane and cetane numbers, whose understanding has large economic consequences.

2021 Sept. 27: GMT 270

Combustion Integrated Rack/Advanced Combustion via Microgravity Experiments/Cool Flames Investigation with Gasses (CIR/ACME/CFI-G part 2): The crew removed an empty butane fuel bottle and installed a new butane fuel bottle, allowing the continuation of the experiment series. Spherical Cool Diffusion Flames Burning Gaseous Fuels (Cool Flames Investigation with Gases) observes the chemical reactions of cool diffusion flames. Diffusion flames are created by supplying fuel to a burner and mixing in ambient air for combustion. Cool flames, which burn at relatively low temperatures of ~600 deg C, are nearly impossible to create in Earth's gravity, but are easily created in microgravity; studying them may improve the understanding of combustion and fires on Earth.

2021 Oct. 4: GMT 277

Cool Flames Investigation with Gases: A crewmember replaced the bottle in CIR Manifold #4 with 100 percent C₄H₁₀ (Butane). Cool diffusion flames were discovered during droplet combustion experiments aboard the ISS in 2012, and this initiated a rapidly growing field of combustion research. A cool flame is one that burns at about 600 degrees Celsius. A typical candle is about two times hotter, burning at around 1,400 degrees Celsius. Most internal combustion engines are designed using computer models that neglect cool flame chemistry, but ignition and flame propagation in engines depend on cool flame

chemistry. Cool flame chemistry also has a significant impact on fuel octane and cetane numbers, whose understanding has large economic consequences.

2021 Oct. 12: GMT 285

Advanced Combustion via Microgravity Experiments (ACME): A crewmember replaced the ACME Thermocouple Array with the ACME Chamber Insert partially installed in the chamber. The ACME Chamber was re-Inserted into the chamber, and the crew replaced three ACME controllers. The ACME project is a set of six independent studies of gaseous flames to be conducted in the Combustion Integrated Rack (CIR). ACME's primary and secondary goals are (1) improve fuel efficiency and reduce pollutant production in practical combustion on Earth, and (2) spacecraft fire prevention through innovative research focused on materials flammability.

2021 Oct. 13: GMT 286

Combustion Integrated Rack (CIR): A crewmember replace the CIR Manifold #2 Bottle with a new bottle containing 40% O₂ and 60% N₂. The Manifold #4 Bottle was also replaced with a new bottle containing 30% C₃H₈ and 70% N₂. These bottle changes are in preparation of continued Cool Flame experiment runs. The CIR includes an optics bench, combustion chamber, fuel, and oxidizer control, and five different cameras for performing combustion investigations in microgravity.

2021 Oct. 14: GMT 287

Combustion Integrated Rack (CIR): A crewmember replaced the CIR Manifold #4 Bottle with a new bottle containing 30% C₄H₁₀ and 70% N₂. The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion investigations in microgravity.

Appendix I. Crew Activity per the NASA Space Station Blog

This section consists of excerpts from <https://blogs.nasa.gov/spacestation/>. Hyperlinks were generally not included.

2021 (for all of CFI-G)

First round (normal flames)

2021 April 29 - [Crew-1 Splashdown Waved Off as Station Teems With Science](#)

Fuel bottles supporting combustion experiments were swapped out ...

Second round (inverse flames except for the final test day)

2021 October 14 - [Russian Trio Nears Departure, Rest of Crew Busy with Research, Lab Upkeep](#)

The station's three NASA flight engineers had their hands full today with a host of research and lab upkeep activities in the orbiting lab's U.S. segment. Megan McArthur swapped fuel bottles inside the Combustion Integrated Rack

Appendix J. Crew Activity in Publicly Available NASA Photos

These photos of CFI-G operations are available from <https://images.nasa.gov/>. Additional photos from the ACME project, CIR facility, etc. can be found by searching that website.

Photo ID	Date taken	Summary description
iss064e040953	2021-03-09	Kate Rubins (NASA) with the ACME chamber insert on the workbench, where the burner cannot be seen
iss064e040971	2021-03-09	Kate Rubins (NASA) with the ACME insert within the open Combustion Integrated Rack (CIR) chamber
iss065e047568	2021-05-18	Megan McArthur (NASA) with the Combustion Integrated Rack (CIR) pulled out but not rotated
iss065e156007	2021-07-02	Mark Vande Hei (NASA) with the ACME insert within open Combustion Integrated Rack (CIR) chamber, but barely visible. Akihiko Hoshide (JAXA) is also visible, but was not working on CFI-G
iss065e156020	2021-07-02	Mark Vande Hei (NASA) with the ACME insert partially withdrawn from the open Combustion Integrated Rack (CIR) chamber. Akihiko Hoshide (JAXA) is visible in the background, but was not working on CFI-G
iss065e369687	2021-09-08	Mark Vande Hei (NASA) with the ACME insert within the open Combustion Integrated Rack (CIR) chamber
iss065e369648	2021-09-10	Mark Vande Hei (NASA) with the Combustion Integrated Rack (CIR) pulled out and rotated, but only seen in part and out of focus
iss065e369799	2021-09-10	Mark Vande Hei (NASA) with the Combustion Integrated Rack (CIR) pulled out and rotated

Appendix K. ACME Operations per @ISS_Research on Twitter

These tweets (from https://twitter.com/ISS_Research/) were from the period of the CFI-G experiment.

2021 May 24: GMT 144

 ISS Research 
@ISS_Research

Get a closer look at one of the flames ignited for the @Space_Station ACME CFI-G study! This image shows a hot flame of nitrogen-diluted propane in a microgravity atmosphere of elevated oxygen. Without buoyancy, soot remained in the flame longer and formed the large clusters.



 NASA Glenn Research and International Space Station

 Megan McArthur  @Astro_Megan · May 24
Last week's fire science was cool! 🧡 I worked on our Combustion Integrated Rack, getting it ready for a continued study of cool flame chemistry. What we learn could lead to cleaner burning and more efficient internal combustion engines. go.nasa.gov/3onDwai



5:38 PM · May 24, 2021 · Twitter Web App

2021 June 29: GMT 180

ISS Research ✓
@ISS_Research

...

Things are heating up in space! 🔥

Aboard the @Space_Station, researchers are creating flames in an attempt to study cool flames, which burn at extremely low temperatures, and are nearly impossible to create in Earth's gravity.

go.nasa.gov/3onOwai



0:24 9.2K views

Flame Lit for Cool Flames Investigation with Gases on Space Station

Aboard the space station, researchers are creating flames in an attempt to study cool flames, which burn at extremely low temperatures, and are nearly impossible to create in Earth's gravity.

11:10 AM · Jun 29, 2021 · Twitter Media Studio



ISS Research ✓
@ISS_Research

...

In this clip, soot that forms at ignition escapes from the flame, increasing the heat loss experienced. This causes it to locally quench, creating a hole in the flame and enhancing flammability. As a result, the flame reforms closing the hole, but it opens again due to heat loss.

11:16 AM · Jun 29, 2021 · Twitter Web App

2021 July 16: GMT 197



ISS Research ✓
@ISS_Research

...

Researchers achieved a first aboard the [@Space_Station](#): creating the first gas burning spherical cool flames. Improving our understanding of cool flames, flames that burn at extremely low temperatures, can have important economic consequences on Earth. go.nasa.gov/3AJmhc4



0:21 24.2K views

Space Station Cool Flame Study

The hot flame created during a test for the Cool Flames Investigation with Gases experiment is shown burning. After the flame appears to extinguish, a cool flame is created.

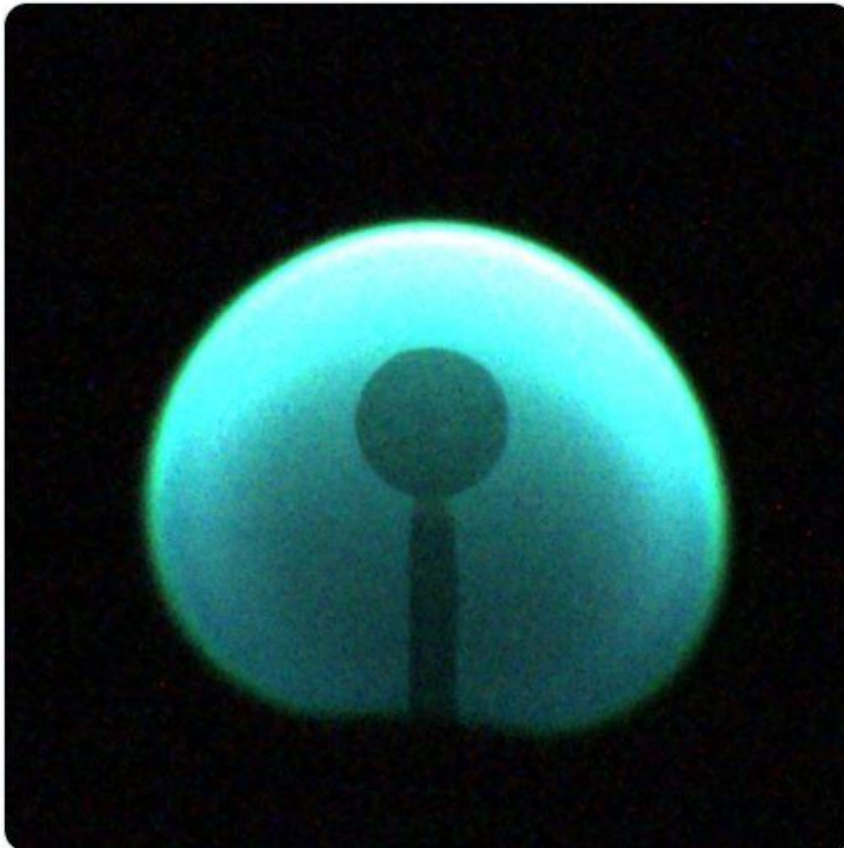
4:30 PM · Jul 16, 2021 · Twitter Media Studio

2021 July 22: GMT 203

ISS Research 
@ISS_Research

...

Testing for @Space_Station combustion research has shifted from normal flames to inverse flames, a first for flame experiments in space! Oxidizer flows from the burner into a combustion chamber filled with gaseous fuel. As a result, these flames are effectively inside out.



 International Space Station and NASA Glenn Research

3:47 PM · Jul 22, 2021 · Twitter Web App



ISS Research 
@ISS_Research

...

Learn more about the Cool Flames Investigation with Gases (CFI-G) experiment which is currently using these inverse flames: go.nasa.gov/3onOwai

3:47 PM · Jul 22, 2021 · Twitter Web App

Appendix L. ACME Operations per Astronauts on Twitter

This tweet is from an ISS crew member who conducted operations in support of the CFI-G experiment. There may be others which are not included here.

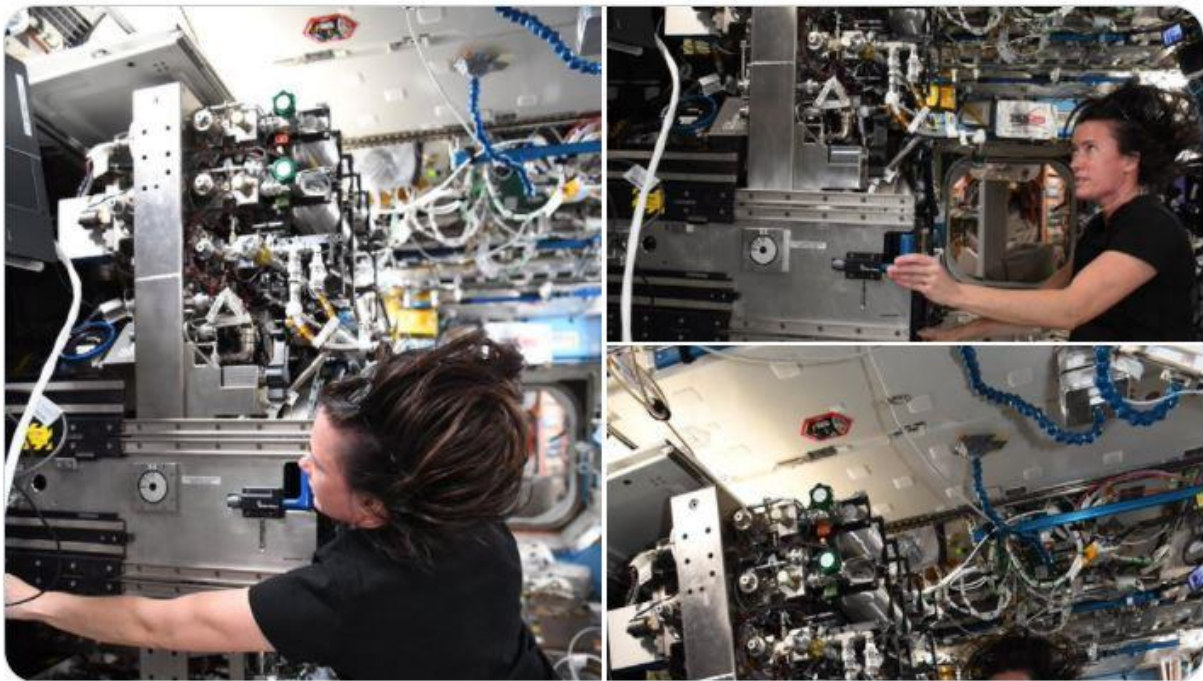
Megan K. McArthur (@Astro_Megan) on Twitter, https://twitter.com/astro_megan/
2021 May 24: GMT 144



Megan McArthur ✓
@Astro_Megan

...

Last week's fire science was cool! 🔥 I worked on our Combustion Integrated Rack, getting it ready for a continued study of cool flame chemistry. What we learn could lead to cleaner burning and more efficient internal combustion engines. go.nasa.gov/3onOwai



5:28 PM · May 24, 2021 · Twitter Web App

Appendix M. Summary of the ISS data collection

The CFI-G data is all digital and includes both image and non-image data, where this is true for the other ACME experiments. To be clear, there was no collection of soot or other physical samples that were subsequently transported to the ground.

The CFI-G image data includes images from the following cameras:

1. ACME DataCam, used with a clear fused-silica ‘filter’ and various zoom and focus settings (but those settings were consistent within each individual test)
2. CFI intensified camera, used always (or at least nearly so) with a formaldehyde filter for CFI-G
3. High-Bitdepth Multispectral (HiBMs) camera, equipped with a red filter corresponding to the opposing CIR illumination package, which was sometimes left off

The raw DataCam images are in a Halcon Iconic Object (hobj) format, while the images from the CFI, and HiBMs cameras are provided in a Tagged Image File (tif) format. In contrast, the analog video was recorded on the ground in a Transport Stream (ts) format.

The key non-imaging data from a test is a comma-delimited file that is described in detail in the *Software Interface Design Description* (ACME-DOC-010 Rev. D).

The raw data is currently archived at NASA and work is underway to make it available publicly through NASA’s Physical Science Informatics System ([PSI](#)).

Appendix N. Publications, presentations, awards, and outreach

Papers, etc. listed in italics are general to ACME rather than specific to the CFI-G experiment.

N.1 Publications and Presentations with ISS Results

Refereed Journal Articles

1. M. Kim, K.A. Waddell, P.B. Sunderland, V. Nayagam, D.P. Stocker, D.L. Dietrich, Y. Ju, F.A. Williams, P. H. Irace, R.L. Axelbaum, *Spherical Gas-Fueled Cool Diffusion Flames*, Proceedings of the Combustion Institute (2022) in press.
<https://doi.org/10.1016/j.proci.2022.07.015>

Conference Proceedings, Abstracts, Posters, Etc.

1. Dennis P. Stocker, *Advanced Combustion via Microgravity Experiments on the International Space Station, Work-in-Progress Poster, 12th U.S. National Combustion Meeting, virtual, 24-26 May 2021.*
2. Dennis P. Stocker, *Recent Research on Flames of Gaseous Fuel Aboard the International Space Station, Conference of the Japanese Society of Microgravity Application (JASMAC-33), virtual, 13-15 Oct. 2021, 11 pp.*
3. Peter B. Sunderland, *Transformative Combustion Research in the ACME Suite of ISS Experiments, 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore MD, 3-6 Nov. 2021.*
4. Dennis P. Stocker, *Laminar Non-Premixed Flames of Gaseous Fuel Aboard the International Space Station, Spring Technical Meeting of The Central States Section of the Combustion Institute, Detroit, Mi, 15-17 May 2022, 10 pp.*
5. M. Kim, K.A. Waddell, P.B. Sunderland, V. Nayagam, D.P. Stocker, D.L. Dietrich, Y. Ju, F.A. Williams., P. H. Irace, R.L. Axelbaum, *The First Observations of Spherical Gas-Fueled Cool Diffusion Flames, Spring Technical Meeting of the Eastern States Section of The Combustion Institute, Orlando, 6 pp., 2022.*

Theses and Dissertations

1. M. Kim, *Spherical Gas-Fueled Cool Diffusion Flames*, MS Thesis, University of Maryland, 60 pp., May 2022.

N.2 Publications and Presentations (comprehensive list)

Papers, etc. in this section which include ISS results are enclosed in borders like this.

Refereed Journal Articles

1. M. Kim, K.A. Waddell, P.B. Sunderland, V. Nayagam, D.P. Stocker, D.L. Dietrich, Y. Ju, F.A. Williams, P. H. Irace, R.L. Axelbaum, *Spherical Gas-Fueled Cool Diffusion Flames*, Proceedings of the Combustion Institute (2022) in press.
<https://doi.org/10.1016/j.proci.2022.07.015>
2. K.A. Waddell, H. Lee, V. Nayagam, R.L. Axelbaum, P.B. Sunderland, *Cool Diffusion Flames in a Stably Stratified Stagnation Flow*, Combustion and Flame, submitted.

Conference Proceedings, Abstracts, Posters, Etc.

1. Dennis P. Stocker, *Advanced Combustion via Microgravity Experiments on the International Space Station, Work-in-Progress Poster, 12th U.S. National Combustion Meeting, virtual, 24-26 May 2021.*
2. Dennis P. Stocker, *Recent Research on Flames of Gaseous Fuel Aboard the International Space Station, Conference of the Japanese Society of Microgravity Application (JASMAC-33), virtual, 13-15 Oct. 2021, 11 pp.*
3. Peter B. Sunderland, *Transformative Combustion Research in the ACME Suite of ISS Experiments, 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore MD, 3-6 Nov. 2021.*
4. Dennis P. Stocker, *Laminar Non-Premixed Flames of Gaseous Fuel Aboard the International Space Station, Spring Technical Meeting of The Central States Section of the Combustion Institute, Detroit, Mi, 15-17 May 2022, 10 pp.*

Theses and Dissertations

1. M. Kim, *Spherical Gas-Fueled Cool Diffusion Flames*, MS Thesis, University of Maryland, 60 pp., May 2022.
2. K.A. Waddell, *Cool Diffusion Flames in Stagnation Flow*, Ph.D. Dissertation, University of Maryland, in preparation.

N.3 Awards

In 2021, CFI-G Principal Investigator Peter Sunderland was inducted as a Fellow of the Combustion Institute for his “significant contributions to understanding laminar flame dynamics and soot formation and oxidation.”

While the awards proceed the CFI-G experiment, CFI-G Co-Investigator Forman Williams was recognized by the Combustion Institute with the Silver Combustion Medal (1978) and the Bernard Lewis Gold Medal (in 1990). He furthermore was inducted in the inaugural 2018 class of Fellows of the Combustion Institute for his “pioneering investigations in energy and combustion including nitric oxide emissions, spray and turbulent combustion, chemical kinetics and fire research.”

N.4 Outreach

NASA Project Scientist Dennis Stocker created the Space Flames Facebook page (www.facebook.com/space.flames) and Flickr site (<https://www.flickr.com/photos/space-flames/>), where some entries are specific to CFI-G. He continues to administer these sites, where most content thus far has been about the ACME project of which CFI-G is a part. As an example of educational outreach, a simple board game related to the ACME’s spherical flames experiments can be downloaded from <https://www1.grc.nasa.gov/space/iss-research/iss-fcf/cir/acme/downloads/>. ACME fact sheets, papers, posters, and presentations providing a project overview and status can also be downloaded from that page. Printed copies of fact sheets have been distributed at meetings of The Combustion Institute and the American Society for Gravitational and Space Research ([ASGSR](https://www.asgsr.org/)).

The University of Maryland and Washington University in St. Louis promoted awareness of the CFI-G investigation online, as can be seen via these links:

- [‘Cool Flame’ Discovered Aboard International Space Station](#)
- [‘Cool flames’ ignited in space](#)

- [Fire Protection Experts Available to the Media](#)
- [Fires in Microgravity](#)
- [FPE Professor Peter Sunderland Designated PI on Upcoming NIST and NSF Projects](#)
- [FPE Prof Peter Sunderland discusses BRE experiments, currently aboard the ISS, with media.](#)
- [Sunderland Gives Invited Talk at ASGSR 2021 Symposium](#)
- [Unlocking the Mysteries of Fire ... in Space](#)
- [Unusual "cool flames" discovered aboard International Space Station](#)

CASIS and NSF, which selected the experiment for spaceflight, promoted awareness of the research as can be seen through these links:

- [Gravity's grip on heat and fire to be studied in space](#)
- [Science in Space: NSF & ISS Briefing](#)
- [The National Science Foundation and the ISS National Lab Announce Three Selected Projects from Combustion and Thermal Transport Funding Opportunity Onboard the International Space Station](#)
- [Unusual "cool flames" discovered aboard International Space Station](#)

Tweets and Facebook posts from the investigators' organizations and NSF are found on subsequent pages, where there could easily be more.

Meanwhile, NASA promoted awareness of ISS combustion research with the video '[The Power of Studying Combustion on the ISS](#)' which was posted on March 13, 2018.

Awareness of ISS combustion research, including the ACME project, was promoted on Oct. 15, 2018 by The Combustion Institute in their online news article, "[International Space Station \(ISS\) Combustion Experiments.](#)"

A photo collage and 1-paragraph article on ACME by Michael Greshko titled "Cosmic Combustion" was published on page 22 of the Oct. 2022 issue of the National Geographic Magazine, for which there is an [online version](#). While general to ACME, the article addresses CFI-G with a statement about the flames: "In microgravity they can swell into ethereal ... orbs that can burn at a surprisingly cool 900 degrees Fahrenheit (a gas stove burner's high setting is about 3100°F)." Despite its brevity, Sunderland and Stocker were both interviewed for the article.

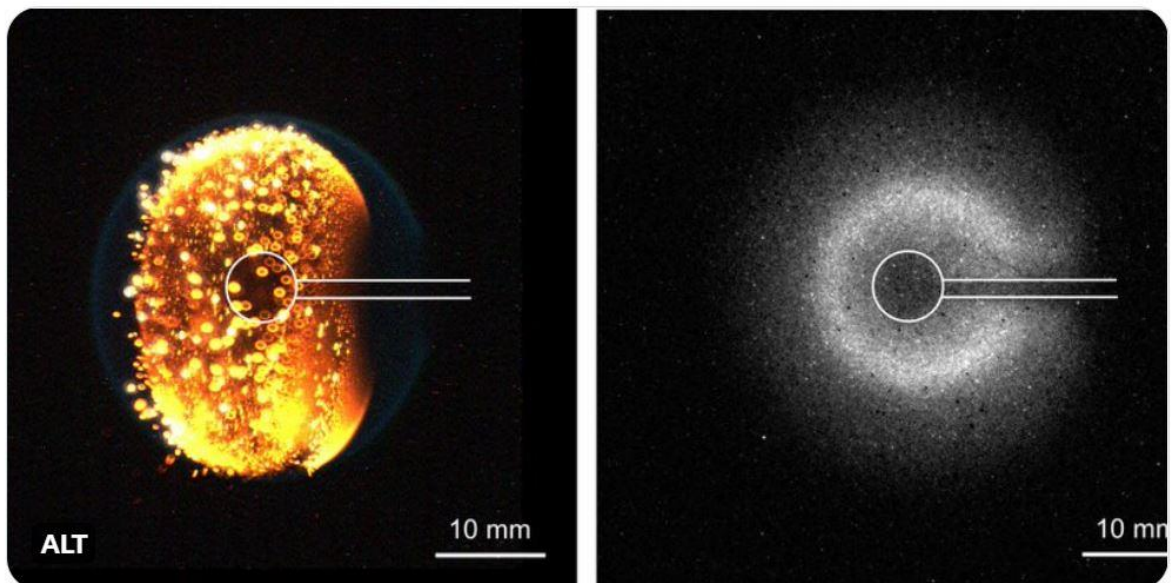
Stocker is involved in a broad range of NASA educational outreach to students and teachers, Scouts of both genders, and the general public. His primary outreach focus is the fundamentals of microgravity and human spaceflight, for example through [microgravity design challenges](#) which he helps create and administer. But he periodically addresses other topics, e.g., where he sometimes speaks on 'NASA and the Science of Superman' – for example at the [Superman Celebration](#) in Metropolis, IL. In this regard, his NASA outreach is carried out locally, regionally, nationally, and online. For example, he regularly presents on microgravity at Space Center Houston's Space Exploration Educators Conference ([SEEC](#)). He has exhibited the last five National Scout Jamborees, five national Boy Scout conferences, the last three national Girl Scout conventions, and the 2019 World Scout Jamboree (which was held in the U.S.). Stocker has been a key lead for the agency in NASA's participation in these Scouting events, where as an example the latter outreach is visually described at www.nasa.gov/feature/stem/world-scout-jamboree.html.

2021 July 15: GMT 196

Maryland Engineering ✓
@ClarkSchool

...

A new class of fire, spherical cool diffusion flames or “cool flames,” has been documented aboard the International @Space_Station. The discovery of these flames—led by UMD's Peter Sunderland—could ignite development of the combustion engine of the future.
go.umd.edu/cfiss



ISS Research and Univ. of Maryland

8:59 AM · Jul 15, 2021 · Twitter Web App

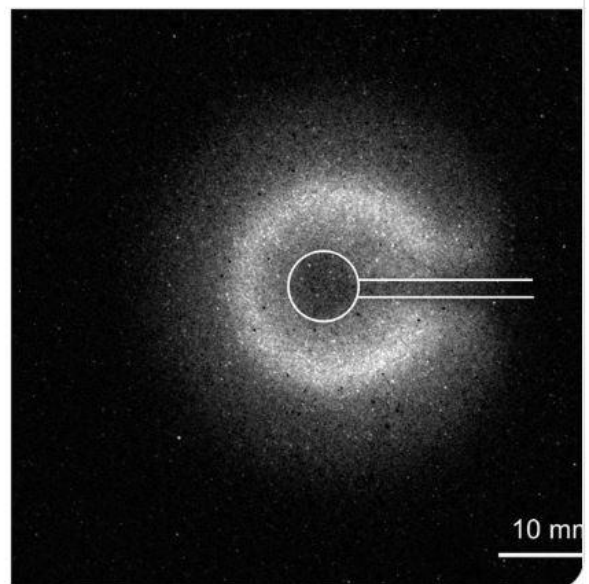
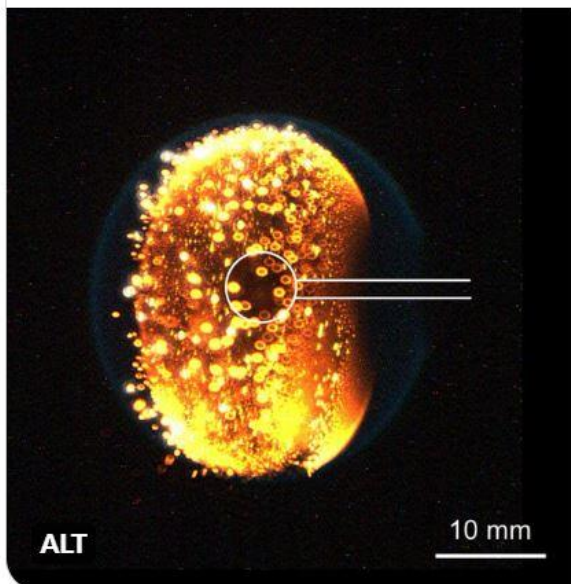
2021 July 15: GMT 196**UMD Right Now**
@UMDRightNow

...

What is a "cool flame?" Researchers from [@ClarkSchool](#) have discovered a new class of fire aboard the International [@Space_Station](#). Click below to read more:

**Maryland Engineering** @ClarkSchool · Jul 15, 2021

A new class of fire, spherical cool diffusion flames or “cool flames,” has been documented aboard the International @Space_Station. The discovery of these flames—led by UMD’s Peter Sunderland—could ignite development of the combustion engine of the future. go.umd.edu/cfiss



11:55 AM · Jul 19, 2021 · Twitter Web App

2021 July 20: GMT 201

WashU Tech Transfer
@WashUOTM

...

“Being able to perform experiments in microgravity allowed us to observe a flame that just would not exist on Earth. This type of discovery is what makes the [@Space_Station](#) so invaluable to scientific exploration.” — Richard Axelbaum [@WashUengineers](#)



source.wustl.edu

'Cool flames' ignited in space - The Source - Washington U...
Richard Axelbaum in the McKelvey School of Engineering at Washington University in St. Louis and collaborators have ...

9:24 AM · Jul 20, 2021 · Twitter Web App

2021 July 27: GMT 208

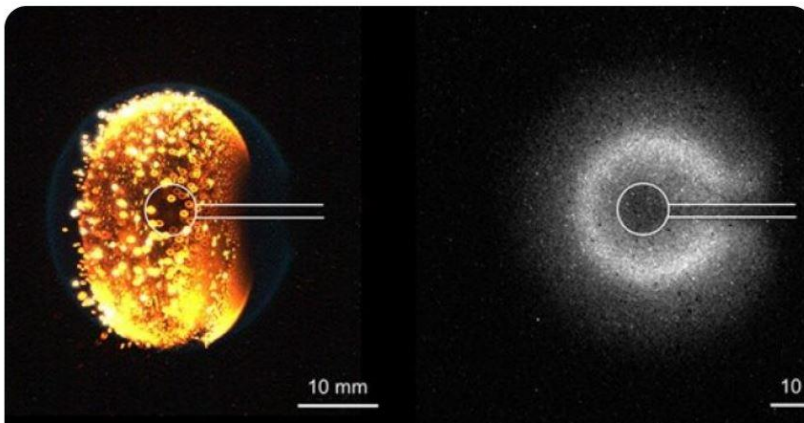
National Science Foundation
@NSF

...

“Cool flames” in space? 🔥

[@UofMaryland](#) discovered super low-temperature fire on the [@Space_Station](#) - & changed our understanding of what fire is. It could also lead to a new combustion engine. bit.ly/3x4kZ8n

[#NSFfunded](#)



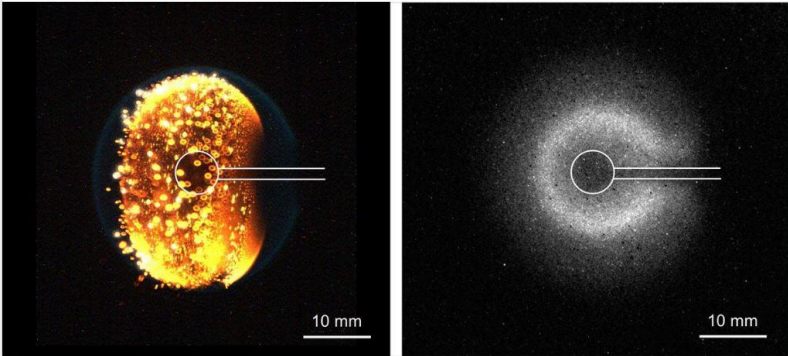
beta.nsf.gov

Unusual "cool flames" discovered aboard International Space Station

10:19 AM · Jul 27, 2021 · Twitter Web App

2021 Aug. 2: GMT 214

University of Maryland Department of Fire Protection Engineering
August 2, 2021 · 🌐



A. James Clark School of Engineering
July 15, 2021 · 🌐

An entirely new class of fire, spherical cool diffusion flames or simply "cool flames," has been documented aboard the International Space Station. The discovery of these dim, quiet flames—led by Professor Peter Sunderland—could literally ignite development of the combustion engine of the future.

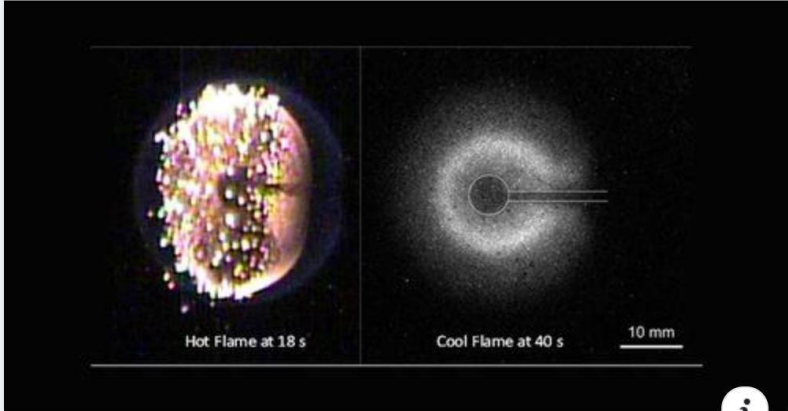
Read more: <https://go.umd.edu/cfiss>

👍 16 2 Comments

2021 Aug. 3: GMT 215

Washington University in St. Louis McKelvey School of Engineering
August 3, 2021 · 🌐

"Being able to perform experiments in microgravity allowed us to observe a flame that just would not exist on Earth. This type of discovery is what makes the Space Station so invaluable to scientific exploration." - Richard Axelbaum, the Stifel & Quinette Jens Professor of Environmental Engineering Science



ENGINEERING.WUSTL.EDU

'Cool flames' ignited in space
Experiments in microgravity allow Axelbaum, collaborators to obser...

👍 2